Finite projective local models for constructible pro-etale complexes

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Editorial note. Complete original mathematical chapter and shared bibliography retained. External reference numbers are not invented; exact source labels are displayed as pointers. Original assumptions, statement and author proof unchanged.

Allowed standard prerequisites (editorial).

- Sheaf cohomology, derived completion, constructible complexes and the pro-etale site. The complete original chapter supplies local definitions and support; standard external pointers and shared bibliography remain.. The selected projective model construction and amplitude induction remain part of the proof.

1 Introduction

The material in this chapter and more can be found in the preprint [2].

The goal of this chapter is to introduce the pro-étale topology and to develop the basic theory of cohomology of abelian sheaves in this topology. A secondary goal is to show how using the pro-étale topology simplifies the introduction of ℓ-adic cohomology in algebraic geometry.

Here is a brief overview of the history of ℓ-adic étale cohomology as we have understood it. In [5, Exposés V and VI] Grothendieck et al developed a theory for dealing with ℓ-adic sheaves as inverse systems of sheaves of $\mathrm{Z}/\ell^n\mathrm{Z}$-modules. In his second paper on the Weil conjectures ([3]) Deligne introduced a derived category of ℓ-adic sheaves as a certain 2-limit of categories of complexes of sheaves of $\mathrm{Z}/\ell^n\mathrm{Z}$-modules on the étale site of a scheme X. This approach is used in the paper by Beilinson, Bernstein, and Deligne ([1]) as the basis for their beautiful theory of perverse sheaves. In a paper entitled "Continuous Étale Cohomology" ([6]) Uwe Jannsen discusses an important variant of the cohomology of a ℓ-adic sheaf on a variety over a field. His paper is followed up by a paper of Torsten Ekedahl ([4]) who discusses the adic formalism needed to work comfortably with derived categories defined as limits.

It turns out that, working with the pro-étale site of a scheme, one can avoid some of the technicalities these authors encountered. This comes at the expense of having to work with non-Noetherian schemes, even when one is only interested in working with ℓ-adic sheaves and cohomology of such on varieties over an algebraically closed field.

A very important and remarkable feature of the (small) pro-étale site of a scheme is that it has enough quasi-compact w-contractible objects. The existence of these objects implies a number of useful and (perhaps) unusual consequences for the derived category of abelian sheaves and for inverse systems of sheaves. This is exactly the feature that will allow us to handle the intricacies of working with ℓ-adic sheaves, but as we will see it has a number of other benefits as well.

2 Some topology

Some preliminaries. We have defined spectral spaces and spectral maps of spectral spaces in Topology, Section [original label: `topology-section-spectral`]. The spectrum of a ring is a spectral space, see Algebra, Lemma [original label: `algebra-lemma-spec-spectral`].

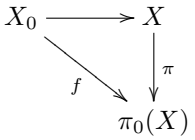
Lemma 2.1. Let X be a spectral space. Let $X_0 \subset X$ be the set of closed points. The following are equivalent

1. Every open covering of X can be refined by a finite disjoint union decomposition $X = \coprod U_i$ with U_i open and closed in X.

2. The composition $X_0 \to X \to \pi_0(X)$ is bijective.

Moreover, if X_0 is closed in X and every point of X specializes to a unique point of X_0, then these conditions are satisfied.

Proof. We will use without further mention that X_0 is quasi-compact (Topology, Lemma [original label: `topology-lemma-closed-points-quasi-compact`]) and $\pi_0(X)$ is profinite (Topology, Lemma [original label: `topology-lemma-spectral-pi0`]). Picture



If (2) holds, the continuous bijective map $f : X_0 \to \pi_0(X)$ is a homeomorphism by Topology, Lemma [original label: `topology-lemma-bijective-map`]. Given an open covering $X = \bigcup U_i$, we get an open covering $\pi_0(X) = \bigcup f(X_0 \cap U_i)$. By Topology, Lemma [original label: `topology-lemma-profinite-refine-open-covering`] we can find a finite open covering of the form $\pi_0(X) = \coprod V_j$ which refines this covering. Since $X_0 \to \pi_0(X)$ is bijective each connected component of X has a unique closed point, whence is equal to the set of points specializing to this closed point. Hence $\pi^{-1}(V_j)$ is the set of points specializing to the points of $f^{-1}(V_j)$. Now, if $f^{-1}(V_j) \subset X_0 \cap U_i \subset U_i$, then it follows that $\pi^{-1}(V_j) \subset U_i$ (because the open set U_i is closed under generalizations). In this way we see that the open covering $X = \coprod \pi^{-1}(V_j)$ refines the covering we started out with. In this way we see that (2) implies (1).

Assume (1). Let $x, y \in X$ be closed points. Then we have the open covering $X = (X \setminus \{x\}) \cup (X \setminus \{y\})$. It follows from (1) that there exists a disjoint union decomposition $X = U \amalg V$ with U and V open (and closed) and $x \in U$ and $y \in V$. In particular we see that every connected component of X has at most one closed point. By Topology, Lemma [original label: `topology-lemma-quasi-compact-closed-point`] every connected component (being closed) also does have a closed point. Thus $X_0 \to \pi_0(X)$ is bijective. In this way we see that (1) implies (2).

Assume X_0 is closed in X and every point specializes to a unique point of X_0. Then X_0 is a spectral space (Topology, Lemma [original label: `topology-lemma-spectral-sub`]) consisting of closed points, hence profinite (Topology, Lemma [original label: `topology-lemma-characterize-profinite-spectral`]). Let $x, y \in X_0$ be distinct. By Topology, Lemma [original label: `topology-lemma-profinite-refine-open-covering`] we can find a disjoint union decomposition $X_0 = U_0 \amalg V_0$ with U_0 and V_0 open and closed and $x \in U_0$ and $y \in V_0$. Let $U \subset X$, resp. $V \subset X$ be the set of points specializing to U_0, resp. V_0. Observe that $X = U \amalg V$. By Topology, Lemma [original label: `topology-lemma-make-spectral-space`] we see that U is an intersection of quasi-compact open subsets. Hence U is closed in the constructible topology. Since U is closed under specialization, we see that U is closed by Topology, Lemma [original label: `topology-lemma-constructible-stable-specialization-closed`]. By symmetry V is closed and hence U and V are both open and closed. This proves that x, y are not in the same connected component of X. In other words, $X_0 \to \pi_0(X)$ is injective. The map is also surjective by Topology, Lemma [original label: `topology-lemma-quasi-compact-closed-point`] and the fact that connected components are closed. In this way we see that the final condition implies (2). □

Example 2.2. Let T be a profinite space. Let $t \in T$ be a point and assume that $T \setminus \{t\}$ is not quasi-compact. Let $X = T \times \{0, 1\}$. Consider the topology on X with a subbase given by the sets $U \times \{0, 1\}$ for $U \subset T$ open, $X \setminus \{(t, 0)\}$, and $U \times \{1\}$ for $U \subset T$ open with $t \notin U$. The set of closed points of X is $X_0 = T \times \{0\}$ and $(t, 1)$ is in the closure of X_0. Moreover, $X_0 \to \pi_0(X)$ is a bijection. This example shows that conditions (1) and (2) of Lemma 2.1 do no imply the set of closed points is closed.

It turns out it is more convenient to work with spectral spaces which have the slightly stronger property mentioned in the final statement of Lemma 2.1. We give this property a name.

Definition 2.3. A spectral space X is w-local if the set of closed points X_0 is closed and every point of X specializes to a unique closed point. A continuous map $f : X \to Y$ of w-local spaces is w-local if it is spectral and maps any closed point of X to a closed point of Y.

We have seen in the proof of Lemma 2.1 that in this case $X_0 \to \pi_0(X)$ is a homeomorphism and that $X_0 \cong \pi_0(X)$ is a profinite space. Moreover, a connected component of X is exactly the set of points specializing to a given $x \in X_0$.

Lemma 2.4. Let X be a w-local spectral space. If $Y \subset X$ is closed, then Y is w-local.

Proof. The subset $Y_0 \subset Y$ of closed points is closed because $Y_0 = X_0 \cap Y$. Since X is w-local, every $y \in Y$ specializes to a unique point of X_0. This specialization is in Y, and hence also in Y_0, because $\overline{\{y\}} \subset Y$. In conclusion, Y is w-local. □

Lemma 2.5. Let X be a spectral space. Let

$$\begin{array}{ccc} Y & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

be a cartesian diagram in the category of topological spaces with T profinite. Then Y is spectral and $T = \pi_0(Y)$. If moreover X is w-local, then Y is w-local, $Y \to X$ is w-local, and the set of closed points of Y is the inverse image of the set of closed points of X.

Proof. Note that Y is a closed subspace of $X \times T$ as $\pi_0(X)$ is a profinite space hence Hausdorff (use Topology, Lemmas [original label: `topology-lemma-spectral-pi0`] and [original label: `topology-lemma-fibre-product-closed`]). Since $X \times T$ is spectral (Topology, Lemma [original label: `topology-lemma-product-spectral-spaces`]) it follows that Y is spectral (Topology, Lemma [original label: `topology-lemma-spectral-sub`]). Let $Y \to \pi_0(Y) \to T$ be the canonical factorization (Topology, Lemma [original label: `topology-lemma-space-connected-components`]). It is clear that $\pi_0(Y) \to T$ is surjective. The fibres of $Y \to T$ are homeomorphic to the fibres of $X \to \pi_0(X)$. Hence these fibres are connected. It follows that $\pi_0(Y) \to T$ is injective. We conclude that $\pi_0(Y) \to T$ is a homeomorphism by Topology, Lemma [original label: `topology-lemma-bijective-map`].

Next, assume that X is w-local and let $X_0 \subset X$ be the set of closed points. The inverse image $Y_0 \subset Y$ of X_0 in Y maps bijectively onto T as $X_0 \to \pi_0(X)$ is a bijection by Lemma 2.1. Moreover, Y_0 is quasi-compact as a closed subset of the spectral space Y. Hence $Y_0 \to \pi_0(Y) = T$ is a homeomorphism by Topology, Lemma [original label: `topology-lemma-bijective-map`]. It follows that all points of Y_0 are closed in Y. Conversely, if $y \in Y$ is a closed point, then it is closed in the fibre of $Y \to \pi_0(Y) = T$ and hence its image x in X is closed in the (homeomorphic) fibre of $X \to \pi_0(X)$. This implies $x \in X_0$ and hence $y \in Y_0$. Thus Y_0 is the collection of closed points of Y and for each $y \in Y_0$ the set of generalizations of y is the fibre of $Y \to \pi_0(Y)$. The lemma follows. □

3 Local isomorphisms

We start with a definition.

Definition 3.1. Let $\varphi : A \to B$ be a ring map.

1. We say $A \to B$ is a local isomorphism if for every prime $\mathfrak{q} \subset B$ there exists a $g \in B$, $g \notin \mathfrak{q}$ such that $A \to B_g$ induces an open immersion $\mathrm{Spec}(B_g) \to \mathrm{Spec}(A)$.
2. We say $A \to B$ identifies local rings if for every prime $\mathfrak{q} \subset B$ the canonical map $A_{\varphi^{-1}(\mathfrak{q})} \to B_{\mathfrak{q}}$ is an isomorphism.

We list some elementary properties.

Lemma 3.2. Let $A \to B$ and $A \to A'$ be ring maps. Let $B' = B \otimes_A A'$ be the base change of B.

1. If $A \to B$ is a local isomorphism, then $A' \to B'$ is a local isomorphism.
2. If $A \to B$ identifies local rings, then $A' \to B'$ identifies local rings.

Proof. Omitted. □

Lemma 3.3. Let $A \to B$ and $B \to C$ be ring maps.

1. If $A \to B$ and $B \to C$ are local isomorphisms, then $A \to C$ is a local isomorphism.
2. If $A \to B$ and $B \to C$ identify local rings, then $A \to C$ identifies local rings.

Proof. Omitted. □

Lemma 3.4. Let A be a ring. Let $B \to C$ be an A-algebra homomorphism.

1. If $A \to B$ and $A \to C$ are local isomorphisms, then $B \to C$ is a local isomorphism.

2. If $A \to B$ and $A \to C$ identify local rings, then $B \to C$ identifies local rings.

Proof. Omitted. □

Lemma 3.5. Let $A \to B$ be a local isomorphism. Then

1. $A \to B$ is étale,
2. $A \to B$ identifies local rings,
3. $A \to B$ is quasi-finite.

Proof. Omitted. □

Lemma 3.6. Let $A \to B$ be a local isomorphism. Then there exist $n \geq 0$, $g_1, \ldots, g_n \in B$, $f_1, \ldots, f_n \in A$ such that $(g_1, \ldots, g_n) = B$ and $A_{f_i} \cong B_{g_i}$.

Proof. Omitted. □

Lemma 3.7. Let $p : (Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ and $q : (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$ be morphisms of locally ringed spaces. If $\mathcal{O}_Y = p^{-1}\mathcal{O}_X$, then

$$\mathrm{Mor}_{\mathrm{LRS}/(X,\mathcal{O}_X)}((Z, \mathcal{O}_Z), (Y, \mathcal{O}_Y)) \longrightarrow \mathrm{Mor}_{\mathrm{Top}/X}(Z, Y), \quad (f, f^\sharp) \longmapsto f$$

is bijective. Here $\mathrm{LRS}/(X, \mathcal{O}_X)$ is the category of locally ringed spaces over X and Top/X is the category of topological spaces over X.

Proof. This is immediate from the definitions. □

Lemma 3.8. Let A be a ring. Set $X = \mathrm{Spec}(A)$. The functor

$$B \longmapsto \mathrm{Spec}(B)$$

from the category of A-algebras B such that $A \to B$ identifies local rings to the category of topological spaces over X is fully faithful.

Proof. This follows from Lemma 3.7 and the fact that if $A \to B$ identifies local rings, then the pullback of the structure sheaf of $\mathrm{Spec}(A)$ via $p : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is equal to the structure sheaf of $\mathrm{Spec}(B)$. □

4 Ind-Zariski algebra

We start with a definition; please see Remark 6.9 for a comparison with the corresponding definition of the article [2].

Definition 4.1. A ring map $A \to B$ is said to be ind-Zariski if B can be written as a filtered colimit $B = \mathrm{colim}\, B_i$ with each $A \to B_i$ a local isomorphism.

An example of an Ind-Zariski map is a localization $A \to S^{-1}A$, see Algebra, Lemma [original label: `algebra-lemma-localization-colimit`]. The category of ind-Zariski algebras is closed under several natural operations.

Lemma 4.2. Let $A \to B$ and $A \to A'$ be ring maps. Let $B' = B \otimes_A A'$ be the base change of B. If $A \to B$ is ind-Zariski, then $A' \to B'$ is ind-Zariski.

Proof. Omitted. □

Lemma 4.3. Let $A \to B$ and $B \to C$ be ring maps. If $A \to B$ and $B \to C$ are ind-Zariski, then $A \to C$ is ind-Zariski.

Proof. Omitted. □

Lemma 4.4. Let A be a ring. Let $B \to C$ be an A-algebra homomorphism. If $A \to B$ and $A \to C$ are ind-Zariski, then $B \to C$ is ind-Zariski.

Proof. Omitted. □

Lemma 4.5. A filtered colimit of ind-Zariski A-algebras is ind-Zariski over A.

Proof. Omitted. □

Lemma 4.6. Let $A \to B$ be ind-Zariski. Then $A \to B$ identifies local rings,

Proof. Omitted. □

5 Constructing w-local affine schemes

An affine scheme X is called w-local if its underlying topological space is w-local (Definition 2.3). It turns out given any ring A there is a canonical faithfully flat ind-Zariski ring map $A \to A_w$ such that $\mathrm{Spec}(A_w)$ is w-local. The key to constructing A_w is the following simple lemma.

Lemma 5.1. Let A be a ring. Set $X = \mathrm{Spec}(A)$. Let $Z \subset X$ be a locally closed subscheme which is of the form $D(f) \cap V(I)$ for some $f \in A$ and ideal $I \subset A$. Then

1. there exists a multiplicative subset $S \subset A$ such that $\mathrm{Spec}(S^{-1}A)$ maps by a homeomorphism to the set of points of X specializing to Z,
2. the A-algebra $A^\sim_Z = S^{-1}A$ depends only on the underlying locally closed subset $Z \subset X$,
3. Z is a closed subscheme of $\mathrm{Spec}(A^\sim_Z)$,

If $A \to A'$ is a ring map and $Z' \subset X' = \mathrm{Spec}(A')$ is a locally closed subscheme of the same form which maps into Z, then there is a unique A-algebra map $A^\sim_Z \to (A')^\sim_{Z'}$.

Proof. Let $S \subset A$ be the multiplicative set of elements which map to invertible elements of $\Gamma(Z, \mathcal{O}_Z) = (A/I)_f$. If $\mathfrak{p}$ is a prime of A which does not specialize to Z, then $\mathfrak{p}$ generates the unit ideal in $(A/I)_f$. Hence we can write $f^n = g + h$ for some $n \geq 0$, $g \in \mathfrak{p}$, $h \in I$. Then $g \in S$ and we see that $\mathfrak{p}$ is not in the spectrum of $S^{-1}A$. Conversely, if $\mathfrak{p}$ does specialize to Z, say $\mathfrak{p} \subset \mathfrak{q} \supset I$ with $f \notin \mathfrak{q}$, then we see that $S^{-1}A$ maps to $A_\mathfrak{q}$ and hence $\mathfrak{p}$ is in the spectrum of $S^{-1}A$. This proves (1).

The isomorphism class of the localization $S^{-1}A$ depends only on the corresponding subset $\mathrm{Spec}(S^{-1}A) \subset \mathrm{Spec}(A)$, whence (2) holds. By construction $S^{-1}A$ maps surjectively onto $(A/I)_f$, hence (3). The final statement follows as the multiplicative subset $S' \subset A'$ corresponding to Z' contains the image of the multiplicative subset S. □

Let A be a ring. Let $E \subset A$ be a finite subset. We get a stratification of $X = \mathrm{Spec}(A)$ into locally closed subschemes by looking at the vanishing behaviour of the elements of E. More precisely, given a disjoint union decomposition $E = E' \amalg E''$ we set

$$Z(E', E'') = \bigcap\nolimits_{f \in E'} D(f) \cap \bigcap\nolimits_{f \in E''} V(f) = D(\prod\nolimits_{f \in E'} f) \cap V(\sum\nolimits_{f \in E''} fA) \tag{5.1.1}$$

The points of $Z(E', E'')$ are exactly those $x \in X$ such that $f \in E'$ maps to a nonzero element in $\kappa(x)$ and $f \in E''$ maps to zero in $\kappa(x)$. Thus it is clear that

$$X = \coprod\nolimits_{E = E' \amalg E''} Z(E', E'') \tag{5.1.2}$$

set theoretically. Observe that each stratum is constructible.

Lemma 5.2. Let $X = \mathrm{Spec}(A)$ as above. Given any finite stratification $X = \coprod T_i$ by constructible subsets, there exists a finite subset $E \subset A$ such that the stratification (5.1.2) refines $X = \coprod T_i$.

Proof. We may write $T_i = \bigcup_j U_{i,j} \cap V_{i,j}^c$ as a finite union for some $U_{i,j}$ and $V_{i,j}$ quasi-compact open in X. Then we may write $U_{i,j} = \bigcup D(f_{i,j,k})$ and $V_{i,j} = \bigcup D(g_{i,j,l})$. Then we set $E = \{f_{i,j,k}\} \cup \{g_{i,j,l}\}$. This does the job, because the stratification (5.1.2) is the one whose strata are labeled by the vanishing pattern of the elements of E which clearly refines the given stratification. □

We continue the discussion. Given a finite subset $E \subset A$ we set

$$A_E = \prod\nolimits_{E = E' \amalg E''} A^\sim_{Z(E', E'')} \tag{5.2.1}$$

with notation as in Lemma 5.1. This makes sense because (5.1.1) shows that each $Z(E', E'')$ has the correct shape. We take the spectrum of this ring and denote it

$$X_E = \mathrm{Spec}(A_E) = \coprod\nolimits_{E = E' \amalg E''} X_{E', E''} \tag{5.2.2}$$

with $X_{E', E''} = \mathrm{Spec}(A^\sim_{Z(E', E'')})$. Note that

$$Z_E = \coprod\nolimits_{E = E' \amalg E''} Z(E', E'') \longrightarrow X_E \tag{5.2.3}$$

is a closed subscheme. By construction the closed subscheme Z_E contains all the closed points of the affine scheme X_E as every point of $X_{E', E''}$ specializes to a point of $Z(E', E'')$.

Let $I(A)$ be the partially ordered set of all finite subsets of A. This is a directed partially ordered set. For $E_1 \subset E_2$ there is a canonical transition map $A_{E_1} \to A_{E_2}$ of A-algebras. Namely, given a decomposition $E_2 = E_2' \amalg E_2''$ we set $E_1' = E_1 \cap E_2'$ and $E_1'' = E_1 \cap E_2''$. Then observe that $Z(E_2', E_2'') \subset Z(E_1', E_1'')$ hence a unique A-algebra map $A^\sim_{Z(E_1', E_1'')} \to A^\sim_{Z(E_2', E_2'')}$ by Lemma 5.1. Using these maps collectively we obtain the desired ring map $A_{E_1} \to A_{E_2}$. Observe that the corresponding map of affine schemes

$$X_{E_2} \longrightarrow X_{E_1} \tag{5.2.4}$$

maps Z_{E_2} into Z_{E_1}. By uniqueness we obtain a system of A-algebras over $I(A)$ and we set

$$A_w = \operatorname{colim}_{E \in I(A)} A_E \tag{5.2.5}$$

This A-algebra is ind-Zariski and faithfully flat over A. Finally, we set $X_w = \operatorname{Spec}(A_w)$ and endow it with the closed subscheme $Z = \lim_{E \in I(A)} Z_E$. In a formula

$$X_w = \lim_{E \in I(A)} X_E \supset Z = \lim_{E \in I(A)} Z_E \tag{5.2.6}$$

Lemma 5.3. Let $X = \operatorname{Spec}(A)$ be an affine scheme. With $A \to A_w$, $X_w = \operatorname{Spec}(A_w)$, and $Z \subset X_w$ as above.

1. $A \to A_w$ is ind-Zariski and faithfully flat,
2. $X_w \to X$ induces a bijection $Z \to X$,
3. Z is the set of closed points of X_w,
4. Z is a reduced scheme, and
5. every point of X_w specializes to a unique point of Z.

In particular, X_w is w-local (Definition 2.3).

Proof. The map $A \to A_w$ is ind-Zariski by construction. For every E the morphism $Z_E \to X$ is a bijection, hence (2). As $Z \subset X_w$ we conclude $X_w \to X$ is surjective and $A \to A_w$ is faithfully flat by Algebra, Lemma [original label: `algebra-lemma-ff-rings`]. This proves (1).

Suppose that $y \in X_w$, $y \notin Z$. Then there exists an E such that the image of y in X_E is not contained in Z_E. Then for all $E \subset E'$ also y maps to an element of $X_{E'}$ not contained in $Z_{E'}$. Let $T_{E'} \subset X_{E'}$ be the reduced closed subscheme which is the closure of the image of y. It is clear that $T = \lim_{E \subset E'} T_{E'}$ is the closure of y in X_w. For every $E \subset E'$ the scheme $T_{E'} \cap Z_{E'}$ is nonempty by construction of $X_{E'}$. Hence $\lim T_{E'} \cap Z_{E'}$ is nonempty and we conclude that $T \cap Z$ is nonempty. Thus y is not a closed point. It follows that every closed point of X_w is in Z.

Suppose that $y \in X_w$ specializes to $z, z' \in Z$. We will show that $z = z'$ which will finish the proof of (3) and will imply (5). Let $x, x' \in X$ be the images of z and z'. Since $Z \to X$ is bijective it suffices to show that $x = x'$. If $x \neq x'$, then there exists an $f \in A$ such that $x \in D(f)$ and $x' \in V(f)$ (or vice versa). Set $E = \{f\}$ so that

$$X_E = \operatorname{Spec}(A_f) \amalg \operatorname{Spec}(A^\sim_{V(f)})$$

Then we see that z and z' map x_E and x'_E which are in different parts of the given decomposition of X_E above. But then it impossible for x_E and x'_E to be specializations of a common point. This is the desired contradiction.

Recall that given a finite subset $E \subset A$ we have Z_E is a disjoint union of the locally closed subschemes $Z(E', E'')$ each isomorphic to the spectrum of $(A/I)_f$ where I is the ideal generated by E'' and f the product of the elements of E'. Any nilpotent element b of $(A/I)_f$ is the class of g/f^n for some $g \in A$. Then setting $E' = E \cup \{g\}$ the reader verifies that b is pulls back to zero under the transition map $Z_{E'} \to Z_E$ of the system. This proves (4). □

Remark 5.4. Let A be a ring. Let κ be an infinite cardinal bigger or equal than the cardinality of A. Then the cardinality of A_w (Lemma 5.3) is at most κ. Namely, each A_E has cardinality at most κ and the set of finite subsets of A has cardinality at most κ as well. Thus the result follows as $\kappa \otimes \kappa = \kappa$, see Sets, Section [original label: `sets-section-cardinals`].

Lemma 5.5 (Universal property of the construction). Let A be a ring. Let $A \to A_w$ be the ring map constructed in Lemma 5.3. For any ring map $A \to B$ such that $\operatorname{Spec}(B)$ is w-local, there is a unique factorization $A \to A_w \to B$ such that $\operatorname{Spec}(B) \to \operatorname{Spec}(A_w)$ is w-local.

Proof. Denote $Y = \mathrm{Spec}(B)$ and $Y_0 \subset Y$ the set of closed points. Denote $f : Y \to X$ the given morphism. Recall that Y_0 is profinite, in particular every constructible subset of Y_0 is open and closed. Let $E \subset A$ be a finite subset. Recall that $A_w = \mathrm{colim}\, A_E$ and that the set of closed points of $\mathrm{Spec}(A_w)$ is the limit of the closed subsets $Z_E \subset X_E = \mathrm{Spec}(A_E)$. Thus it suffices to show there is a unique factorization $A \to A_E \to B$ such that $Y \to X_E$ maps Y_0 into Z_E. Since $Z_E \to X = \mathrm{Spec}(A)$ is bijective, and since the strata $Z(E', E'')$ are constructible we see that

$$Y_0 = \coprod f^{-1}(Z(E', E'')) \cap Y_0$$

is a disjoint union decomposition into open and closed subsets. As $Y_0 = \pi_0(Y)$ we obtain a corresponding decomposition of Y into open and closed pieces. Thus it suffices to construct the factorization in case $f(Y_0) \subset Z(E', E'')$ for some decomposition $E = E' \amalg E''$. In this case $f(Y)$ is contained in the set of points of X specializing to $Z(E', E'')$ which is homeomorphic to $X_{E',E''}$. Thus we obtain a unique continuous map $Y \to X_{E',E''}$ over X. By Lemma 3.7 this corresponds to a unique morphism of schemes $Y \to X_{E',E''}$ over X. This finishes the proof. □

Recall that the spectrum of a ring is profinite if and only if every point is closed. There are in fact a whole slew of equivalent conditions that imply this. See Algebra, Lemma [original label: `algebra-lemma-ring-with-only-minimal-primes`] or Topology, Lemma [original label: `topology-lemma-characterize-profinite-spectral`].

Lemma 5.6. Let A be a ring such that $\mathrm{Spec}(A)$ is profinite. Let $A \to B$ be a ring map. Then $\mathrm{Spec}(B)$ is profinite in each of the following cases:

1. if $\mathfrak{q}, \mathfrak{q}' \subset B$ lie over the same prime of A, then neither $\mathfrak{q} \subset \mathfrak{q}'$, nor $\mathfrak{q}' \subset \mathfrak{q}$,
2. $A \to B$ induces algebraic extensions of residue fields,
3. $A \to B$ is a local isomorphism,
4. $A \to B$ identifies local rings,
5. $A \to B$ is weakly étale,
6. $A \to B$ is quasi-finite,
7. $A \to B$ is unramified,
8. $A \to B$ is étale,
9. B is a filtered colimit of A-algebras as in (1) – (8),
10. etc.

Proof. By the references mentioned above (Algebra, Lemma [original label: `algebra-lemma-ring-with-only-minimal-primes`] or Topology, Lemma [original label: `topology-lemma-characterize-profinite-spectral`]) there are no specializations between distinct points of $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$ is profinite if and only if there are no specializations between distinct points of $\mathrm{Spec}(B)$. These specializations can only happen in the fibres of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$. In this way we see that (1) is true.

The assumption in (2) implies all primes of B are maximal by Algebra, Lemma [original label: `algebra-lemma-finite-residue-extension-closed`]. Thus (2) holds. If $A \to B$ is a local isomorphism or identifies local rings, then the residue field extensions are trivial, so (3) and (4) follow from (2). If $A \to B$ is weakly étale, then More on Algebra, Lemma [original label: `more-algebra-lemma-weakly-etale-residue-field-extensions`] tells us it induces separable algebraic residue field extensions, so (5) follows from (2). If $A \to B$ is quasi-finite, then the fibres are finite discrete topological spaces. Hence (6) follows from (1). Hence (3) follows from (1). Cases (7) and (8) follow from this as unramified and étale ring map are quasi-finite (Algebra, Lemmas [original label: `algebra-lemma-unramified-quasi-finite`] and [original label: `algebra-lemma-etale-quasi-finite`]). If $B = \mathrm{colim}\, B_i$ is a filtered colimit of A-algebras, then $\mathrm{Spec}(B) = \lim \mathrm{Spec}(B_i)$ in the category of topological spaces by Limits, Lemma [original label: `limits-lemma-inverse-limit-top`]. Hence if each $\mathrm{Spec}(B_i)$ is profinite, so is $\mathrm{Spec}(B)$ by Topology, Lemma [original label: `topology-lemma-directed-inverse-limit-profinite`]. This proves (9). □

Lemma 5.7. Let A be a ring. Let $V(I) \subset \mathrm{Spec}(A)$ be a closed subset which is a profinite topological space. Then there exists an ind-Zariski ring map $A \to B$ such that $\mathrm{Spec}(B)$ is w-local, the set of closed points is $V(IB)$, and $A/I \cong B/IB$.

Proof. Let $A \to A_w$ and $Z \subset Y = \mathrm{Spec}(A_w)$ as in Lemma 5.3. Let $T \subset Z$ be the inverse image of $V(I)$. Then $T \to V(I)$ is a homeomorphism by Topology, Lemma [original label: `topology-lemma-bijective-map`]. Let $B = (A_w)^\sim_T$, see Lemma 5.1. It is clear that B is w-local with closed points $V(IB)$. The ring map $A/I \to B/IB$ is ind-Zariski and induces a homeomorphism on underlying topological spaces. Hence it is an isomorphism by Lemma 3.8. □

Lemma 5.8. Let A be a ring such that $X = \mathrm{Spec}(A)$ is w-local. Let $I \subset A$ be the radical ideal cutting out the set X_0 of closed points in X. Let $A \to B$ be a ring map inducing algebraic extensions on residue fields at primes. Then

1. every point of $Z = V(IB)$ is a closed point of $\mathrm{Spec}(B)$,
2. there exists an ind-Zariski ring map $B \to C$ such that
   (a) $B/IB \to C/IC$ is an isomorphism,
   (b) the space $Y = \mathrm{Spec}(C)$ is w-local,
   (c) the induced map $p : Y \to X$ is w-local, and
   (d) $p^{-1}(X_0)$ is the set of closed points of Y.

Proof. By Lemma 5.6 applied to $A/I \to B/IB$ all points of $Z = V(IB) = \mathrm{Spec}(B/IB)$ are closed, in fact $\mathrm{Spec}(B/IB)$ is a profinite space. To finish the proof we apply Lemma 5.7 to $IB \subset B$. □

6 Identifying local rings versus ind-Zariski

An ind-Zariski ring map $A \to B$ identifies local rings (Lemma 4.6). The converse does not hold (Examples, Section [original label: `examples-section-not-ind-etale`]). However, it turns out that there is a kind of structure theorem for ring maps which identify local rings in terms of ind-Zariski ring maps, see Proposition 6.6.

Let A be a ring. Let $X = \mathrm{Spec}(A)$. The space of connected components $\pi_0(X)$ is a profinite space by Topology, Lemma [original label: `topology-lemma-spectral-pi0`] (and Algebra, Lemma [original label: `algebra-lemma-spec-spectral`]).

Lemma 6.1. Let A be a ring. Let $X = \mathrm{Spec}(A)$. Let $T \subset \pi_0(X)$ be a closed subset. There exists a surjective ind-Zariski ring map $A \to B$ such that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ induces a homeomorphism of $\mathrm{Spec}(B)$ with the inverse image of T in X.

Proof. Let $Z \subset X$ be the inverse image of T. Then Z is the intersection $Z = \bigcap Z_\alpha$ of the open and closed subsets of X containing Z, see Topology, Lemma [original label: `topology-lemma-closed-union-connected-components`]. For each α we have $Z_\alpha = \mathrm{Spec}(A_\alpha)$ where $A \to A_\alpha$ is a local isomorphism (a localization at an idempotent). Setting $B = \mathrm{colim}\, A_\alpha$ proves the lemma. □

Lemma 6.2. Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be a profinite space and let $T \to \pi_0(X)$ be a continuous map. There exists an ind-Zariski ring map $A \to B$ such that with $Y = \mathrm{Spec}(B)$ the diagram

$$\begin{array}{ccc} Y & \longrightarrow & \pi_0(Y) \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

is cartesian in the category of topological spaces and such that $\pi_0(Y) = T$ as spaces over $\pi_0(X)$.

Proof. Namely, write $T = \lim T_i$ as the limit of an inverse system finite discrete spaces over a directed set (see Topology, Lemma [original label: `topology-lemma-profinite`]). For each i let $Z_i = \mathrm{Im}(T \to \pi_0(X) \times T_i)$. This is a closed subset. Observe that $X \times T_i$ is the spectrum of $A_i = \prod_{t \in T_i} A$ and that $A \to A_i$ is a local isomorphism. By Lemma 6.1 we see that $Z_i \subset \pi_0(X \times T_i) = \pi_0(X) \times T_i$ corresponds to a surjection $A_i \to B_i$ which is ind-Zariski such that $\mathrm{Spec}(B_i) = X \times_{\pi_0(X)} Z_i$ as subsets of $X \times T_i$. The transition maps $T_i \to T_{i'}$ induce maps $Z_i \to Z_{i'}$ and $X \times_{\pi_0(X)} Z_i \to X \times_{\pi_0(X)} Z_{i'}$. Hence ring maps $B_{i'} \to B_i$ (Lemmas 3.8 and 4.6). Set $B = \mathrm{colim}\, B_i$. Because $T = \lim Z_i$ we have $X \times_{\pi_0(X)} T = \lim X \times_{\pi_0(X)} Z_i$ and hence $Y = \mathrm{Spec}(B) = \lim \mathrm{Spec}(B_i)$ fits into the cartesian diagram

$$\begin{array}{ccc} Y & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

of topological spaces. By Lemma 2.5 we conclude that $T = \pi_0(Y)$. □

Example 6.3. Let k be a field. Let T be a profinite topological space. There exists an ind-Zariski ring map $k \to A$ such that $\mathrm{Spec}(A)$ is homeomorphic to T. Namely, just apply Lemma 6.2 to $T \to \pi_0(\mathrm{Spec}(k)) = \{*\}$. In fact, in this case we have

$$A = \mathrm{colim}\, \mathrm{Map}(T_i, k)$$

whenever we write $T = \lim T_i$ as a filtered limit with each T_i finite.

Lemma 6.4. Let $A \to B$ be ring map such that

1. $A \to B$ identifies local rings,
2. the topological spaces $\mathrm{Spec}(B)$, $\mathrm{Spec}(A)$ are w-local,
3. $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is w-local, and
4. $\pi_0(\mathrm{Spec}(B)) \to \pi_0(\mathrm{Spec}(A))$ is bijective.

Then $A \to B$ is an isomorphism

Proof. Let $X_0 \subset X = \mathrm{Spec}(A)$ and $Y_0 \subset Y = \mathrm{Spec}(B)$ be the sets of closed points. By assumption Y_0 maps into X_0 and the induced map $Y_0 \to X_0$ is a bijection. As a space $\mathrm{Spec}(A)$ is the disjoint union of the spectra of the local rings of A at closed points. Similarly for B. Hence $X \to Y$ is a bijection. Since $A \to B$ is flat we have going down (Algebra, Lemma [original label: `algebra-lemma-flat-going-down`]). Thus Algebra, Lemma [original label: `algebra-lemma-unique-prime-over-localize-below`] shows for any prime $\mathfrak{q} \subset B$ lying over $\mathfrak{p} \subset A$ we have $B_\mathfrak{q} = B_\mathfrak{p}$. Since $B_\mathfrak{q} = A_\mathfrak{p}$ by assumption, we see that $A_\mathfrak{p} = B_\mathfrak{p}$ for all primes $\mathfrak{p}$ of A. Thus $A = B$ by Algebra, Lemma [original label: `algebra-lemma-characterize-zero-local`]. □

Lemma 6.5. Let $A \to B$ be ring map such that

1. $A \to B$ identifies local rings,
2. the topological spaces $\mathrm{Spec}(B)$, $\mathrm{Spec}(A)$ are w-local, and
3. $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is w-local.

Then $A \to B$ is ind-Zariski.

Proof. Set $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. Let $X_0 \subset X$ and $Y_0 \subset Y$ be the set of closed points. Let $A \to A'$ be the ind-Zariski morphism of affine schemes such that with $X' = \mathrm{Spec}(A')$ the diagram

$$\begin{array}{ccc} X' & \longrightarrow & \pi_0(X') \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

is cartesian in the category of topological spaces and such that $\pi_0(X') = \pi_0(Y)$ as spaces over $\pi_0(X)$, see Lemma 6.2. By Lemma 2.5 we see that X' is w-local and the set of closed points $X'_0 \subset X'$ is the inverse image of X_0.

We obtain a continuous map $Y \to X'$ of underlying topological spaces over X identifying $\pi_0(Y)$ with $\pi_0(X')$. By Lemma 3.8 (and Lemma 4.6) this corresponds to a morphism of affine schemes $Y \to X'$ over X. Since $Y \to X$ maps Y_0 into X_0 we see that $Y \to X'$ maps Y_0 into X'_0, i.e., $Y \to X'$ is w-local. By Lemma 6.4 we see that $Y \cong X'$ and we win. □

The following proposition is a warm up for the type of result we will prove later.

Proposition 6.6. Let $A \to B$ be a ring map which identifies local rings. Then there exists a faithfully flat, ind-Zariski ring map $B \to B'$ such that $A \to B'$ is ind-Zariski.

Proof. Let $A \to A_w$, resp. $B \to B_w$ be the faithfully flat, ind-Zariski ring map constructed in Lemma 5.3 for A, resp. B. Since $\mathrm{Spec}(B_w)$ is w-local, there exists a unique factorization $A \to A_w \to B_w$ such that $\mathrm{Spec}(B_w) \to \mathrm{Spec}(A_w)$ is w-local by Lemma 5.5. Note that $A_w \to B_w$ identifies local rings, see Lemma 3.4. By Lemma 6.5 this means $A_w \to B_w$ is ind-Zariski. Since $B \to B_w$ is faithfully flat, ind-Zariski (Lemma 5.3) and the composition $A \to B \to B_w$ is ind-Zariski (Lemma 4.3) the proposition is proved. □

The proposition above allows us to characterize the affine, weakly contractible objects in the pro-Zariski site of an affine scheme.

Lemma 6.7. Let A be a ring. The following are equivalent

1. every faithfully flat ring map $A \to B$ identifying local rings has a retraction,
2. every faithfully flat ind-Zariski ring map $A \to B$ has a retraction, and
3. A satisfies
   (a) $\mathrm{Spec}(A)$ is w-local, and
   (b) $\pi_0(\mathrm{Spec}(A))$ is extremally disconnected.

Proof. The equivalence of (1) and (2) follows immediately from Proposition 6.6.

Assume (3)(a) and (3)(b). Let $A \to B$ be faithfully flat and ind-Zariski. We will use without further mention the fact that a flat map $A \to B$ is faithfully flat if and only if every closed point of $\mathrm{Spec}(A)$ is in the image of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$. We will show that $A \to B$ has a retraction.

Let $I \subset A$ be an ideal such that $V(I) \subset \mathrm{Spec}(A)$ is the set of closed points of $\mathrm{Spec}(A)$. We may replace B by the ring C constructed in Lemma 5.8 for $A \to B$ and $I \subset A$. Thus we may assume $\mathrm{Spec}(B)$ is w-local such that the set of closed points of $\mathrm{Spec}(B)$ is $V(IB)$.

Assume $\mathrm{Spec}(B)$ is w-local and the set of closed points of $\mathrm{Spec}(B)$ is $V(IB)$. Choose a continuous section to the surjective continuous map $V(IB) \to V(I)$. This is possible as $V(I) \cong \pi_0(\mathrm{Spec}(A))$ is extremally disconnected, see Topology, Proposition [original label: `topology-proposition-projective-in-category-hausdorff-qc`]. The image is a closed subspace $T \subset \pi_0(\mathrm{Spec}(B)) \cong V(IB)$ mapping homeomorphically onto $\pi_0(A)$. Replacing B by the ind-Zariski quotient ring constructed in Lemma 6.1 we see that we may assume $\pi_0(\mathrm{Spec}(B)) \to \pi_0(\mathrm{Spec}(A))$ is bijective. At this point $A \to B$ is an isomorphism by Lemma 6.4.

Assume (1) or equivalently (2). Let $A \to A_w$ be the ring map constructed in Lemma 5.3. By (1) there is a retraction $A_w \to A$. Thus $\mathrm{Spec}(A)$ is homeomorphic to a closed subset of $\mathrm{Spec}(A_w)$. By Lemma 2.4 we see (3)(a) holds. Finally, let $T \to \pi_0(A)$ be a surjective map with T an extremally disconnected, quasi-compact, Hausdorff topological space (Topology, Lemma [original label: `topology-lemma-existence-projective-cover`]). Choose $A \to B$ as in Lemma 6.2 adapted to $T \to \pi_0(\mathrm{Spec}(A))$. By (1) there is a retraction $B \to A$. Thus we see that $T = \pi_0(\mathrm{Spec}(B)) \to \pi_0(\mathrm{Spec}(A))$ has a section. A formal categorical argument, using Topology, Proposition [original label: `topology-proposition-projective-in-category-hausdorff-qc`], implies that $\pi_0(\mathrm{Spec}(A))$ is extremally disconnected. □

Lemma 6.8. Let A be a ring. There exists a faithfully flat, ind-Zariski ring map $A \to B$ such that B satisfies the equivalent conditions of Lemma 6.7.

Proof. We first apply Lemma 5.3 to see that we may assume that $\mathrm{Spec}(A)$ is w-local. Choose an extremally disconnected space T and a surjective continuous map $T \to \pi_0(\mathrm{Spec}(A))$, see Topology, Lemma [original label: `topology-lemma-existence-projective-cover`]. Note that T is profinite. Apply Lemma 6.2 to find an ind-Zariski ring map $A \to B$ such that $\pi_0(\mathrm{Spec}(B)) \to \pi_0(\mathrm{Spec}(A))$ realizes $T \to \pi_0(\mathrm{Spec}(A))$ and such that

$$\begin{array}{ccc} \mathrm{Spec}(B) & \longrightarrow & \pi_0(\mathrm{Spec}(B)) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & \pi_0(\mathrm{Spec}(A)) \end{array}$$

is cartesian in the category of topological spaces. Note that $\mathrm{Spec}(B)$ is w-local, that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is w-local, and that the set of closed points of $\mathrm{Spec}(B)$ is the inverse image of the set of closed points of $\mathrm{Spec}(A)$, see Lemma 2.5. Thus condition (3) of Lemma 6.7 holds for B. □

Remark 6.9. In each of Lemmas 6.1, 6.2, Proposition 6.6, and Lemma 6.8 we find an ind-Zariski ring map with some properties. In the paper [2] the authors use the notion of an ind-(Zariski localization) which is a filtered colimit of finite products of principal localizations. It is possible to replace ind-Zariski by ind-(Zariski localization) in each of the results listed above. However, we do not need this and the notion of an ind-Zariski homomorphism of rings as defined here has slightly better formal properties. Moreover, the notion of an ind-Zariski ring map is the natural analogue of the notion of an ind-étale ring map defined in the next section.

7 Ind-étale algebra

We start with a definition.

Definition 7.1. A ring map $A \to B$ is said to be ind-étale if B can be written as a filtered colimit of étale A-algebras.

The category of ind-étale algebras is closed under a number of natural operations.

Lemma 7.2. Let $A \to B$ and $A \to A'$ be ring maps. Let $B' = B \otimes_A A'$ be the base change of B. If $A \to B$ is ind-étale, then $A' \to B'$ is ind-étale.

Proof. This is Algebra, Lemma [original label: `algebra-lemma-base-change-colimit-etale`]. □

Lemma 7.3. Let $A \to B$ and $B \to C$ be ring maps. If $A \to B$ and $B \to C$ are ind-étale, then $A \to C$ is ind-étale.

Proof. This is Algebra, Lemma [original label: `algebra-lemma-composition-colimit-etale`]. □

Lemma 7.4. A filtered colimit of ind-étale A-algebras is ind-étale over A.

Proof. This is Algebra, Lemma [original label: `algebra-lemma-colimit-colimit-etale`]. □

Lemma 7.5. Let A be a ring. Let $B \to C$ be an A-algebra map of ind-étale A-algebras. Then C is an ind-étale B-algebra.

Proof. This is Algebra, Lemma [original label: `algebra-lemma-colimits-of-etale`]. □

Lemma 7.6. Let $A \to B$ be ind-étale. Then $A \to B$ is weakly étale (More on Algebra, Definition [original label: `more-algebra-definition-weakly-etale`]).

Proof. This follows from More on Algebra, Lemma [original label: `more-algebra-lemma-when-weakly-etale`]. □

Lemma 7.7. Let A be a ring and let $I \subset A$ be an ideal. The base change functor

$$\text{ind-étale } A\text{-algebras} \longrightarrow \text{ind-étale } A/I\text{-algebras}, \quad C \longmapsto C/IC$$

has a fully faithful right adjoint v. In particular, given an ind-étale A/I-algebra $\overline{C}$ there exists an ind-étale A-algebra $C = v(\overline{C})$ such that $\overline{C} = C/IC$.

Proof. Let $\overline{C}$ be an ind-étale A/I-algebra. Consider the category $\mathcal{C}$ of factorizations $A \to B \to \overline{C}$ where $A \to B$ is étale. (We ignore some set theoretical issues in this proof.) We will show that this category is directed and that $C = \operatorname{colim}_{\mathcal{C}} B$ is an ind-étale A-algebra such that $\overline{C} = C/IC$.

We first prove that $\mathcal{C}$ is directed (Categories, Definition [original label: `categories-definition-directed`]). The category is nonempty as $A \to A \to \overline{C}$ is an object. Suppose that $A \to B \to \overline{C}$ and $A \to B' \to \overline{C}$ are two objects of $\mathcal{C}$. Then $A \to B \otimes_A B' \to \overline{C}$ is another (use Algebra, Lemma [original label: `algebra-lemma-etale`]). Suppose that $f, g : B \to B'$ are two maps between objects $A \to B \to \overline{C}$ and $A \to B' \to \overline{C}$ of $\mathcal{C}$. Then a coequalizer is $A \to B' \otimes_{f,B,g} B' \to \overline{C}$. This is an object of $\mathcal{C}$ by Algebra, Lemmas [original label: `algebra-lemma-etale`] and [original label: `algebra-lemma-map-between-etale`]. Thus the category $\mathcal{C}$ is directed.

Write $\overline{C} = \operatorname{colim} \overline{B_i}$ as a filtered colimit with $\overline{B_i}$ étale over A/I. For every i there exists $A \to B_i$ étale with $\overline{B_i} = B_i/IB_i$, see Algebra, Lemma [original label: `algebra-lemma-lift-etale`]. Thus $C \to \overline{C}$ is surjective. Since $C/IC \to \overline{C}$ is ind-étale (Lemma 7.5) we see that it is flat. Hence $\overline{C}$ is a localization of C/IC at some multiplicative subset $S \subset C/IC$ (Algebra, Lemma [original label: `algebra-lemma-pure`]). Take an $f \in C$ mapping to an element of $S \subset C/IC$. Choose $A \to B \to \overline{C}$ in $\mathcal{C}$ and $g \in B$ mapping to f in the colimit. Then we see that $A \to B_g \to \overline{C}$ is an object of $\mathcal{C}$ as well. Thus f is an invertible element of C. It follows that $C/IC = \overline{C}$.

Next, we claim that for an ind-étale algebra D over A we have

$$\operatorname{Mor}_A(D, C) = \operatorname{Mor}_{A/I}(D/ID, \overline{C})$$

Namely, let $D/ID \to \overline{C}$ be an A/I-algebra map. Write $D = \operatorname{colim}_{i \in I} D_i$ as a colimit over a directed set I with D_i étale over A. By choice of $\mathcal{C}$ we obtain a transformation $I \to \mathcal{C}$ and hence a map $D \to C$ compatible with maps to $\overline{C}$. Whence the claim.

It follows that the functor v defined by the rule

$$\overline{C} \longmapsto v(\overline{C}) = \operatorname{colim}_{A \to B \to \overline{C}} B$$

is a right adjoint to the base change functor u as required by the lemma. The functor v is fully faithful because $u \circ v = \mathrm{id}$ by construction, see Categories, Lemma [original label: `categories-lemma-adjoint-fully-faithful`]. □

8 Constructing ind-étale algebras

Let A be a ring. Recall that any étale ring map $A \to B$ is isomorphic to a standard smooth ring map of relative dimension 0. Such a ring map is of the form

$$A \longrightarrow A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$$

where the determinant of the $n \times n$-matrix with entries $\partial f_i/\partial x_j$ is invertible in the quotient ring. See Algebra, Lemma [original label: `algebra-lemma-etale-standard-smooth`].

Let $S(A)$ be the set of all faithfully flat[1] standard smooth A-algebras of relative dimension 0. Let $I(A)$ be the partially ordered (by inclusion) set of finite subsets E of $S(A)$. Note that $I(A)$ is a directed partially ordered set. For $E = \{A \to B_1, \ldots, A \to B_n\}$ set

$$B_E = B_1 \otimes_A \ldots \otimes_A B_n$$

Observe that B_E is a faithfully flat étale A-algebra. For $E \subset E'$, there is a canonical transition map $B_E \to B_{E'}$ of étale A-algebras. Namely, say $E = \{A \to B_1, \ldots, A \to B_n\}$ and $E' = \{A \to B_1, \ldots, A \to B_{n+m}\}$ then $B_E \to B_{E'}$ sends $b_1 \otimes \ldots \otimes b_n$ to the element $b_1 \otimes \ldots \otimes b_n \otimes 1 \otimes \ldots \otimes 1$ of $B_{E'}$. This construction defines a system of faithfully flat étale A-algebras over $I(A)$ and we set

$$T(A) = \operatorname{colim}_{E \in I(A)} B_E$$

Observe that $T(A)$ is a faithfully flat ind-étale A-algebra (Algebra, Lemma [original label: `algebra-lemma-colimit-faithfully-flat`]). By construction given any faithfully flat étale A-algebra B there is a (non-unique) A-algebra map $B \to T(A)$. Namely, pick some $(A \to B_0) \in S(A)$ and an isomorphism $B \cong B_0$. Then the canonical coprojection

$$B \to B_0 \to T(A) = \operatorname{colim}_{E \in I(A)} B_E$$

is the desired map.

Lemma 8.1. Given a ring A there exists a faithfully flat ind-étale A-algebra C such that every faithfully flat étale ring map $C \to B$ has a retraction.

Proof. Set $T^1(A) = T(A)$ and $T^{n+1}(A) = T(T^n(A))$. Let

$$C = \operatorname{colim} T^n(A)$$

This algebra is faithfully flat over each $T^n(A)$ and in particular over A, see Algebra, Lemma [original label: `algebra-lemma-colimit-faithfully-flat`]. Moreover, C is ind-étale over A by Lemma 7.4. If $C \to B$ is étale, then there exists an n and an étale ring map $T^n(A) \to B'$ such that $B = C \otimes_{T^n(A)} B'$, see Algebra, Lemma [original label: `algebra-lemma-etale`]. If $C \to B$ is faithfully flat, then $\operatorname{Spec}(B) \to \operatorname{Spec}(C) \to \operatorname{Spec}(T^n(A))$ is surjective, hence $\operatorname{Spec}(B') \to \operatorname{Spec}(T^n(A))$ is surjective. In other words, $T^n(A) \to B'$ is faithfully flat. By our construction, there is a $T^n(A)$-algebra map $B' \to T^{n+1}(A)$. This induces a C-algebra map $B \to C$ which finishes the proof. □

Remark 8.2. Let A be a ring. Let κ be an infinite cardinal bigger or equal than the cardinality of A. Then the cardinality of $T(A)$ is at most κ. Namely, each B_E has cardinality at most κ and the index set $I(A)$ has cardinality at most κ as well. Thus the result follows as $\kappa \otimes \kappa = \kappa$, see Sets, Section [original label: `sets-section-cardinals`]. It follows that the ring constructed in the proof of Lemma 8.1 has cardinality at most κ as well.

Remark 8.3. The construction $A \mapsto T(A)$ is functorial in the following sense: If $A \to A'$ is a ring map, then we can construct a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & T(A) \\ \downarrow & & \downarrow \\ A' & \longrightarrow & T(A') \end{array}$$

Namely, given $(A \to A[x_1, \ldots, x_n]/(f_1, \ldots, f_n))$ in $S(A)$ we can use the ring map $\varphi : A \to A'$ to obtain a corresponding element $(A' \to A'[x_1, \ldots, x_n]/(f_1^\varphi, \ldots, f_n^\varphi))$ of $S(A')$ where f^φ means the polynomial obtained by applying φ to the coefficients of the polynomial f. Moreover, there is a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & A[x_1, \ldots, x_n]/(f_1, \ldots, f_n) \\ \downarrow & & \downarrow \\ A' & \longrightarrow & A'[x_1, \ldots, x_n]/(f_1^\varphi, \ldots, f_n^\varphi) \end{array}$$

[1] In the presence of flatness, e.g., for smooth or étale ring maps, this just means that the induced map on spectra is surjective. See Algebra, Lemma [original label: `algebra-lemma-ff-rings`].

in the category of rings. For $E \subset S(A)$ finite, set $E' = \varphi(E)$ and define $B_E \to B_{E'}$ in the obvious manner. Taking the colimit gives the desired map $T(A) \to T(A')$, see Categories, Lemma [original label: `categories-lemma-functorial-colimit`].

Lemma 8.4. Let A be a ring such that every faithfully flat étale ring map $A \to B$ has a retraction. Then the same is true for every quotient ring A/I.

Proof. Let $A/I \to \overline{B}$ be faithfully flat étale. By Algebra, Lemma [original label: `algebra-lemma-lift-etale`] we can write $\overline{B} = B/IB$ for some étale ring map $A \to B$. The image U of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is open and contains $V(I)$. Hence the complement $Z = \mathrm{Spec}(A) \setminus U$ is quasi-compact and disjoint from $V(I)$. Hence $Z \subset D(f_1) \cup \ldots \cup D(f_r)$ for some $r \geq 0$ and $f_i \in I$. Then $A \to B' = B \times \prod A_{f_i}$ is faithfully flat étale and $\overline{B} = B'/IB'$. Hence the retraction $B' \to A$ to $A \to B'$, induces a retraction to $A/I \to \overline{B}$. □

Lemma 8.5. Let A be a ring such that every faithfully flat étale ring map $A \to B$ has a retraction. Then every local ring of A at a maximal ideal is strictly henselian.

Proof. Let $\mathfrak{m}$ be a maximal ideal of A. Let $A \to B$ be an étale ring map and let $\mathfrak{q} \subset B$ be a prime lying over $\mathfrak{m}$. By the description of the strict henselization $A^{sh}_{\mathfrak{m}}$ in Algebra, Lemma [original label: `algebra-lemma-strict-henselization-different`] it suffices to show that $A_{\mathfrak{m}} = B_{\mathfrak{q}}$. Note that there are finitely many primes $\mathfrak{q} = \mathfrak{q}_1, \mathfrak{q}_2, \ldots, \mathfrak{q}_n$ lying over $\mathfrak{m}$ and there are no specializations between them as an étale ring map is quasi-finite, see Algebra, Lemma [original label: `algebra-lemma-etale-quasi-finite`]. Thus $\mathfrak{q}_i$ is a maximal ideal and we can find $g \in \mathfrak{q}_2 \cap \ldots \cap \mathfrak{q}_n$, $g \notin \mathfrak{q}$ (Algebra, Lemma [original label: `algebra-lemma-silly`]). After replacing B by B_g we see that $\mathfrak{q}$ is the only prime of B lying over $\mathfrak{m}$. The image $U \subset \mathrm{Spec}(A)$ of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is open (Algebra, Proposition [original label: `algebra-proposition-fppf-open`]). Thus the complement $\mathrm{Spec}(A) \setminus U$ is closed and we can find $f \in A$, $f \notin \mathfrak{p}$ such that $\mathrm{Spec}(A) = U \cup D(f)$. The ring map $A \to B \times A_f$ is faithfully flat and étale, hence has a retraction $\sigma : B \times A_f \to A$ by assumption on A. Observe that σ is étale, hence flat as a map between étale A-algebras (Algebra, Lemma [original label: `algebra-lemma-map-between-etale`]). Since $\mathfrak{q}$ is the only prime of $B \times A_f$ lying over A we find that $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ has a retraction which is also flat. Thus $A_{\mathfrak{p}} \to B_{\mathfrak{q}} \to A_{\mathfrak{p}}$ are flat local ring maps whose composition is the identity. Since a flat local homomorphism of local rings is injective we conclude these maps are isomorphisms as desired. □

Lemma 8.6. Let A be a ring such that every faithfully flat étale ring map $A \to B$ has a retraction. Let $Z \subset \mathrm{Spec}(A)$ be a closed subscheme. Let $A \to A^\sim_Z$ be as constructed in Lemma 5.1. Then every faithfully flat étale ring map $A^\sim_Z \to C$ has a retraction.

Proof. There exists an étale ring map $A \to B'$ such that $C = B' \otimes_A A^\sim_Z$ as $A^\sim_Z$-algebras. The image $U' \subset \mathrm{Spec}(A)$ of $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is open and contains $V(I)$, hence we can find $f \in I$ such that $\mathrm{Spec}(A) = U' \cup D(f)$. Then $A \to B' \times A_f$ is étale and faithfully flat. By assumption there is a retraction $B' \times A_f \to A$. Localizing we obtain the desired retraction $C \to A^\sim_Z$. □

Lemma 8.7. Let $A \to B$ be a ring map inducing algebraic extensions on residue fields. There exists a commutative diagram

$$\begin{array}{ccc} B & \longrightarrow & D \\ \uparrow & & \uparrow \\ A & \longrightarrow & C \end{array}$$

with the following properties:

1. $A \to C$ is faithfully flat and ind-étale,
2. $B \to D$ is faithfully flat and ind-étale,
3. $\mathrm{Spec}(C)$ is w-local,
4. $\mathrm{Spec}(D)$ is w-local,
5. $\mathrm{Spec}(D) \to \mathrm{Spec}(C)$ is w-local,
6. the set of closed points of $\mathrm{Spec}(D)$ is the inverse image of the set of closed points of $\mathrm{Spec}(C)$,
7. the set of closed points of $\mathrm{Spec}(C)$ surjects onto $\mathrm{Spec}(A)$,
8. the set of closed points of $\mathrm{Spec}(D)$ surjects onto $\mathrm{Spec}(B)$,
9. for $\mathfrak{m} \subset C$ maximal the local ring $C_{\mathfrak{m}}$ is strictly henselian.

Proof. There is a faithfully flat, ind-Zariski ring map $A \to A'$ such that $\mathrm{Spec}(A')$ is w-local and such that the set of closed points of $\mathrm{Spec}(A')$ maps onto $\mathrm{Spec}(A)$, see Lemma 5.3. Let $I \subset A'$ be the ideal such that $V(I)$ is the set of closed points of $\mathrm{Spec}(A')$. Choose $A' \to C'$ as in Lemma 8.1. Note that the local rings $C'_{\mathfrak{m}'}$ at maximal ideals $\mathfrak{m}' \subset C'$ are strictly henselian by Lemma 8.5. We apply Lemma 5.8 to $A' \to C'$ and $I \subset A'$ to get $C' \to C$ with $C'/IC' \cong C/IC$. Note that since $A' \to C'$ is faithfully flat, $\mathrm{Spec}(C'/IC')$ surjects onto the set of closed points of A' and in particular onto $\mathrm{Spec}(A)$. Moreover, as $V(IC) \subset \mathrm{Spec}(C)$ is the set of closed points of C and $C' \to C$ is ind-Zariski (and identifies local rings) we obtain properties (1), (3), (7), and (9).

Denote $J \subset C$ the ideal such that $V(J)$ is the set of closed points of $\mathrm{Spec}(C)$. Set $D' = B \otimes_A C$. The ring map $C \to D'$ induces algebraic residue field extensions. Keep in mind that since $V(J) \to \mathrm{Spec}(A)$ is surjective the map $T = V(JD') \to \mathrm{Spec}(B)$ is surjective too. Apply Lemma 5.8 to $C \to D'$ and $J \subset C$ to get $D' \to D$ with $D'/JD' \cong D/JD$. All of the remaining properties given in the lemma are immediate from the results of Lemma 5.8. □

9 Weakly étale versus pro-étale

Recall that a ring homomorphism $A \to B$ is weakly étale if $A \to B$ is flat and $B \otimes_A B \to B$ is flat. We have proved some properties of such ring maps in More on Algebra, Section [original label: `more-algebra-section-weakly-etale`]. In particular, if $A \to B$ is a local homomorphism, and A is a strictly henselian local rings, then $A = B$, see More on Algebra, Theorem [original label: `more-algebra-theorem-olivier`]. Using this theorem and the work we've done above we obtain the following structure theorem for weakly étale ring maps.

Proposition 9.1. Let $A \to B$ be a weakly étale ring map. Then there exists a faithfully flat, ind-étale ring map $B \to B'$ such that $A \to B'$ is ind-étale.

Proof. The ring map $A \to B$ induces (separable) algebraic extensions of residue fields, see More on Algebra, Lemma [original label: `more-algebra-lemma-weakly-etale-residue-field-extensions`]. Thus we may apply Lemma 8.7 and choose a diagram

$$\begin{array}{ccc} B & \longrightarrow & D \\ \uparrow & & \uparrow \\ A & \longrightarrow & C \end{array}$$

with the properties as listed in the lemma. Note that $C \to D$ is weakly étale by More on Algebra, Lemma [original label: `more-algebra-lemma-weakly-etale-permanence`]. Pick a maximal ideal $\mathfrak{m} \subset D$. By construction this lies over a maximal ideal $\mathfrak{m}' \subset C$. By More on Algebra, Theorem [original label: `more-algebra-theorem-olivier`] the ring map $C_{\mathfrak{m}'} \to D_{\mathfrak{m}}$ is an isomorphism. As every point of $\mathrm{Spec}(C)$ specializes to a closed point we conclude that $C \to D$ identifies local rings. Thus Proposition 6.6 applies to the ring map $C \to D$. Pick $D \to D'$ faithfully flat and ind-Zariski such that $C \to D'$ is ind-Zariski. Then $B \to D'$ is a solution to the problem posed in the proposition. □

10 The V topology and the pro-h topology

The V topology was introduced in Topologies, Section [original label: `topologies-section-V`]. The h topology was introduced in More on Flatness, Section [original label: `flat-section-h`]. A kind of intermediate topology, namely the ph topology, was introduced in Topologies, Section [original label: `topologies-section-ph`].

Given a topology τ on a suitable category $\mathcal{C}$ of schemes, we can introduce a "pro-τ topology" on $\mathcal{C}$ as follows. Recall that for X in $\mathcal{C}$ we use h_X to denote the representable presheaf associated to X. Let us temporarily say a morphism $X \to Y$ of $\mathcal{C}$ is a τ-cover[2] if the τ-sheafification of $h_X \to h_Y$ is surjective. Then we can define the pro-τ topology as the coarsest topology such that

1. the pro-τ topology is finer than the τ topology, and
2. $X \to Y$ is a pro-τ-cover if Y is affine and $X = \lim X_\lambda$ is a directed limit of affine schemes X_λ over Y such that $h_{X_\lambda} \to h_Y$ is a τ-cover for all λ.

We use this pedantic formulation because we do not want to specify a choice of pro-τ coverings: for different τ different choices of collections of coverings are suitable. For example, in Section 12 we will see that in order to define the pro-étale topology looking at families of weakly étale morphisms with some finiteness property works

[2]This should not be confused with the notion of a covering. For example if $\tau = \textit{étale}$, any morphism $X \to Y$ which has a section is a τ-covering. But our definition of étale coverings $\{V_i \to Y\}_{i \in I}$ forces each $V_i \to Y$ to be étale.

well. More generally, the proposed construction given in this paragraph is meant mainly to motivate the results in this section and we will never implicitly define a pro-τ topology using this method.

The following lemma tells us that the pro-V topology is equal to the V topology.

Lemma 10.1. Let Y be an affine scheme. Let $X = \lim X_i$ be a directed limit of affine schemes over Y. The following are equivalent

1. $\{X \to Y\}$ is a standard V covering (Topologies, Definition [original label: `topologies-definition-standard-V-covering`]), and
2. $\{X_i \to Y\}$ is a standard V covering for all i.

Proof. A singleton $\{X \to Y\}$ is a standard V covering if and only if given a morphism $g : \mathrm{Spec}(V) \to Y$ there is an extension of valuation rings $V \subset W$ and a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(W) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(V) & \xrightarrow{g} & Y \end{array}$$

Thus (1) $\Rightarrow$ (2) is immediate from the definition. Conversely, assume (2) and let $g : \mathrm{Spec}(V) \to Y$ as above be given. Write $\mathrm{Spec}(V) \times_Y X_i = \mathrm{Spec}(A_i)$. Since $\{X_i \to Y\}$ is a standard V covering, we may choose a valuation ring W_i and a ring map $A_i \to W_i$ such that the composition $V \to A_i \to W_i$ is an extension of valuation rings. In particular, the quotient A_i' of A_i by its V-torsion is a faitfhully flat V-algebra. Flatness by More on Algebra, Lemma [original label: `more-algebra-lemma-valuation-ring-torsion-free-flat`] and surjectivity on spectra because $A_i \to W_i$ factors through A_i'. Thus

$$A = \mathrm{colim}\, A_i'$$

is a faithfully flat V-algebra (Algebra, Lemma [original label: `algebra-lemma-colimit-faithfully-flat`]). Since $\{\mathrm{Spec}(A) \to \mathrm{Spec}(V)\}$ is a standard fpqc cover, it is a standard V cover (Topologies, Lemma [original label: `topologies-lemma-standard-fpqc-standard-V`]) and hence we can choose $\mathrm{Spec}(W) \to \mathrm{Spec}(A)$ such that $V \to W$ is an extension of valuation rings. Since we can compose with the morphism $\mathrm{Spec}(A) \to X = \mathrm{Spec}(\mathrm{colim}\, A_i)$ the proof is complete. □

The following lemma tells us that the pro-h topology is equal to the pro-ph topology is equal to the V topology.

Lemma 10.2. Let $X \to Y$ be a morphism of affine schemes. The following are equivalent

1. $\{X \to Y\}$ is a standard V covering (Topologies, Definition [original label: `topologies-definition-standard-V-covering`]),
2. $X = \lim X_i$ is a directed limit of affine schemes over Y such that $\{X_i \to Y\}$ is a ph covering for each i, and
3. $X = \lim X_i$ is a directed limit of affine schemes over Y such that $\{X_i \to Y\}$ is an h covering for each i.

Proof. Proof of (2) $\Rightarrow$ (1). Recall that a V covering given by a single arrow between affines is a standard V covering, see Topologies, Definition [original label: `topologies-definition-V-covering`] and Lemma [original label: `topologies-lemma-refine-standard-V`]. Recall that any ph covering is a V covering, see Topologies, Lemma [original label: `topologies-lemma-zariski-etale-smooth-syntomic-fppf-fpqc-ph-V`]. Hence if $X = \lim X_i$ as in (2), then $\{X_i \to Y\}$ is a standard V covering for each i. Thus by Lemma 10.1 we see that (1) is true.

Proof of (3) $\Rightarrow$ (2). This is clear because an h covering is always a ph covering, see More on Flatness, Definition [original label: `flat-definition-h-covering`].

Proof of (1) $\Rightarrow$ (3). This is the interesting direction, but the interesting content in this proof is hidden in More on Flatness, Lemma [original label: `flat-lemma-equivalence-h-v-locally-finite-presentation`]. Write $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(R)$. We can write $A = \mathrm{colim}\, A_i$ with A_i of finite presentation over R, see Algebra, Lemma [original label: `algebra-lemma-ring-colimit-fp`]. Set $X_i = \mathrm{Spec}(A_i)$. Then $\{X_i \to Y\}$ is a standard V covering for all i by (1) and Topologies, Lemma [original label: `topologies-lemma-refine-standard-V`]. Hence $\{X_i \to Y\}$ is an h covering by More on Flatness, Definition [original label: `flat-definition-h-covering`]. This finishes the proof. □

The following lemma tells us, roughly speaking, that an h sheaf which is limit preserving satisfies the sheaf condition for V coverings. Please also compare with Remark 10.4.

Lemma 10.3. Let S be a scheme. Let F be a contravariant functor defined on the category of all schemes over S. If

1. F satisfies the sheaf property for the h topology, and
2. F is limit preserving (Limits, Remark [original label: `limits-remark-limit-preserving`]),

then F satisfies the sheaf property for the V topology.

Proof. We will prove this by verifying (1) and (2') of Topologies, Lemma [original label: `topologies-lemma-sheaf-property-V`]. The sheaf property for Zariski coverings follows from the fact that F has the sheaf property for all h coverings. Finally, suppose that $X \to Y$ is a morphism of affine schemes over S such that $\{X \to Y\}$ is a V covering. By Lemma 10.2 we can write $X = \lim X_i$ as a directed limit of affine schemes over Y such that $\{X_i \to Y\}$ is an h covering for each i. We obtain

$$\begin{aligned}
&\text{Equalizer}(\ F(X) \rightrightarrows F(X \times_Y X)\ ) \\
&= \text{Equalizer}(\ \operatorname{colim} F(X_i) \rightrightarrows \operatorname{colim} F(X_i \times_Y X_i)\ ) \\
&= \operatorname{colim} \text{Equalizer}(\ F(X_i) \rightrightarrows F(X_i \times_Y X_i)\ ) \\
&= \operatorname{colim} F(Y) = F(Y)
\end{aligned}$$

which is what we wanted to show. The first equality because F is limit preserving and $X = \lim X_i$ and $X \times_Y X = \lim X_i \times_Y X_i$. The second equality because filtered colimits are exact. The third equality because F satisfies the sheaf property for h coverings. □

Remark 10.4. Let S be a scheme contained in a big site Sch_h. Let F be a sheaf of sets on $(\mathrm{Sch}/S)_h$ such that $F(T) = \operatorname{colim} F(T_i)$ whenever $T = \lim T_i$ is a directed limit of affine schemes in $(\mathrm{Sch}/S)_h$. In this situation F extends uniquely to a contravariant functor F' on the category of all schemes over S such that (a) F' satisfies the sheaf property for the h topology and (b) F' is limit preserving. See More on Flatness, Lemma [original label: `flat-lemma-extend-sheaf-h`]. In this situation Lemma 10.3 tells us that F' satisfies the sheaf property for the V topology.

11 Constructing w-contractible covers

In this section we construct w-contractible covers of affine schemes.

Definition 11.1. Let A be a ring. We say A is w-contractible if every faithfully flat weakly étale ring map $A \to B$ has a retraction.

We remark that by Proposition 9.1 an equivalent definition would be to ask that every faithfully flat, ind-étale ring map $A \to B$ has a retraction. Here is a key observation that will allow us to construct w-contractible rings.

Lemma 11.2. Let A be a ring. The following are equivalent

1. A is w-contractible,
2. every faithfully flat, ind-étale ring map $A \to B$ has a retraction, and
3. A satisfies
   (a) $\mathrm{Spec}(A)$ is w-local,
   (b) $\pi_0(\mathrm{Spec}(A))$ is extremally disconnected, and
   (c) for every maximal ideal $\mathfrak{m} \subset A$ the local ring $A_{\mathfrak{m}}$ is strictly henselian.

Proof. The equivalence of (1) and (2) follows immediately from Proposition 9.1.

Assume (3)(a), (3)(b), and (3)(c). Let $A \to B$ be faithfully flat and ind-étale. We will use without further mention the fact that a flat map $A \to B$ is faithfully flat if and only if every closed point of $\mathrm{Spec}(A)$ is in the image of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ We will show that $A \to B$ has a retraction.

Let $I \subset A$ be an ideal such that $V(I) \subset \mathrm{Spec}(A)$ is the set of closed points of $\mathrm{Spec}(A)$. We may replace B by the ring C constructed in Lemma 5.8 for $A \to B$ and $I \subset A$. Thus we may assume $\mathrm{Spec}(B)$ is w-local such that

the set of closed points of $\mathrm{Spec}(B)$ is $V(IB)$. In this case $A \to B$ identifies local rings by condition (3)(c) as it suffices to check this at maximal ideals of B which lie over maximal ideals of A. Thus $A \to B$ has a retraction by Lemma 6.7.

Assume (1) or equivalently (2). We have (3)(c) by Lemma 8.5. Properties (3)(a) and (3)(b) follow from Lemma 6.7. □

Proposition 11.3. For every ring A there exists a faithfully flat, ind-étale ring map $A \to D$ such that D is w-contractible.

Proof. Applying Lemma 8.7 to $\mathrm{id}_A : A \to A$ we find a faithfully flat, ind-étale ring map $A \to C$ such that C is w-local and such that every local ring at a maximal ideal of C is strictly henselian. Choose an extremally disconnected space T and a surjective continuous map $T \to \pi_0(\mathrm{Spec}(C))$, see Topology, Lemma [original label: `topology-lemma-existence-projective-cover`]. Note that T is profinite. Apply Lemma 6.2 to find an ind-Zariski ring map $C \to D$ such that $\pi_0(\mathrm{Spec}(D)) \to \pi_0(\mathrm{Spec}(C))$ realizes $T \to \pi_0(\mathrm{Spec}(C))$ and such that

$$\begin{array}{ccc} \mathrm{Spec}(D) & \longrightarrow & \pi_0(\mathrm{Spec}(D)) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(C) & \longrightarrow & \pi_0(\mathrm{Spec}(C)) \end{array}$$

is cartesian in the category of topological spaces. Note that $\mathrm{Spec}(D)$ is w-local, that $\mathrm{Spec}(D) \to \mathrm{Spec}(C)$ is w-local, and that the set of closed points of $\mathrm{Spec}(D)$ is the inverse image of the set of closed points of $\mathrm{Spec}(C)$, see Lemma 2.5. Thus it is still true that the local rings of D at its maximal ideals are strictly henselian (as they are isomorphic to the local rings at the corresponding maximal ideals of C). It follows from Lemma 11.2 that D is w-contractible. □

Remark 11.4. Let A be a ring. Let κ be an infinite cardinal bigger or equal than the cardinality of A. Then the cardinality of the ring D constructed in Proposition 11.3 is at most

$$\kappa^{2^{2^{2^\kappa}}}.$$

Namely, the ring map $A \to D$ is constructed as a composition

$$A \to A_w = A' \to C' \to C \to D.$$

Here the first three steps of the construction are carried out in the first paragraph of the proof of Lemma 8.7. For the first step we have $|A_w| \leq \kappa$ by Remark 5.4. We have $|C'| \leq \kappa$ by Remark 8.2. Then $|C| \leq \kappa$ because C is a localization of $(C')_w$ (it is constructed from C' by an application of Lemma 5.7 in the proof of Lemma 5.8). Thus C has at most 2^κ maximal ideals. Finally, the ring map $C \to D$ identifies local rings and the cardinality of the set of maximal ideals of D is at most $2^{2^{2^\kappa}}$ by Topology, Remark [original label: `topology-remark-size-projective-cover`]. Since $D \subset \prod_{\mathfrak{m} \subset D} D_{\mathfrak{m}}$ we see that D has at most the size displayed above.

Lemma 11.5. Let $A \to B$ be a quasi-finite and finitely presented ring map. If the residue fields of A are separably algebraically closed and $\mathrm{Spec}(A)$ is Hausdorff and extremally disconnected, then $\mathrm{Spec}(B)$ is extremally disconnected.

Proof. Set $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. Choose a finite partition $X = \coprod X_i$ and $X_i' \to X_i$ as in Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-decompose-quasi-finite-morphism`]. The map of topological spaces $\coprod X_i \to X$ (where the source is the disjoint union in the category of topological spaces) has a section by Topology, Proposition [original label: `topology-proposition-projective-in-category-hausdorff-qc`]. Hence we see that X is topologically the disjoint union of the strata X_i. Thus we may replace X by the X_i and assume there exists a surjective finite locally free morphism $X' \to X$ such that $(X' \times_X Y)_{red}$ is isomorphic to a finite disjoint union of copies of X'_{red}. Picture

$$\begin{array}{ccc} \coprod_{i=1,\ldots,r} X' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

The assumption on the residue fields of A implies that this diagram is a fibre product diagram on underlying sets of points (details omitted). Since X is extremally disconnected and X' is Hausdorff (Lemma 5.6), the continuous map $X' \to X$ has a continuous section σ. Then $\coprod_{i=1,\ldots,r} \sigma(X) \to Y$ is a bijective continuous map. By Topology, Lemma [original label: `topology-lemma-bijective-map`] we see that it is a homeomorphism and the proof is done. □

Lemma 11.6. Let $A \to B$ be a finite and finitely presented ring map. If A is w-contractible, so is B.

Proof. We will use the criterion of Lemma 11.2. Set $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$ and denote $f : Y \to X$ the induced morphism. As $f : Y \to X$ is a finite morphism, we see that the set of closed points Y_0 of Y is the inverse image of the set of closed points X_0 of X. Let $y \in Y$ with image $x \in X$. Then x specializes to a unique closed point $x_0 \in X$. Say $f^{-1}(\{x_0\}) = \{y_1, \ldots, y_n\}$ with y_i closed in Y. Since $R = \mathcal{O}_{X,x_0}$ is strictly henselian and since f is finite, we see that $Y \times_{f,X} \mathrm{Spec}(R)$ is equal to $\coprod_{i=1,\ldots,n} \mathrm{Spec}(R_i)$ where each R_i is a local ring finite over R whose maximal ideal corresponds to y_i, see Algebra, Lemma [original label: `algebra-lemma-characterize-henselian`] part (10). Then y is a point of exactly one of these $\mathrm{Spec}(R_i)$ and we see that y specializes to exactly one of the y_i. In other words, every point of Y specializes to a unique point of Y_0. Thus Y is w-local. For every $y \in Y_0$ with image $x \in X_0$ we see that $\mathcal{O}_{Y,y}$ is strictly henselian by Algebra, Lemma [original label: `algebra-lemma-finite-over-henselian`] applied to $\mathcal{O}_{X,x} \to B \otimes_A \mathcal{O}_{X,x}$. It remains to show that Y_0 is extremally disconnected. To do this we look at $X_0 \times_X Y \to X_0$ where $X_0 \subset X$ is the reduced induced scheme structure. Note that the underlying topological space of $X_0 \times_X Y$ agrees with Y_0. Now the desired result follows from Lemma 11.5. □

Lemma 11.7. Let A be a ring. Let $Z \subset \mathrm{Spec}(A)$ be a closed subset of the form $Z = V(f_1, \ldots, f_r)$. Set $B = A^\sim_Z$, see Lemma 5.1. If A is w-contractible, so is B.

Proof. Let $A^\sim_Z \to B$ be a weakly étale faithfully flat ring map. Consider the ring map

$$A \longrightarrow A_{f_1} \times \ldots \times A_{f_r} \times B$$

this is faithful flat and weakly étale. If A is w-contractible, then there is a retraction σ. Consider the morphism

$$\mathrm{Spec}(A^\sim_Z) \to \mathrm{Spec}(A) \xrightarrow{\mathrm{Spec}(\sigma)} \coprod \mathrm{Spec}(A_{f_i}) \amalg \mathrm{Spec}(B)$$

Every point of $Z \subset \mathrm{Spec}(A^\sim_Z)$ maps into the component $\mathrm{Spec}(B)$. Since every point of $\mathrm{Spec}(A^\sim_Z)$ specializes to a point of Z we find a morphism $\mathrm{Spec}(A^\sim_Z) \to \mathrm{Spec}(B)$ as desired. □

12 The pro-étale site

In this section we only discuss the actual definition and construction of the various pro-étale sites and the morphisms between them. The existence of weakly contractible objects will be done in Section 13.

The pro-étale topology is a bit like the fpqc topology (see Topologies, Section [original label: `topologies-section-fpqc`]) in that the topos of sheaves on the small pro-étale site of a scheme depends on the choice of the underlying category of schemes. Thus we cannot speak of the pro-étale topos of a scheme. However, it will be true that the cohomology groups of a sheaf are unchanged if we enlarge our underlying category of schemes, see Section 31.

We will define pro-étale coverings using weakly étale morphisms of schemes, see More on Morphisms, Section [original label: `more-morphisms-section-weakly-etale`]. The reason is that, on the one hand, it is somewhat awkward to define the notion of a pro-étale morphism of schemes, and on the other, Proposition 9.1 assures us that we obtain the same sheaves[3] with the definition that follows.

Definition 12.1. Let T be a scheme. A pro-étale covering of T is a family of morphisms $\{f_i : T_i \to T\}_{i \in I}$ of schemes such that each f_i is weakly-étale and such that for every affine open $U \subset T$ there exists $n \geq 0$, a map $a : \{1, \ldots, n\} \to I$ and affine opens $V_j \subset T_{a(j)}$, $j = 1, \ldots, n$ with $\bigcup_{j=1}^n f_{a(j)}(V_j) = U$.

To be sure this condition implies that $T = \bigcup f_i(T_i)$. Here is a lemma that will allow us to recognize pro-étale coverings. It will also allow us to reduce many lemmas about pro-étale coverings to the corresponding results for fpqc coverings.

Lemma 12.2. Let T be a scheme. Let $\{f_i : T_i \to T\}_{i \in I}$ be a family of morphisms of schemes with target T. The following are equivalent

1. $\{f_i : T_i \to T\}_{i \in I}$ is a pro-étale covering,
2. each f_i is weakly étale and $\{f_i : T_i \to T\}_{i \in I}$ is an fpqc covering,
3. each f_i is weakly étale and for every affine open $U \subset T$ there exist quasi-compact opens $U_i \subset T_i$ which are almost all empty, such that $U = \bigcup f_i(U_i)$,

[3] To be precise the pro-étale topology we obtain using our choice of coverings is the same as the one gotten from the general procedure explained in Section 10 starting with $\tau = \textit{étale}$.

4. each f_i is weakly étale and there exists an affine open covering $T = \bigcup_{\alpha \in A} U_\alpha$ and for each $\alpha \in A$ there exist $i_{\alpha,1}, \ldots, i_{\alpha,n(\alpha)} \in I$ and quasi-compact opens $U_{\alpha,j} \subset T_{i_{\alpha,j}}$ such that $U_\alpha = \bigcup_{j=1,\ldots,n(\alpha)} f_{i_{\alpha,j}}(U_{\alpha,j})$.

If T is quasi-separated, these are also equivalent to

(5) each f_i is weakly étale, and for every $t \in T$ there exist $i_1, \ldots, i_n \in I$ and quasi-compact opens $U_j \subset T_{i_j}$ such that $\bigcup_{j=1,\ldots,n} f_{i_j}(U_j)$ is a (not necessarily open) neighbourhood of t in T.

Proof. The equivalence of (1) and (2) is immediate from the definitions. Hence the lemma follows from Topologies, Lemma [original label: `topologies-lemma-recognize-fpqc-covering`]. □

Lemma 12.3. Any étale covering and any Zariski covering is a pro-étale covering.

Proof. This follows from the corresponding result for fpqc coverings (Topologies, Lemma [original label: `topologies-lemma-zariski-etale-smooth-syntomic-fppf-fpqc`]), Lemma 12.2, and the fact that an étale morphism is a weakly étale morphism, see More on Morphisms, Lemma [original label: `more-morphisms-lemma-when-weakly-etale`]. □

Lemma 12.4. Let T be a scheme.

1. If $T' \to T$ is an isomorphism then $\{T' \to T\}$ is a pro-étale covering of T.
2. If $\{T_i \to T\}_{i \in I}$ is a pro-étale covering and for each i we have a pro-étale covering $\{T_{ij} \to T_i\}_{j \in J_i}$, then $\{T_{ij} \to T\}_{i \in I, j \in J_i}$ is a pro-étale covering.
3. If $\{T_i \to T\}_{i \in I}$ is a pro-étale covering and $T' \to T$ is a morphism of schemes then $\{T' \times_T T_i \to T'\}_{i \in I}$ is a pro-étale covering.

Proof. This follows from the fact that composition and base changes of weakly étale morphisms are weakly étale (More on Morphisms, Lemmas [original label: `more-morphisms-lemma-composition-weakly-etale`] and [original label: `more-morphisms-lemma-base-change-weakly-etale`]), Lemma 12.2, and the corresponding results for fpqc coverings, see Topologies, Lemma [original label: `topologies-lemma-fpqc`]. □

Lemma 12.5. Let T be an affine scheme. Let $\{T_i \to T\}_{i \in I}$ be a pro-étale covering of T. Then there exists a pro-étale covering $\{U_j \to T\}_{j=1,\ldots,n}$ which is a refinement of $\{T_i \to T\}_{i \in I}$ such that each U_j is an affine scheme. Moreover, we may choose each U_j to be open affine in one of the T_i.

Proof. This follows directly from the definition. □

Thus we define the corresponding standard coverings of affines as follows.

Definition 12.6. Let T be an affine scheme. A standard pro-étale covering of T is a family $\{f_i : T_i \to T\}_{i=1,\ldots,n}$ where each T_j is affine, each f_i is weakly étale, and $T = \bigcup f_i(T_i)$.

We follow the general outline given in Topologies, Section [original label: `topologies-section-procedure`] for constructing the big pro-étale site we will be working with. However, because we need a bit larger rings to accommodate for the size of certain constructions we modify the constructions slightly.

Definition 12.7. A big pro-étale site is any site $\mathrm{Sch}_{pro\text{-}étale}$ as in Sites, Definition [original label: `sites-definition-site`] constructed as follows:

1. Choose any set of schemes S_0, and any set of pro-étale coverings Cov_0 among these schemes.
2. Change the function *Bound* of Sets, Equation ([original label: `sets-equation-bound`]) into
$$Bound(\kappa) = \max\{\kappa^{2^{2^{2^\kappa}}}, \kappa^{\aleph_0}, \kappa^+\}.$$
3. As underlying category take any category Sch_α constructed as in Sets, Lemma [original label: `sets-lemma-construct-category`] starting with the set S_0 and the function *Bound*.
4. Choose any set of coverings as in Sets, Lemma [original label: `sets-lemma-coverings-site`] starting with the category Sch_α and the class of pro-étale coverings, and the set Cov_0 chosen above.

See the remarks following Topologies, Definition [original label: `topologies-definition-big-zariski-site`] for motivation and explanation regarding the definition of big sites.

It will turn out, see Lemma 31.1, that the topology on a big pro-étale site $\mathrm{Sch}_{pro\text{-}étale}$ is in some sense induced from the pro-étale topology on the category of all schemes.

Definition 12.8. Let S be a scheme. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site containing S.

1. The big pro-étale site of S, denoted $(\mathrm{Sch}/S)_{pro\text{-}étale}$, is the site $\mathrm{Sch}_{pro\text{-}étale}/S$ introduced in Sites, Section [original label: `sites-section-localize`].
2. The small pro-étale site of S, which we denote $S_{pro\text{-}étale}$, is the full subcategory of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ whose objects are those U/S such that $U \to S$ is weakly étale. A covering of $S_{pro\text{-}étale}$ is any covering $\{U_i \to U\}$ of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ with $U \in \mathrm{Ob}(S_{pro\text{-}étale})$.
3. The big affine pro-étale site of S, denoted $(\mathrm{Aff}/S)_{pro\text{-}étale}$, is the full subcategory of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ whose objects are affine U/S. A covering of $(\mathrm{Aff}/S)_{pro\text{-}étale}$ is any covering $\{U_i \to U\}$ of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ which is a standard pro-étale covering.

It is not completely clear that the small pro-étale site and the big affine pro-étale site are sites. We check this now.

Lemma 12.9. Let S be a scheme. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site containing S. Both $S_{pro\text{-}étale}$ and $(\mathrm{Aff}/S)_{pro\text{-}étale}$ are sites.

Proof. Let us show that $S_{pro\text{-}étale}$ is a site. It is a category with a given set of families of morphisms with fixed target. Thus we have to show properties (1), (2) and (3) of Sites, Definition [original label: `sites-definition-site`]. Since $(\mathrm{Sch}/S)_{pro\text{-}étale}$ is a site, it suffices to prove that given any covering $\{U_i \to U\}$ of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ with $U \in \mathrm{Ob}(S_{pro\text{-}étale})$ we also have $U_i \in \mathrm{Ob}(S_{pro\text{-}étale})$. This follows from the definitions as the composition of weakly étale morphisms is weakly étale.

To show that $(\mathrm{Aff}/S)_{pro\text{-}étale}$ is a site, reasoning as above, it suffices to show that the collection of standard pro-étale coverings of affines satisfies properties (1), (2) and (3) of Sites, Definition [original label: `sites-definition-site`]. This follows from Lemma 12.2 and the corresponding result for standard fpqc coverings (Topologies, Lemma [original label: `topologies-lemma-fpqc-affine-axioms`]). □

Lemma 12.10. Let S be a scheme. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site containing S. Let Sch be the category of all schemes.

1. The categories $\mathrm{Sch}_{pro\text{-}étale}$, $(\mathrm{Sch}/S)_{pro\text{-}étale}$, $S_{pro\text{-}étale}$, and $(\mathrm{Aff}/S)_{pro\text{-}étale}$ have fibre products agreeing with fibre products in Sch.
2. The categories $\mathrm{Sch}_{pro\text{-}étale}$, $(\mathrm{Sch}/S)_{pro\text{-}étale}$, $S_{pro\text{-}étale}$ have equalizers agreeing with equalizers in Sch.
3. The categories $(\mathrm{Sch}/S)_{pro\text{-}étale}$, and $S_{pro\text{-}étale}$ both have a final object, namely S/S.
4. The category $\mathrm{Sch}_{pro\text{-}étale}$ has a final object agreeing with the final object of Sch, namely $\mathrm{Spec}(\mathrm{Z})$.

Proof. The category $\mathrm{Sch}_{pro\text{-}étale}$ contains $\mathrm{Spec}(\mathrm{Z})$ and is closed under products and fibre products by construction, see Sets, Lemma [original label: `sets-lemma-what-is-in-it`]. Suppose we have $U \to S$, $V \to U$, $W \to U$ morphisms of schemes with $U, V, W \in \mathrm{Ob}(\mathrm{Sch}_{pro\text{-}étale})$. The fibre product $V \times_U W$ in $\mathrm{Sch}_{pro\text{-}étale}$ is a fibre product in Sch and is the fibre product of V/S with W/S over U/S in the category of all schemes over S, and hence also a fibre product in $(\mathrm{Sch}/S)_{pro\text{-}étale}$. This proves the result for $(\mathrm{Sch}/S)_{pro\text{-}étale}$. If $U \to S$, $V \to U$ and $W \to U$ are weakly étale then so is $V \times_U W \to S$ (see More on Morphisms, Section [original label: `more-morphisms-section-weakly-etale`]) and hence we get fibre products for $S_{pro\text{-}étale}$. If U, V, W are affine, so is $V \times_U W$ and hence we get fibre products for $(\mathrm{Aff}/S)_{pro\text{-}étale}$.

Let $a, b : U \to V$ be two morphisms in $\mathrm{Sch}_{pro\text{-}étale}$. In this case the equalizer of a and b (in the category of schemes) is

$$V \times_{\Delta_{V/\mathrm{Spec}(\mathrm{Z})}, V \times_{\mathrm{Spec}(\mathrm{Z})} V, (a,b)} (U \times_{\mathrm{Spec}(\mathrm{Z})} U)$$

which is an object of $\mathrm{Sch}_{pro\text{-}étale}$ by what we saw above. Thus $\mathrm{Sch}_{pro\text{-}étale}$ has equalizers. If a and b are morphisms over S, then the equalizer (in the category of schemes) is also given by

$$V \times_{\Delta_{V/S}, V \times_S V, (a,b)} (U \times_S U)$$

hence we see that $(\mathrm{Sch}/S)_{pro\text{-}étale}$ has equalizers. Moreover, if U and V are weakly-étale over S, then so is the equalizer above as a fibre product of schemes weakly étale over S. Thus $S_{pro\text{-}étale}$ has equalizers. The statements on final objects is clear. □

Next, we check that the big affine pro-étale site defines the same topos as the big pro-étale site.

Lemma 12.11. Let S be a scheme. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site containing S. The functor $(\mathrm{Aff}/S)_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$ is a special cocontinuous functor. Hence it induces an equivalence of topoi from $\mathrm{Sh}((\mathrm{Aff}/S)_{pro\text{-}étale})$ to $\mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})$.

Proof. The notion of a special cocontinuous functor is introduced in Sites, Definition [original label: `sites-definition-special-cocontinuous-functor`]. Thus we have to verify assumptions (1) – (5) of Sites, Lemma [original label: `sites-lemma-equivalence`]. Denote the inclusion functor $u : (\mathrm{Aff}/S)_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$. Being cocontinuous just means that any pro-étale covering of T/S, T affine, can be refined by a standard pro-étale covering of T. This is the content of Lemma 12.5. Hence (1) holds. We see u is continuous simply because a standard pro-étale covering is a pro-étale covering. Hence (2) holds. Parts (3) and (4) follow immediately from the fact that u is fully faithful. And finally condition (5) follows from the fact that every scheme has an affine open covering. □

Lemma 12.12. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site. Let $f : T \to S$ be a morphism in $\mathrm{Sch}_{pro\text{-}étale}$. The functor $T_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$ is cocontinuous and induces a morphism of topoi

$$i_f : \mathrm{Sh}(T_{pro\text{-}étale}) \longrightarrow \mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})$$

For a sheaf $\mathcal{G}$ on $(\mathrm{Sch}/S)_{pro\text{-}étale}$ we have the formula $(i_f^{-1}\mathcal{G})(U/T) = \mathcal{G}(U/S)$. The functor i_f^{-1} also has a left adjoint $i_{f,!}$ which commutes with fibre products and equalizers.

Proof. Denote the functor $u : T_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$. In other words, given a weakly étale morphism $j : U \to T$ corresponding to an object of $T_{pro\text{-}étale}$ we set $u(U \to T) = (f \circ j : U \to S)$. This functor commutes with fibre products, see Lemma 12.10. Moreover, $T_{pro\text{-}étale}$ has equalizers and u commutes with them by Lemma 12.10. It is clearly cocontinuous. It is also continuous as u transforms coverings to coverings and commutes with fibre products. Hence the lemma follows from Sites, Lemmas [original label: `sites-lemma-when-shriek`] and [original label: `sites-lemma-preserve-equalizers`]. □

Lemma 12.13. Let S be a scheme. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site containing S. The inclusion functor $S_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$ satisfies the hypotheses of Sites, Lemma [original label: `sites-lemma-bigger-site`] and hence induces a morphism of sites

$$\pi_S : (\mathrm{Sch}/S)_{pro\text{-}étale} \longrightarrow S_{pro\text{-}étale}$$

and a morphism of topoi

$$i_S : \mathrm{Sh}(S_{pro\text{-}étale}) \longrightarrow \mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})$$

such that $\pi_S \circ i_S = \mathrm{id}$. Moreover, $i_S = i_{\mathrm{id}_S}$ with i_{id_S} as in Lemma 12.12. In particular the functor $i_S^{-1} = \pi_{S,*}$ is described by the rule $i_S^{-1}(\mathcal{G})(U/S) = \mathcal{G}(U/S)$.

Proof. In this case the functor $u : S_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$, in addition to the properties seen in the proof of Lemma 12.12 above, also is fully faithful and transforms the final object into the final object. The lemma follows from Sites, Lemma [original label: `sites-lemma-bigger-site`]. □

Definition 12.14. In the situation of Lemma 12.13 the functor $i_S^{-1} = \pi_{S,*}$ is often called the restriction to the small pro-étale site, and for a sheaf $\mathcal{F}$ on the big pro-étale site we denote $\mathcal{F}|_{S_{pro\text{-}étale}}$ this restriction.

With this notation in place we have for a sheaf $\mathcal{F}$ on the big site and a sheaf $\mathcal{G}$ on the big site that

$$\mathrm{Mor}_{\mathrm{Sh}(S_{pro\text{-}étale})}(\mathcal{F}|_{S_{pro\text{-}étale}}, \mathcal{G}) = \mathrm{Mor}_{\mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})}(\mathcal{F}, i_{S,*}\mathcal{G})$$
$$\mathrm{Mor}_{\mathrm{Sh}(S_{pro\text{-}étale})}(\mathcal{G}, \mathcal{F}|_{S_{pro\text{-}étale}}) = \mathrm{Mor}_{\mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})}(\pi_S^{-1}\mathcal{G}, \mathcal{F})$$

Moreover, we have $(i_{S,*}\mathcal{G})|_{S_{pro\text{-}étale}} = \mathcal{G}$ and we have $(\pi_S^{-1}\mathcal{G})|_{S_{pro\text{-}étale}} = \mathcal{G}$.

Lemma 12.15. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site. Let $f : T \to S$ be a morphism in $\mathrm{Sch}_{pro\text{-}étale}$. The functor

$$u : (\mathrm{Sch}/T)_{pro\text{-}étale} \longrightarrow (\mathrm{Sch}/S)_{pro\text{-}étale}, \quad V/T \longmapsto V/S$$

is cocontinuous, and has a continuous right adjoint

$$v : (\mathrm{Sch}/S)_{pro\text{-}étale} \longrightarrow (\mathrm{Sch}/T)_{pro\text{-}étale}, \quad (U \to S) \longmapsto (U \times_S T \to T).$$

They induce the same morphism of topoi

$$f_{big} : \mathrm{Sh}((\mathrm{Sch}/T)_{pro\text{-}étale}) \longrightarrow \mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})$$

We have $f_{big}^{-1}(\mathcal{G})(U/T) = \mathcal{G}(U/S)$. We have $f_{big,*}(\mathcal{F})(U/S) = \mathcal{F}(U \times_S T/T)$. Also, f_{big}^{-1} has a left adjoint $f_{big!}$ which commutes with fibre products and equalizers.

Proof. The functor u is cocontinuous, continuous, and commutes with fibre products and equalizers (details omitted; compare with proof of Lemma 12.12). Hence Sites, Lemmas [original label: `sites-lemma-when-shriek`] and [original label: `sites-lemma-preserve-equalizers`] apply and we deduce the formula for f_{big}^{-1} and the existence of $f_{big!}$. Moreover, the functor v is a right adjoint because given U/T and V/S we have $\mathrm{Mor}_S(u(U), V) = \mathrm{Mor}_T(U, V \times_S T)$ as desired. Thus we may apply Sites, Lemmas [original label: `sites-lemma-have-functor-other-way`] and [original label: `sites-lemma-have-functor-other-way-morphism`] to get the formula for $f_{big,*}$. □

Lemma 12.16. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site. Let $f : T \to S$ be a morphism in $\mathrm{Sch}_{pro\text{-}étale}$.

1. We have $i_f = f_{big} \circ i_T$ with i_f as in Lemma 12.12 and i_T as in Lemma 12.13.
2. The functor $S_{pro\text{-}étale} \to T_{pro\text{-}étale}$, $(U \to S) \mapsto (U \times_S T \to T)$ is continuous and induces a morphism of topoi
$$f_{small} : \mathrm{Sh}(T_{pro\text{-}étale}) \longrightarrow \mathrm{Sh}(S_{pro\text{-}étale}).$$
We have $f_{small,*}(\mathcal{F})(U/S) = \mathcal{F}(U \times_S T/T)$.
3. We have a commutative diagram of morphisms of sites
$$\begin{array}{ccc} T_{pro\text{-}étale} & \xleftarrow{\pi_T} & (\mathrm{Sch}/T)_{pro\text{-}étale} \\ {\scriptstyle f_{small}}\downarrow & & \downarrow{\scriptstyle f_{big}} \\ S_{pro\text{-}étale} & \xleftarrow{\pi_S} & (\mathrm{Sch}/S)_{pro\text{-}étale} \end{array}$$
so that $f_{small} \circ \pi_T = \pi_S \circ f_{big}$ as morphisms of topoi.
4. We have $f_{small} = \pi_S \circ f_{big} \circ i_T = \pi_S \circ i_f$.

Proof. The equality $i_f = f_{big} \circ i_T$ follows from the equality $i_f^{-1} = i_T^{-1} \circ f_{big}^{-1}$ which is clear from the descriptions of these functors above. Thus we see (1).

The functor $u : S_{pro\text{-}étale} \to T_{pro\text{-}étale}$, $u(U \to S) = (U \times_S T \to T)$ transforms coverings into coverings and commutes with fibre products, see Lemmas 12.4 and 12.10. Moreover, both $S_{pro\text{-}étale}, T_{pro\text{-}étale}$ have final objects, namely S/S and T/T and $u(S/S) = T/T$. Hence by Sites, Proposition [original label: `sites-proposition-get-morphism`] the functor u corresponds to a morphism of sites $T_{pro\text{-}étale} \to S_{pro\text{-}étale}$. This in turn gives rise to the morphism of topoi, see Sites, Lemma [original label: `sites-lemma-morphism-sites-topoi`]. The description of the pushforward is clear from these references.

Part (3) follows because π_S and π_T are given by the inclusion functors and f_{small} and f_{big} by the base change functors $U \mapsto U \times_S T$.

Statement (4) follows from (3) by precomposing with i_T. □

In the situation of the lemma, using the terminology of Definition 12.14 we have: for $\mathcal{F}$ a sheaf on the big pro-étale site of T
$$(f_{big,*}\mathcal{F})|_{S_{pro\text{-}étale}} = f_{small,*}(\mathcal{F}|_{T_{pro\text{-}étale}}), \tag{12.16.1}$$
This equality is clear from the commutativity of the diagram of sites of the lemma, since restriction to the small pro-étale site of T, resp. S is given by $\pi_{T,*}$, resp. $\pi_{S,*}$. A similar formula involving pullbacks and restrictions is false.

Lemma 12.17. Given schemes X, Y, Y in $\mathrm{Sch}_{pro\text{-}étale}$ and morphisms $f : X \to Y$, $g : Y \to Z$ we have $g_{big} \circ f_{big} = (g \circ f)_{big}$ and $g_{small} \circ f_{small} = (g \circ f)_{small}$.

Proof. This follows from the simple description of pushforward and pullback for the functors on the big sites from Lemma 12.15. For the functors on the small sites this follows from the description of the pushforward functors in Lemma 12.16. □

Lemma 12.18. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site. Consider a cartesian diagram
$$\begin{array}{ccc} T' & \xrightarrow{g'} & T \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$
in $\mathrm{Sch}_{pro\text{-}étale}$. Then $i_g^{-1} \circ f_{big,*} = f'_{small,*} \circ (i_{g'})^{-1}$ and $g_{big}^{-1} \circ f_{big,*} = f'_{big,*} \circ (g'_{big})^{-1}$.

Proof. Since the diagram is cartesian, we have for U'/S' that $U' \times_{S'} T' = U' \times_S T$. Hence both $i_g^{-1} \circ f_{big,*}$ and $f'_{small,*} \circ (i_{g'})^{-1}$ send a sheaf $\mathcal{F}$ on $(\mathrm{Sch}/T)_{pro\text{-}étale}$ to the sheaf $U' \mapsto \mathcal{F}(U' \times_{S'} T')$ on $S'_{pro\text{-}étale}$ (use Lemmas 12.12 and 12.15). The second equality can be proved in the same manner or can be deduced from the very general Sites, Lemma [original label: `sites-lemma-localize-morphism`]. □

We can think about a sheaf on the big pro-étale site of S as a collection of sheaves on the small pro-étale site on schemes over S.

Lemma 12.19. Let S be a scheme contained in a big pro-étale site $\mathrm{Sch}_{pro\text{-}étale}$. A sheaf $\mathcal{F}$ on the big pro-étale site $(\mathrm{Sch}/S)_{pro\text{-}étale}$ is given by the following data:

1. for every $T/S \in \mathrm{Ob}((\mathrm{Sch}/S)_{pro\text{-}étale})$ a sheaf $\mathcal{F}_T$ on $T_{pro\text{-}étale}$,
2. for every $f : T' \to T$ in $(\mathrm{Sch}/S)_{pro\text{-}étale}$ a map $c_f : f_{small}^{-1}\mathcal{F}_T \to \mathcal{F}_{T'}$.

These data are subject to the following conditions:

(a) given any $f : T' \to T$ and $g : T'' \to T'$ in $(\mathrm{Sch}/S)_{pro\text{-}étale}$ the composition $c_g \circ g_{small}^{-1} c_f$ is equal to $c_{f \circ g}$, and

(b) if $f : T' \to T$ in $(\mathrm{Sch}/S)_{pro\text{-}étale}$ is weakly étale then c_f is an isomorphism.

Proof. Identical to the proof of Topologies, Lemma [original label: `topologies-lemma-characterize-sheaf-big-etale`]. □

Lemma 12.20. Let S be a scheme. Let $S_{affine,pro\text{-}étale}$ denote the full subcategory of $S_{pro\text{-}étale}$ consisting of affine objects. A covering of $S_{affine,pro\text{-}étale}$ will be a standard pro-étale covering, see Definition 12.6. Then restriction

$$\mathcal{F} \longmapsto \mathcal{F}|_{S_{affine,étale}}$$

defines an equivalence of topoi $\mathrm{Sh}(S_{pro\text{-}étale}) \cong \mathrm{Sh}(S_{affine,pro\text{-}étale})$.

Proof. This you can show directly from the definitions, and is a good exercise. But it also follows immediately from Sites, Lemma [original label: `sites-lemma-equivalence`] by checking that the inclusion functor $S_{affine,pro\text{-}étale} \to S_{pro\text{-}étale}$ is a special cocontinuous functor (see Sites, Definition [original label: `sites-definition-special-cocontinuous-functor`]). □

Lemma 12.21. Let S be an affine scheme. Let S_{app} denote the full subcategory of $S_{pro\text{-}étale}$ consisting of affine objects U such that $\mathcal{O}(S) \to \mathcal{O}(U)$ is ind-étale. A covering of S_{app} will be a standard pro-étale covering, see Definition 12.6. Then restriction

$$\mathcal{F} \longmapsto \mathcal{F}|_{S_{app}}$$

defines an equivalence of topoi $\mathrm{Sh}(S_{pro\text{-}étale}) \cong \mathrm{Sh}(S_{app})$.

Proof. By Lemma 12.20 we may replace $S_{pro\text{-}étale}$ by $S_{affine,pro\text{-}étale}$. The lemma follows from Sites, Lemma [original label: `sites-lemma-equivalence`] by checking that the inclusion functor $S_{app} \to S_{affine,pro\text{-}étale}$ is a special cocontinuous functor, see Sites, Definition [original label: `sites-definition-special-cocontinuous-functor`]. The conditions of Sites, Lemma [original label: `sites-lemma-equivalence`] follow immediately from the definition and the facts (a) any object U of $S_{affine,pro\text{-}étale}$ has a covering $\{V \to U\}$ with V ind-étale over X (Proposition 9.1) and (b) the functor u is fully faithful. □

Lemma 12.22. Let S be a scheme. The topology on each of the pro-étale sites $\mathrm{Sch}_{pro\text{-}étale}$, $S_{pro\text{-}étale}$, $(\mathrm{Sch}/S)_{pro\text{-}étale}$, $S_{affine,pro\text{-}étale}$, and $(\mathrm{Aff}/S)_{pro\text{-}étale}$ is subcanonical.

Proof. Combine Lemma 12.2 and Descent, Lemma [original label: `descent-lemma-fpqc-universal-effective-epimorphisms`]. □

13 Weakly contractible objects

In this section we prove the key fact that our pro-étale sites contain many weakly contractible objects. In fact, the proof of Lemma 13.3 is the reason for the shape of the function *Bound* in Definition 12.7 (although for readers who are ignoring set theoretical questions, this information is without content).
We first express the notion of w-contractible rings in terms of pro-étale coverings.

Lemma 13.1. Let $T = \mathrm{Spec}(A)$ be an affine scheme. The following are equivalent

1. A is w-contractible, and

2. every pro-étale covering of T can be refined by a Zariski covering of the form $T = \coprod_{i=1,\ldots,n} U_i$.

Proof. Assume A is w-contractible. By Lemma 12.5 it suffices to prove we can refine every standard pro-étale covering $\{f_i : T_i \to T\}_{i=1,\ldots,n}$ by a Zariski covering of T. The morphism $\coprod T_i \to T$ is a surjective weakly étale morphism of affine schemes. Hence by Definition 11.1 there exists a morphism $\sigma : T \to \coprod T_i$ over T. Then the Zariski covering $T = \coprod \sigma^{-1}(T_i)$ refines $\{f_i : T_i \to T\}$.

Conversely, assume (2). If $A \to B$ is faithfully flat and weakly étale, then $\{\mathrm{Spec}(B) \to T\}$ is a pro-étale covering. Hence there exists a Zariski covering $T = \coprod U_i$ and morphisms $U_i \to \mathrm{Spec}(B)$ over T. Since $T = \coprod U_i$ we obtain $T \to \mathrm{Spec}(B)$, i.e., an A-algebra map $B \to A$. This means A is w-contractible. □

Lemma 13.2. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site as in Definition 12.7. Let $T = \mathrm{Spec}(A)$ be an affine object of $\mathrm{Sch}_{pro\text{-}étale}$. The following are equivalent

1. A is w-contractible,
2. T is a weakly contractible (Sites, Definition [original label: `sites-definition-w-contractible`]) object of $\mathrm{Sch}_{pro\text{-}étale}$, and
3. every pro-étale covering of T can be refined by a Zariski covering of the form $T = \coprod_{i=1,\ldots,n} U_i$.

Proof. We have seen the equivalence of (1) and (3) in Lemma 13.1.

Assume (3) and let $\mathcal{F} \to \mathcal{G}$ be a surjection of sheaves on $\mathrm{Sch}_{pro\text{-}étale}$. Let $s \in \mathcal{G}(T)$. To prove (2) we will show that s is in the image of $\mathcal{F}(T) \to \mathcal{G}(T)$. We can find a covering $\{T_i \to T\}$ of $\mathrm{Sch}_{pro\text{-}étale}$ such that s lifts to a section of $\mathcal{F}$ over T_i (Sites, Definition [original label: `sites-definition-sheaves-injective-surjective`]). By (3) we may assume we have a finite covering $T = \coprod_{j=1,\ldots,m} U_j$ by open and closed subsets and we have $t_j \in \mathcal{F}(U_j)$ mapping to $s|_{U_j}$. Since Zariski coverings are coverings in $\mathrm{Sch}_{pro\text{-}étale}$ (Lemma 12.3) we conclude that $\mathcal{F}(T) = \prod \mathcal{F}(U_j)$. Thus $t = (t_1, \ldots, t_m) \in \mathcal{F}(T)$ is a section mapping to s.

Assume (2). Let $A \to D$ be as in Proposition 11.3. Then $\{V \to T\}$ is a covering of $\mathrm{Sch}_{pro\text{-}étale}$. (Note that $V = \mathrm{Spec}(D)$ is an object of $\mathrm{Sch}_{pro\text{-}étale}$ by Remark 11.4 combined with our choice of the function *Bound* in Definition 12.7 and the computation of the size of affine schemes in Sets, Lemma [original label: `sets-lemma-bound-size`].) Since the topology on $\mathrm{Sch}_{pro\text{-}étale}$ is subcanonical (Lemma 12.22) we see that $h_V \to h_T$ is a surjective map of sheaves (Sites, Lemma [original label: `sites-lemma-covering-surjective-after-sheafification`]). Since T is assumed weakly contractible, we see that there is an element $f \in h_V(T) = \mathrm{Mor}(T, V)$ whose image in $h_T(T)$ is id_T. Thus $A \to D$ has a retraction $\sigma : D \to A$. Now if $A \to B$ is faithfully flat and weakly étale, then $D \to D \otimes_A B$ has the same properties, hence there is a retraction $D \otimes_A B \to D$ and combined with σ we get a retraction $B \to D \otimes_A B \to D \to A$ of $A \to B$. Thus A is w-contractible and (1) holds. □

Lemma 13.3. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site as in Definition 12.7. For every object T of $\mathrm{Sch}_{pro\text{-}étale}$ there exists a covering $\{T_i \to T\}$ in $\mathrm{Sch}_{pro\text{-}étale}$ with each T_i affine and the spectrum of a w-contractible ring. In particular, T_i is weakly contractible in $\mathrm{Sch}_{pro\text{-}étale}$.

Proof. For those readers who do not care about set-theoretical issues this lemma is a trivial consequence of Lemma 13.2 and Proposition 11.3. Here are the details. Choose an affine open covering $T = \bigcup U_i$. Write $U_i = \mathrm{Spec}(A_i)$. Choose faithfully flat, ind-étale ring maps $A_i \to D_i$ such that D_i is w-contractible as in Proposition 11.3. The family of morphisms $\{\mathrm{Spec}(D_i) \to T\}$ is a pro-étale covering. If we can show that $\mathrm{Spec}(D_i)$ is isomorphic to an object, say T_i, of $\mathrm{Sch}_{pro\text{-}étale}$, then $\{T_i \to T\}$ will be combinatorially equivalent to a covering of $\mathrm{Sch}_{pro\text{-}étale}$ by the construction of $\mathrm{Sch}_{pro\text{-}étale}$ in Definition 12.7 and more precisely the application of Sets, Lemma [original label: `sets-lemma-coverings-site`] in the last step. To prove $\mathrm{Spec}(D_i)$ is isomorphic to an object of $\mathrm{Sch}_{pro\text{-}étale}$, it suffices to prove that $|D_i| \leq \mathit{Bound}(\mathrm{size}(T))$ by the construction of $\mathrm{Sch}_{pro\text{-}étale}$ in Definition 12.7 and more precisely the application of Sets, Lemma [original label: `sets-lemma-construct-category`] in step (3). Since $|A_i| \leq \mathrm{size}(U_i) \leq \mathrm{size}(T)$ by Sets, Lemmas [original label: `sets-lemma-bound-affine`] and [original label: `sets-lemma-bound-finite-type`] we get $|D_i| \leq \kappa^{2^{2^{2^\kappa}}}$ where $\kappa = \mathrm{size}(T)$ by Remark 11.4. Thus by our choice of the function *Bound* in Definition 12.7 we win. □

Lemma 13.4. Let S be a scheme. The pro-étale sites $S_{pro\text{-}étale}$, $(\mathrm{Sch}/S)_{pro\text{-}étale}$, $S_{affine,pro\text{-}étale}$, and $(\mathrm{Aff}/S)_{pro\text{-}étale}$ and if S is affine S_{app} have enough (affine) quasi-compact, weakly contractible objects, see Sites, Definition [original label: `sites-definition-w-contractible`].

Proof. Follows immediately from Lemma 13.3. □

Lemma 13.5. Let S be a scheme. The pro-étale sites $\mathrm{Sch}_{pro\text{-}étale}$, $S_{pro\text{-}étale}$, $(\mathrm{Sch}/S)_{pro\text{-}étale}$ have the following property: for any object U there exists a covering $\{V \to U\}$ with V a weakly contractible object. If U is quasi-compact, then we may choose V affine and weakly contractible.

Proof. Suppose that $V = \coprod_{j \in J} V_j$ is an object of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ which is the disjoint union of weakly contractible objects V_j. Since a disjoint union decomposition is a pro-étale covering we see that $\mathcal{F}(V) = \prod_{j \in J} \mathcal{F}(V_j)$ for any pro-étale sheaf $\mathcal{F}$. Let $\mathcal{F} \to \mathcal{G}$ be a surjective map of sheaves of sets. Since V_j is weakly contractible, the map $\mathcal{F}(V_j) \to \mathcal{G}(V_j)$ is surjective, see Sites, Definition [original label: `sites-definition-w-contractible`]. Thus $\mathcal{F}(V) \to \mathcal{G}(V)$ is surjective as a product of surjective maps of sets and we conclude that V is weakly contractible.

Choose a covering $\{U_i \to U\}_{i \in I}$ with U_i affine and weakly contractible as in Lemma 13.3. Take $V = \coprod_{i \in I} U_i$ (there is a set theoretic issue here which we will address below). Then $\{V \to U\}$ is the desired pro-étale covering by a weakly contractible object (to check it is a covering use Lemma 12.2). If U is quasi-compact, then it follows immediately from Lemma 12.2 that we can choose a finite subset $I' \subset I$ such that $\{U_i \to U\}_{i \in I'}$ is still a covering and then $\{\coprod_{i \in I'} U_i \to U\}$ is the desired covering by an affine and weakly contractible object.

In this paragraph, which we urge the reader to skip, we address set theoretic problems. In order to know that the disjoint union lies in our partial universe, we need to bound the cardinality of the index set I. It is seen immediately from the construction of the covering $\{U_i \to U\}_{i \in I}$ in the proof of Lemma 13.3 that $|I| \leq \mathrm{size}(U)$ where the size of a scheme is as defined in Sets, Section [original label: `sets-section-categories-schemes`]. Moreover, for each i we have $\mathrm{size}(U_i) \leq \mathit{Bound}(\mathrm{size}(U))$; this follows for the bound of the cardinality of $\Gamma(U_i, \mathcal{O}_{U_i})$ in the proof of Lemma 13.3 and Sets, Lemma [original label: `sets-lemma-bound-affine`]. Thus $\mathrm{size}(\coprod_{i \in I} U_i)) \leq \mathit{Bound}(\mathrm{size}(U))$ by Sets, Lemma [original label: `sets-lemma-bound-size`]. Hence by construction of the big pro-étale site through Sets, Lemma [original label: `sets-lemma-construct-category`] we see that $\coprod_{i \in I} U_i$ is isomorphic to an object of our site and the proof is complete. □

14 Weakly contractible hypercoverings

The results of Section 13 leads to the existence of hypercoverings made up out weakly contractible objects.

Lemma 14.1. Let X be a scheme.

1. For every object U of $X_{pro\text{-}étale}$ there exists a hypercovering K of U in $X_{pro\text{-}étale}$ such that each term K_n consists of a single weakly contractible object of $X_{pro\text{-}étale}$ covering U.
2. For every quasi-compact and quasi-separated object U of $X_{pro\text{-}étale}$ there exists a hypercovering K of U in $X_{pro\text{-}étale}$ such that each term K_n consists of a single affine and weakly contractible object of $X_{pro\text{-}étale}$ covering U.

Proof. Let $\mathcal{B} \subset \mathrm{Ob}(X_{pro\text{-}étale})$ be the set of weakly contractible objects of $X_{pro\text{-}étale}$. Every object T of $X_{pro\text{-}étale}$ has a covering $\{T_i \to T\}_{i \in I}$ with I finite and $T_i \in \mathcal{B}$ by Lemma 13.5. By Hypercoverings, Lemma [original label: `hypercovering-lemma-w-contractible`] we get a hypercovering K of U such that $K_n = \{U_{n,i}\}_{i \in I_n}$ with I_n finite and $U_{n,i}$ weakly contractible. Then we can replace K by the hypercovering of U given by $\{U_n\}$ in degree n where $U_n = \coprod_{i \in I_n} U_{n,i}$ This is allowed by Hypercoverings, Remark [original label: `hypercovering-remark-take-unions-hypercovering-X`].

Let $X_{qcqs,pro\text{-}étale} \subset X_{pro\text{-}étale}$ be the full subcategory consisting of quasi-compact and quasi-separated objects. A covering of $X_{qcqs,pro\text{-}étale}$ will be a finite pro-étale covering. Then $X_{qcqs,pro\text{-}étale}$ is a site, has fibre products, and the inclusion functor $X_{qcqs,pro\text{-}étale} \to X_{pro\text{-}étale}$ is continuous and commutes with fibre products. In particular, if K is a hypercovering of an object U in $X_{qcqs,pro\text{-}étale}$ then K is a hypercovering of U in $X_{pro\text{-}étale}$ by Hypercoverings, Lemma [original label: `hypercovering-lemma-hypercovering-continuous-functor`]. Let $\mathcal{B} \subset \mathrm{Ob}(X_{qcqs,pro\text{-}étale})$ be the set of affine and weakly contractible objects. By Lemma 13.3 and the fact that finite unions of affines are affine, for every object U of $X_{qcqs,pro\text{-}étale}$ there exists a covering $\{V \to U\}$ of $X_{qcqs,pro\text{-}étale}$ with $V \in \mathcal{B}$. By Hypercoverings, Lemma [original label: `hypercovering-lemma-w-contractible`] we get a hypercovering K of U such that $K_n = \{U_{n,i}\}_{i \in I_n}$ with I_n finite and $U_{n,i}$ affine and weakly contractible. Then we can replace K by the hypercovering of U given by $\{U_n\}$ in degree n where $U_n = \coprod_{i \in I_n} U_{n,i}$. This is allowed by Hypercoverings, Remark [original label: `hypercovering-remark-take-unions-hypercovering-X`]. □

In the following lemma we use the Čech complex $s(\mathcal{F}(K))$ associated to a hypercovering K in a site. See Hypercoverings, Section [original label: `hypercovering-section-hyper-cech`]. If K is a hypercovering of U and $K_n = \{U_n \to U\}$, then the Čech complex looks like this:

$$s(\mathcal{F}(K)) = (\mathcal{F}(U_0) \to \mathcal{F}(U_1) \to \mathcal{F}(U_2) \to \ldots)$$

where $s(\mathcal{F}(U_n))$ is placed in cohomological degree n.

Lemma 14.2. Let X be a scheme. Let $E \in D^+(X_{pro\text{-}étale})$ be represented by a bounded below complex $\mathcal{E}^\bullet$ of abelian sheaves. Let K be a hypercovering of $U \in \mathrm{Ob}(X_{pro\text{-}étale})$ with $K_n = \{U_n \to U\}$ where U_n is a weakly contractible object of $X_{pro\text{-}étale}$. Then

$$R\Gamma(U, E) = \mathrm{Tot}(s(\mathcal{E}^\bullet(K)))$$

in $D(\mathrm{Ab})$.

Proof. If $\mathcal{E}$ is an abelian sheaf on $X_{pro\text{-}étale}$, then the spectral sequence of Hypercoverings, Lemma [original label: `hypercovering-lemma-cech-spectral-sequence`] implies that

$$R\Gamma(X_{pro\text{-}étale}, \mathcal{E}) = s(\mathcal{E}(K))$$

because the higher cohomology groups of any sheaf over U_n vanish, see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-w-contractible`].

If $\mathcal{E}^\bullet$ is bounded below, then we can choose an injective resolution $\mathcal{E}^\bullet \to \mathcal{I}^\bullet$ and consider the map of complexes

$$\mathrm{Tot}(s(\mathcal{E}^\bullet(K))) \longrightarrow \mathrm{Tot}(s(\mathcal{I}^\bullet(K)))$$

For every n the map $\mathcal{E}^\bullet(U_n) \to \mathcal{I}^\bullet(U_n)$ is a quasi-isomorphism because taking sections over U_n is exact. Hence the displayed map is a quasi-isomorphism by one of the spectral sequences of Homology, Lemma [original label: `homology-lemma-first-quadrant-ss`]. Using the result of the first paragraph we see that for every p the complex $s(\mathcal{I}^p(K))$ is acyclic in degrees $n > 0$ and computes $\mathcal{I}^p(U)$ in degree 0. Thus the other spectral sequence of Homology, Lemma [original label: `homology-lemma-first-quadrant-ss`] shows $\mathrm{Tot}(s(\mathcal{I}^\bullet(K)))$ computes $R\Gamma(U, E) = \mathcal{I}^\bullet(U)$. □

Lemma 14.3. Let X be a quasi-compact and quasi-separated scheme. The functor $R\Gamma(X, -) : D^+(X_{pro\text{-}étale}) \to D(\mathrm{Ab})$ commutes with direct sums and homotopy colimits.

Proof. The statement means the following: Suppose we have a family of objects E_i of $D^+(X_{pro\text{-}étale})$ such that $\bigoplus E_i$ is an object of $D^+(X_{pro\text{-}étale})$. Then $R\Gamma(X, \bigoplus E_i) = \bigoplus R\Gamma(X, E_i)$. To see this choose a hypercovering K of X with $K_n = \{U_n \to X\}$ where U_n is an affine and weakly contractible scheme, see Lemma 14.1. Let N be an integer such that $H^p(E_i) = 0$ for $p < N$. Choose a complex of abelian sheaves $\mathcal{E}_i^\bullet$ representing E_i with $\mathcal{E}_i^p = 0$ for $p < N$. The termwise direct sum $\bigoplus \mathcal{E}_i^\bullet$ represents $\bigoplus E_i$ in $D(X_{pro\text{-}étale})$, see Injectives, Lemma [original label: `injectives-lemma-derived-products`]. By Lemma 14.2 we have

$$R\Gamma(X, \bigoplus E_i) = \mathrm{Tot}(s((\bigoplus \mathcal{E}_i^\bullet)(K)))$$

and

$$R\Gamma(X, E_i) = \mathrm{Tot}(s(\mathcal{E}_i^\bullet(K)))$$

Since each U_n is quasi-compact we see that

$$\mathrm{Tot}(s((\bigoplus \mathcal{E}_i^\bullet)(K))) = \bigoplus \mathrm{Tot}(s(\mathcal{E}_i^\bullet(K)))$$

by Modules on Sites, Lemma [original label: `sites-modules-lemma-sections-over-quasi-compact`]. The statement on homotopy colimits is a formal consequence of the fact that $R\Gamma$ is an exact functor of triangulated categories and the fact (just proved) that it commutes with direct sums. □

Remark 14.4. Let X be a scheme. Because $X_{pro\text{-}étale}$ has enough weakly contractible objects for all K in $D(X_{pro\text{-}étale})$ we have $K = R\lim \tau_{\geq -n} K$ by Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`]. Since $R\Gamma$ commutes with $R\lim$ by Injectives, Lemma [original label: `injectives-lemma-RF-commutes-with-Rlim`] we see that

$$R\Gamma(X, K) = R\lim R\Gamma(X, \tau_{\geq -n} K)$$

in $D(\mathrm{Ab})$. This will sometimes allow us to extend results from bounded below complexes to all complexes.

15 Compact generation

In this section we prove that various derived categories associated to our pro-étale sites are compactly generated as defined in Derived Categories, Definition [original label: `derived-definition-compactly-generated`].

Lemma 15.1. Let S be a scheme. Let Λ be a ring.

1. $D(S_{pro\text{-}étale})$ is compactly generated,
2. $D(S_{pro\text{-}étale}, \Lambda)$ is compactly generated,
3. $D(S_{pro\text{-}étale}, \mathcal{A})$ is compactly generated for any sheaf of rings $\mathcal{A}$ on $S_{pro\text{-}étale}$,
4. $D((\mathrm{Sch}/S)_{pro\text{-}étale})$ is compactly generated,
5. $D((\mathrm{Sch}/S)_{pro\text{-}étale}, \Lambda)$ is compactly generated, and
6. $D((\mathrm{Sch}/S)_{pro\text{-}étale}, \mathcal{A})$ is compactly generated for any sheaf of rings $\mathcal{A}$ on $(\mathrm{Sch}/S)_{pro\text{-}étale}$,

Proof. Proof of (3). Let U be an affine object of $S_{pro\text{-}étale}$ which is weakly contractible. Then $j_{U!}\mathcal{A}_U$ is a compact object of the derived category $D(S_{pro\text{-}étale}, \mathcal{A})$, see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-quasi-compact-weakly-contractible-compact`]. Choose a set I and for each $i \in I$ an affine weakly contractible object U_i of $S_{pro\text{-}étale}$ such that every affine weakly contractible object of $S_{pro\text{-}étale}$ is isomorphic to one of the U_i. This is possible because $\mathrm{Ob}(S_{pro\text{-}étale})$ is a set. To finish the proof of (3) it suffices to show that $\bigoplus j_{U_i,!}\mathcal{A}_{U_i}$ is a generator of $D(S_{pro\text{-}étale}, \mathcal{A})$, see Derived Categories, Definition [original label: `derived-definition-generators`]. To see this, let K be a nonzero object of $D(S_{pro\text{-}étale}, \mathcal{A})$. Then there exists an object T of our site $S_{pro\text{-}étale}$ and a nonzero element ξ of $H^n(K)(T)$. In other words, ξ is a nonzero section of the nth cohomology sheaf of K. We may assume K is represented by a complex $\mathcal{K}^\bullet$ of sheaves of $\mathcal{A}$-modules and ξ is the class of a section $s \in \mathcal{K}^n(T)$ with $\mathrm{d}(s) = 0$. Namely, ξ is locally represented as the class of a section (so you get the result after replacing T by a member of a covering of T). Next, we choose a covering $\{T_j \to T\}_{j\in J}$ as in Lemma 13.3. Since $H^n(K)$ is a sheaf, we see that for some j the restriction $\xi|_{T_j}$ remains nonzero. Thus $s|_{T_j}$ defines a nonzero map $j_{T_j,!}\mathcal{A}_{T_j} \to K$ in $D(S_{pro\text{-}étale}, \mathcal{A})$. Since $T_j \cong U_i$ for some $i \in I$ we conclude.

The exact same argument works for the big pro-étale site of S. □

16 Comparing topologies

This section is the analogue of Étale Cohomology, Section [original label: `etale-cohomology-section-compare-topologies`].

Lemma 16.1. Let X be a scheme. Let $\mathcal{F}$ be a presheaf of sets on $X_{pro\text{-}étale}$ which sends finite disjoint unions to products. Then $\mathcal{F}^\#(W) = \mathcal{F}(W)$ if W is an affine weakly contractible object of $X_{pro\text{-}étale}$.

Proof. Recall that $\mathcal{F}^\#$ is equal to $(\mathcal{F}^+)^+$, see Sites, Theorem [original label: `sites-theorem-plus`], where $\mathcal{F}^+$ is the presheaf which sends an object U of $X_{pro\text{-}étale}$ to $\mathrm{colim}\, H^0(\mathcal{U}, \mathcal{F})$ where the colimit is over all pro-étale coverings $\mathcal{U}$ of U. Thus it suffices to prove that (a) $\mathcal{F}^+$ sends finite disjoint unions to products and (b) sends W to $\mathcal{F}(W)$. If $U = U_1 \amalg U_2$, then given a pro-étale covering $\mathcal{U} = \{f_j : V_j \to U\}$ of U we obtain pro-étale coverings $\mathcal{U}_i = \{f_j^{-1}(U_i) \to U_i\}$ and we clearly have

$$H^0(\mathcal{U}, \mathcal{F}) = H^0(\mathcal{U}_1, \mathcal{F}) \times H^0(\mathcal{U}_2, \mathcal{F})$$

because $\mathcal{F}$ sends finite disjoint unions to products (this includes the condition that $\mathcal{F}$ sends the empty scheme to the singleton). This proves (a). Finally, any pro-étale covering of W can be refined by a finite disjoint union decomposition $W = W_1 \amalg \ldots W_n$ by Lemma 13.2. Hence $\mathcal{F}^+(W) = \mathcal{F}(W)$ exactly because the value of $\mathcal{F}$ on W is the product of the values of $\mathcal{F}$ on the W_j. This proves (b). □

Lemma 16.2. Let $f : X \to Y$ be a morphism of schemes. Let $\mathcal{F}$ be a sheaf of sets on $Y_{pro\text{-}étale}$. If W is an affine weakly contractible object of $X_{pro\text{-}étale}$, then

$$f_{small}^{-1}\mathcal{F}(W) = \mathrm{colim}_{W\to V}\, \mathcal{F}(V)$$

where the colimit is over morphisms $W \to V$ over Y with $V \in Y_{pro\text{-}étale}$.

Proof. Recall that $f_{small}^{-1}\mathcal{F}$ is the sheaf associated to the presheaf

$$u_p\mathcal{F} : U \mapsto \mathrm{colim}_{U\to V}\, \mathcal{F}(V)$$

on $X_{étale}$, see Sites, Sections [original label: `sites-section-morphism-sites`] and [original label: `sites-section-continuous-functors`]; we've suppressed from the notation that the colimit is over the opposite of the category $\{U \to V, V \in Y_{pro\text{-}étale}\}$. By Lemma 16.1 it suffices to prove that $u_p\mathcal{F}$ sends finite disjoint unions to products. Suppose that $U = U_1 \amalg U_2$ is a disjoint union of open and closed subschemes. There is a functor

$$\{U_1 \to V_1\} \times \{U_2 \to V_2\} \longrightarrow \{U \to V\}, \quad (U_1 \to V_1, U_2 \to V_2) \longmapsto (U \to V_1 \amalg V_2)$$

which is initial (Categories, Definition [original label: categories-definition-initial]). Hence the corresponding functor on opposite categories is cofinal and by Categories, Lemma [original label: categories-lemma-cofinal] we see that $u_p\mathcal{F}$ on U is the colimit of the values $\mathcal{F}(V_1 \amalg V_2)$ over the product category. Since $\mathcal{F}$ is a sheaf it sends disjoint unions to products and we conclude $u_p\mathcal{F}$ does too. □

Lemma 16.3. Let S be a scheme. Consider the morphism

$$\pi_S : (\mathrm{Sch}/S)_{pro\text{-}étale} \longrightarrow S_{pro\text{-}étale}$$

of Lemma 12.13. Let $\mathcal{F}$ be a sheaf on $S_{pro\text{-}étale}$. Then $\pi_S^{-1}\mathcal{F}$ is given by the rule

$$(\pi_S^{-1}\mathcal{F})(T) = \Gamma(T_{pro\text{-}étale}, f_{small}^{-1}\mathcal{F})$$

where $f : T \to S$. Moreover, $\pi_S^{-1}\mathcal{F}$ satisfies the sheaf condition with respect to fpqc coverings.

Proof. Observe that we have a morphism $i_f : \mathrm{Sh}(T_{pro\text{-}étale}) \to \mathrm{Sh}(\mathrm{Sch}/S)_{pro\text{-}étale})$ such that $\pi_S \circ i_f = f_{small}$ as morphisms $T_{pro\text{-}étale} \to S_{pro\text{-}étale}$, see Lemma 12.12. Since pullback is transitive we see that $i_f^{-1}\pi_S^{-1}\mathcal{F} = f_{small}^{-1}\mathcal{F}$ as desired.

Let $\{g_i : T_i \to T\}_{i\in I}$ be an fpqc covering. The final statement means the following: Given a sheaf $\mathcal{G}$ on $T_{pro\text{-}étale}$ and given sections $s_i \in \Gamma(T_i, g_{i,small}^{-1}\mathcal{G})$ whose pullbacks to $T_i \times_T T_j$ agree, there is a unique section s of $\mathcal{G}$ over T whose pullback to T_i agrees with s_i. We will prove this statement when T is affine and the covering is given by a single surjective flat morphism $T' \to T$ of affines and omit the reduction of the general case to this case.

Let $g : T' \to T$ be a surjective flat morphism of affines and let $s' \in g_{small}^{-1}\mathcal{G}(T')$ be a section with $\mathrm{pr}_0^* s' = \mathrm{pr}_1^* s'$ on $T' \times_T T'$. Choose a surjective weakly étale morphism $W \to T'$ with W affine and weakly contractible, see Lemma 13.5. By Lemma 16.2 the restriction $s'|_W$ is an element of $\mathrm{colim}_{W\to U}\mathcal{G}(U)$. Choose $\phi : W \to U_0$ and $s_0 \in \mathcal{G}(U_0)$ corresponding to s'. Choose a surjective weakly étale morphism $V \to W \times_T W$ with V affine and weakly contractible. Denote $a, b : V \to W$ the induced morphisms. Since $a^*(s'|_W) = b^*(s'|_W)$ and since the category $\{V \to U, U \in T_{pro\text{-}étale}\}$ is cofiltered (this is clear but see Sites, Lemma [original label: sites-lemma-directed-morphism] if in doubt), we see that the two morphisms $\phi \circ a, \phi \circ b : V \to U_0$ have to be equal. By the results in Descent, Section [original label: descent-section-fpqc-universal-effective-epimorphisms] (especially Descent, Lemma [original label: descent-lemma-fpqc-universal-effective-epimorphisms]) it follows there is a unique morphism $T \to U_0$ such that ϕ is the composition of this morphism with the structure morphism $W \to T$ (small detail omitted). Then we can let s be the pullback of s_0 by this morphism. We omit the verification that s pulls back to s' on T'. □

17 Comparing big and small topoi

This section is the analogue of Étale Cohomology, Section [original label: etale-cohomology-section-compare]. In the following we will often denote $\mathcal{F} \mapsto \mathcal{F}|_{S_{pro\text{-}étale}}$ the pullback functor i_S^{-1} corresponding to the morphism of topoi $i_S : \mathrm{Sh}(S_{pro\text{-}étale}) \to \mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale})$ of Lemma 12.13.

Lemma 17.1. Let S be a scheme. Let T be an object of $(\mathrm{Sch}/S)_{pro\text{-}étale}$.

1. If $\mathcal{I}$ is injective in $\mathrm{Ab}((\mathrm{Sch}/S)_{pro\text{-}étale})$, then
   (a) $i_f^{-1}\mathcal{I}$ is injective in $\mathrm{Ab}(T_{pro\text{-}étale})$,
   (b) $\mathcal{I}|_{S_{pro\text{-}étale}}$ is injective in $\mathrm{Ab}(S_{pro\text{-}étale})$,
2. If $\mathcal{I}^\bullet$ is a K-injective complex in $\mathrm{Ab}((\mathrm{Sch}/S)_{pro\text{-}étale})$, then
   (a) $i_f^{-1}\mathcal{I}^\bullet$ is a K-injective complex in $\mathrm{Ab}(T_{pro\text{-}étale})$,
   (b) $\mathcal{I}^\bullet|_{S_{pro\text{-}étale}}$ is a K-injective complex in $\mathrm{Ab}(S_{pro\text{-}étale})$,

Proof. Proof of (1)(a) and (2)(a): i_f^{-1} is a right adjoint of an exact functor $i_{f,!}$. Namely, recall that i_f corresponds to a cocontinuous functor $u : T_{pro\text{-}étale} \to (\mathrm{Sch}/S)_{pro\text{-}étale}$ which is continuous and commutes with fibre products and equalizers, see Lemma 12.12 and its proof. Hence we obtain $i_{f,!}$ by Modules on Sites, Lemma [original label: sites-modules-lemma-g-shriek-adjoint]. It is shown in Modules on Sites, Lemma [original label: sites-modules-lemma-exactness-lower-shriek] that it is exact. Then we conclude (1)(a) and (2)(a) hold by Homology, Lemma [original label: homology-lemma-adjoint-preserve-injectives] and Derived Categories, Lemma [original label: derived-lemma-adjoint-preserve-K-injectives].

Parts (1)(b) and (2)(b) are special cases of (1)(a) and (2)(a) as $i_S = i_{\mathrm{id}_S}$. □

Lemma 17.2. Let $f : T \to S$ be a morphism of schemes. For K in $D((\mathrm{Sch}/T)_{pro\text{-}étale})$ we have

$$(Rf_{big,*}K)|_{S_{pro\text{-}étale}} = Rf_{small,*}(K|_{T_{pro\text{-}étale}})$$

in $D(S_{pro\text{-}étale})$. More generally, let $S' \in \mathrm{Ob}((\mathrm{Sch}/S)_{pro\text{-}étale})$ with structure morphism $g : S' \to S$. Consider the fibre product

$$\begin{array}{ccc} T' & \xrightarrow{g'} & T \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

Then for K in $D((\mathrm{Sch}/T)_{pro\text{-}étale})$ we have

$$i_g^{-1}(Rf_{big,*}K) = Rf'_{small,*}(i_{g'}^{-1}K)$$

in $D(S'_{pro\text{-}étale})$ and

$$g_{big}^{-1}(Rf_{big,*}K) = Rf'_{big,*}((g'_{big})^{-1}K)$$

in $D((\mathrm{Sch}/S')_{pro\text{-}étale})$.

Proof. The first equality follows from Lemma 17.1 and (12.16.1) on choosing a K-injective complex of abelian sheaves representing K. The second equality follows from Lemma 17.1 and Lemma 12.18 on choosing a K-injective complex of abelian sheaves representing K. The third equality follows similarly from Cohomology on Sites, Lemmas [original label: `sites-cohomology-lemma-cohomology-of-open`] and [original label: `sites-cohomology-lemma-restrict-K-injective-to-open`] and Lemma 12.18 on choosing a K-injective complex of abelian sheaves representing K. □

Let S be a scheme and let $\mathcal{H}$ be an abelian sheaf on $(\mathrm{Sch}/S)_{pro\text{-}étale}$. Recall that $H^n_{pro\text{-}étale}(U, \mathcal{H})$ denotes the cohomology of $\mathcal{H}$ over an object U of $(\mathrm{Sch}/S)_{pro\text{-}étale}$.

Lemma 17.3. Let $f : T \to S$ be a morphism of schemes. For K in $D(S_{pro\text{-}étale})$ we have

$$H^n_{pro\text{-}étale}(S, \pi_S^{-1}K) = H^n(S_{pro\text{-}étale}, K)$$

and

$$H^n_{pro\text{-}étale}(T, \pi_S^{-1}K) = H^n(T_{pro\text{-}étale}, f_{small}^{-1}K).$$

For M in $D((\mathrm{Sch}/S)_{pro\text{-}étale})$ we have

$$H^n_{pro\text{-}étale}(T, M) = H^n(T_{pro\text{-}étale}, i_f^{-1}M).$$

Proof. To prove the last equality represent M by a K-injective complex of abelian sheaves and apply Lemma 17.1 and work out the definitions. The second equality follows from this as $i_f^{-1} \circ \pi_S^{-1} = f_{small}^{-1}$. The first equality is a special case of the second one. □

Lemma 17.4. Let S be a scheme. For $K \in D(S_{pro\text{-}étale})$ the map

$$K \longrightarrow R\pi_{S,*}\pi_S^{-1}K$$

is an isomorphism.

Proof. This is true because both π_S^{-1} and $\pi_{S,*} = i_S^{-1}$ are exact functors and the composition $\pi_{S,*} \circ \pi_S^{-1}$ is the identity functor. □

18 Points of the pro-étale site

We first apply Deligne's criterion to show that there are enough points.

Lemma 18.1. Let S be a scheme. The pro-étale sites $\mathrm{Sch}_{pro\text{-}étale}$, $S_{pro\text{-}étale}$, $(\mathrm{Sch}/S)_{pro\text{-}étale}$, $S_{affine,pro\text{-}étale}$, and $(\mathrm{Aff}/S)_{pro\text{-}étale}$ have enough points.

Proof. The big pro-étale topos of S is equivalent to the topos defined by $(\mathrm{Aff}/S)_{pro\text{-}étale}$, see Lemma 12.11. The topos of sheaves on $S_{pro\text{-}étale}$ is equivalent to the topos associated to $S_{affine,pro\text{-}étale}$, see Lemma 12.20. The result for the sites $(\mathrm{Aff}/S)_{pro\text{-}étale}$ and $S_{affine,pro\text{-}étale}$ follows immediately from Deligne's result Sites, Lemma [original label: `sites-lemma-criterion-points`]. The case $\mathrm{Sch}_{pro\text{-}étale}$ is handled because it is equal to $(\mathrm{Sch}/\mathrm{Spec}(\mathrm{Z}))_{pro\text{-}étale}$. □

Let S be a scheme. Let $\overline{s} : \mathrm{Spec}(k) \to S$ be a geometric point. We define a pro-étale neighbourhood of $\overline{s}$ to be a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(k) & \xrightarrow{\overline{u}} & U \\ & \searrow^{\overline{s}} & \downarrow \\ & & S \end{array}$$

with $U \to S$ weakly étale.

Lemma 18.2. Let S be a scheme and let $\overline{s} : \mathrm{Spec}(k) \to S$ be a geometric point. The category of pro-étale neighbourhoods of $\overline{s}$ is cofiltered.

Proof. The proof is identitical to the proof of Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-cofinal-etale`] but using the corresponding facts about weakly étale morphisms proven in More on Morphisms, Lemmas [original label: `more-morphisms-lemma-composition-weakly-etale`], [original label: `more-morphisms-lemma-base-change-weakly-etale`], and [original label: `more-morphisms-lemma-weakly-etale-permanence`]. □

Lemma 18.3. Let S be a scheme. Let $\overline{s}$ be a geometric point of S. Let $\mathcal{U} = \{\varphi_i : S_i \to S\}_{i \in I}$ be a pro-étale covering. Then there exist $i \in I$ and geometric point $\overline{s}_i$ of S_i mapping to $\overline{s}$.

Proof. Immediate from the fact that $\coprod \varphi_i$ is surjective and that residue field extensions induced by weakly étale morphisms are separable algebraic (see for example More on Morphisms, Lemma [original label: `more-morphisms-lemma-weakly-etale-strictly-henselian-local-rings`]. □

Let S be a scheme and let $\overline{s}$ be a geometric point of S. For $\mathcal{F}$ in $\mathrm{Sh}(S_{pro\text{-}étale})$ define the stalk of $\mathcal{F}$ at $\overline{s}$ by the formula

$$\mathcal{F}_{\overline{s}} = \mathrm{colim}_{(U,\overline{u})} \mathcal{F}(U)$$

where the colimit is over all pro-étale neighbourhoods $(U, \overline{u})$ of $\overline{s}$ with $U \in \mathrm{Ob}(S_{pro\text{-}étale})$. It follows from the two lemmas above that the functor

$$S_{pro\text{-}étale}\mathrm{Sets}, \quad U \longmapsto \{\overline{u} \text{ geometric point of } U \text{ mapping to } \overline{s}\}$$

defines a point of the site $S_{pro\text{-}étale}$, see Sites, Definition [original label: `sites-definition-point`] and Lemma [original label: `sites-lemma-neighbourhoods-cofiltered`]. Hence the functor $\mathcal{F} \mapsto \mathcal{F}_{\overline{s}}$ defines a point of the topos $\mathrm{Sh}(S_{pro\text{-}étale})$, see Sites, Definition [original label: `sites-definition-point-topos`] and Lemma [original label: `sites-lemma-point-site-topos`]. In particular this functor is exact and commutes with arbitrary colimits. In fact, this functor has another description.

Lemma 18.4. In the situation above the scheme $\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$ is an object of $X_{pro\text{-}étale}$ and there is a canonical isomorphism

$$\mathcal{F}(\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})) = \mathcal{F}_{\overline{s}}$$

functorial in $\mathcal{F}$.

Proof. The first statement is clear from the construction of the strict henselization as a filtered colimit of étale algebras over S, or by the characterization of weakly étale morphisms of More on Morphisms, Lemma [original label: `more-morphisms-lemma-weakly-etale-strictly-henselian-local-rings`]. The second statement follows as by Olivier's theorem (More on Algebra, Theorem [original label: `more-algebra-theorem-olivier`]) the scheme $\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$ is an initial object of the category of pro-étale neighbourhoods of $\overline{s}$. □

Contrary to the situation with the étale topos of S it is not true that every point of $\mathrm{Sh}(S_{pro\text{-}étale})$ is of this form, and it is not true that the collection of points associated to geometric points is conservative. Namely, suppose that $S = \mathrm{Spec}(k)$ where k is an algebraically closed field. Let A be a nonzero abelian group. Consider the sheaf $\mathcal{F}$ on $S_{pro\text{-}étale}$ defined by the

$$\mathcal{F}(U) = \frac{\{\text{functions } U \to A\}}{\{\text{locally constant functions}\}}$$

for U affine and by sheafification in general, see Example 19.12. Then $\mathcal{F}(U) = 0$ if $U = S = \mathrm{Spec}(k)$ but in general $\mathcal{F}$ is not zero. Namely, $S_{pro\text{-}étale}$ contains affine objects with infinitely many points. For example, let $E = \lim E_n$ be an inverse limit of finite sets with surjective transition maps, e.g., $E = \mathrm{Z}_p = \lim \mathrm{Z}/p^n\mathrm{Z}$. The scheme $U = \mathrm{Spec}(\mathrm{colim}\, \mathrm{Map}(E_n, k))$ is an object of $S_{pro\text{-}étale}$ because $\mathrm{colim}\, \mathrm{Map}(E_n, k)$ is weakly étale (even ind-Zariski) over k. Thus $\mathcal{F}(U)$ is nonzero as there exist maps $E \to A$ which aren't locally constant. Thus $\mathcal{F}$ is a nonzero abelian sheaf whose stalk at the unique geometric point of S is zero. Since we know that

$S_{pro\text{-}étale}$ has enough points, we conclude there must be a point of the pro-étale site which does not come from the construction explained above.

The replacement for arguments using points, is to use affine weakly contractible objects. First, there are enough affine weakly contractible objects by Lemma 13.4. Second, if $W \in \mathrm{Ob}(S_{pro\text{-}étale})$ is affine weakly contractible, then the functor

$$\mathrm{Sh}(S_{pro\text{-}étale}) \longrightarrow \mathrm{Sets}, \quad \mathcal{F} \longmapsto \mathcal{F}(W)$$

is an exact functor $\mathrm{Sh}(S_{pro\text{-}étale}) \to \mathrm{Sets}$ which commutes with all limits. The functor

$$\mathrm{Ab}(S_{pro\text{-}étale}) \longrightarrow \mathrm{Ab}, \quad \mathcal{F} \longmapsto \mathcal{F}(W)$$

is exact and commutes with direct sums (as W is quasi-compact, see Sites, Lemma [original label: `sites-lemma-directed-colimits-sections`]), hence commutes with all limits and colimits. Moreover, we can check exactness of a complex of abelian sheaves by evaluation at these affine weakly contractible objects of $S_{pro\text{-}étale}$, see Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`].

A final remark is that the functor $\mathcal{F} \mapsto \mathcal{F}(W)$ for W affine weakly contractible in general isn't a stalk functor of a point of $S_{pro\text{-}étale}$ because it doesn't preserve coproducts of sheaves of sets if W is disconnected. And in fact, W is disconnected as soon as W has more than 1 closed point, i.e., when W is not the spectrum of a strictly henselian local ring (which is the special case discussed above).

19 Comparison with the étale site

Let X be a scheme. With suitable choices of sites[4] the functor $u : X_{étale} \to X_{pro\text{-}étale}$ sending U/X to U/X defines a morphism of sites

$$\epsilon : X_{pro\text{-}étale} \longrightarrow X_{étale}$$

This follows from Sites, Proposition [original label: `sites-proposition-get-morphism`].

Lemma 19.1. With notation as above. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. The rule

$$X_{pro\text{-}étale} \longrightarrow \mathrm{Sets}, \quad (f : Y \to X) \longmapsto \Gamma(Y_{étale}, f_{étale}^{-1}\mathcal{F})$$

is a sheaf and is equal to $\epsilon^{-1}\mathcal{F}$. Here $f_{étale} : Y_{étale} \to X_{étale}$ is the morphism of small étale sites constructed in Étale Cohomology, Section [original label: `etale-cohomology-section-functoriality`].

Proof. By Lemma 12.2 any pro-étale covering is an fpqc covering. Hence the formula defines a sheaf on $X_{pro\text{-}étale}$ by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-describe-pullback`]. Let $a : \mathrm{Sh}(X_{étale}) \to \mathrm{Sh}(X_{pro\text{-}étale})$ be the functor sending $\mathcal{F}$ to the sheaf given by the formula in the lemma. To show that $a = \epsilon^{-1}$ it suffices to show that a is a left adjoint to ϵ_*.

Let $\mathcal{G}$ be an object of $\mathrm{Sh}(X_{pro\text{-}étale})$. Recall that $\epsilon_*\mathcal{G}$ is simply given by the restriction of $\mathcal{G}$ to the full subcategory $X_{étale}$. Let $f : Y \to X$ be an object of $X_{pro\text{-}étale}$. We view $Y_{étale}$ as a subcategory of $X_{pro\text{-}étale}$. The restriction maps of the sheaf $\mathcal{G}$ define a map

$$\epsilon_*\mathcal{G} = \mathcal{G}|_{X_{étale}} \longrightarrow f_{étale,*}(\mathcal{G}|_{Y_{étale}})$$

Namely, for U in $X_{étale}$ the value of $f_{étale,*}(\mathcal{G}|_{Y_{étale}})$ on U is $\mathcal{G}(Y \times_X U)$ and there is a restriction map $\mathcal{G}(U) \to \mathcal{G}(Y \times_X U)$. By adjunction this determines a map

$$f_{étale}^{-1}(\epsilon_*\mathcal{G}) \to \mathcal{G}|_{Y_{étale}}$$

Putting these together for all $f : Y \to X$ in $X_{pro\text{-}étale}$ we obtain a canonical map $a(\epsilon_*\mathcal{G}) \to \mathcal{G}$.

Let $\mathcal{F}$ be an object of $\mathrm{Sh}(X_{étale})$. It is immediately clear that $\mathcal{F} = \epsilon_* a(\mathcal{F})$.

We claim the maps $\mathcal{F} \to \epsilon_* a(\mathcal{F})$ and $a(\epsilon_*\mathcal{G}) \to \mathcal{G}$ are the unit and counit of the adjunction (see Categories, Section [original label: `categories-section-adjoint`]). To see this it suffices to show that the corresponding maps

$$\mathrm{Mor}_{\mathrm{Sh}(X_{pro\text{-}étale})}(a(\mathcal{F}), \mathcal{G}) \to \mathrm{Mor}_{\mathrm{Sh}(X_{étale})}(\mathcal{F}, \epsilon^{-1}\mathcal{G})$$

and

$$\mathrm{Mor}_{\mathrm{Sh}(X_{étale})}(\mathcal{F}, \epsilon^{-1}\mathcal{G}) \to \mathrm{Mor}_{\mathrm{Sh}(X_{pro\text{-}étale})}(a(\mathcal{F}), \mathcal{G})$$

are mutually inverse. We omit the detailed verification. □

[4] Choose a big pro-étale site $\mathrm{Sch}_{pro\text{-}étale}$ containing X as in Definition 12.7. Then let $\mathrm{Sch}_{étale}$ be the site having the same underlying category as $\mathrm{Sch}_{pro\text{-}étale}$ but whose coverings are exactly those pro-étale coverings which are also étale coverings. With these choices let $X_{étale}$ and $X_{pro\text{-}étale}$ be the subcategories defined in Definition 12.8 and Topologies, Definition [original label: `topologies-definition-big-small-etale`]. Compare with Topologies, Remark [original label: `topologies-remark-choice-sites`].

Lemma 19.2. Let X be a scheme. For every sheaf $\mathcal{F}$ on $X_{étale}$ the adjunction map $\mathcal{F} \to \epsilon_*\epsilon^{-1}\mathcal{F}$ is an isomorphism, i.e., $\epsilon^{-1}\mathcal{F}(U) = \mathcal{F}(U)$ for U in $X_{étale}$.

Proof. Follows immediately from the description of ϵ^{-1} in Lemma 19.1. □

Lemma 19.3. Let X be a scheme. Let $Y = \lim Y_i$ be the limit of a directed inverse system of quasi-compact and quasi-separated objects of $X_{pro\text{-}étale}$ with affine transition morphisms. For any sheaf $\mathcal{F}$ on $X_{étale}$ we have

$$\epsilon^{-1}\mathcal{F}(Y) = \operatorname{colim} \epsilon^{-1}\mathcal{F}(Y_i)$$

Moreover, if Y_i is in $X_{étale}$ we have $\epsilon^{-1}\mathcal{F}(Y) = \operatorname{colim} \mathcal{F}(Y_i)$.

Proof. By the description of $\epsilon^{-1}\mathcal{F}$ in Lemma 19.1, the displayed formula is a special case of Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-colimit`]. (When X, Y, and the Y_i are all affine, see the easier to parse Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-directed-colimit-cohomology`].) The final statement follows immediately from this and Lemma 19.2. □

Lemma 19.4. Let X be an affine scheme. For injective abelian sheaf $\mathcal{I}$ on $X_{étale}$ we have $H^p(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{I}) = 0$ for $p > 0$.

Proof. We are going to use Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-cech-vanish-collection`] to prove this. Let $\mathcal{B} \subset \mathrm{Ob}(X_{pro\text{-}étale})$ be the set of affine objects U of $X_{pro\text{-}étale}$ such that $\mathcal{O}(X) \to \mathcal{O}(U)$ is ind-étale. Let Cov be the set of pro-étale coverings $\{U_i \to U\}_{i=1,\ldots,n}$ with $U \in \mathcal{B}$ such that $\mathcal{O}(U) \to \mathcal{O}(U_i)$ is ind-étale for $i = 1, \ldots, n$. Properties (1) and (2) of Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-cech-vanish-collection`] hold for $\mathcal{B}$ and Cov by Lemmas 7.3, 7.2, and 12.5 and Proposition 9.1.

To check condition (3) suppose that $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ is an element of Cov. We have to show that the higher Cech cohomology groups of $\epsilon^{-1}\mathcal{I}$ with respect to $\mathcal{U}$ are zero. First we write $U_i = \lim_{a \in A_i} U_{i,a}$ as a directed inverse limit with $U_{i,a} \to U$ étale and $U_{i,a}$ affine. We think of $A_1 \times \ldots \times A_n$ as a direct set with ordering $(a_1, \ldots, a_n) \geq (a'_1, \ldots, a'_n)$ if and only if $a_i \geq a'_i$ for $i = 1, \ldots, n$. Observe that $\mathcal{U}_{(a_1,\ldots,a_n)} = \{U_{i,a_i} \to U\}_{i=1,\ldots,n}$ is an étale covering for all $a_1, \ldots, a_n \in A_1 \times \ldots \times A_n$. Observe that

$$U_{i_0} \times_U U_{i_1} \times_U \ldots \times_U U_{i_p} = \lim_{(a_1,\ldots,a_n) \in A_1 \times \ldots \times A_n} U_{i_0,a_{i_0}} \times_U U_{i_1,a_{i_1}} \times_U \ldots \times_U U_{i_p,a_{i_p}}$$

for all $i_0, \ldots, i_p \in \{1, \ldots, n\}$ because limits commute with fibred products. Hence by Lemma 19.3 and exactness of filtered colimits we have

$$\check{H}^p(\mathcal{U}, \epsilon^{-1}\mathcal{I}) = \operatorname{colim} \check{H}^p(\mathcal{U}_{(a_1,\ldots,a_n)}, \epsilon^{-1}\mathcal{I})$$

Thus it suffices to prove the vanishing for étale coverings of U!

Let $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ be an étale covering with U_i affine. Write $U = \lim_{b \in B} U_b$ as a directed inverse limit with U_b affine and $U_b \to X$ étale. By Limits, Lemmas [original label: `limits-lemma-descend-finite-presentation`], [original label: `limits-lemma-limit-affine`], and [original label: `limits-lemma-descend-etale`] we can choose a $b_0 \in B$ such that for $i = 1, \ldots, n$ there is an étale morphism $U_{i,b_0} \to U_{b_0}$ of affines such that $U_i = U \times_{U_{b_0}} U_{i,b_0}$. Set $U_{i,b} = U_b \times_{U_{b_0}} U_{i,b_0}$ for $b \geq b_0$. For b large enough the family $\mathcal{U}_b = \{U_{i,b} \to U_b\}_{i=1,\ldots,n}$ is an étale covering, see Limits, Lemma [original label: `limits-lemma-descend-surjective`]. Exactly as before we find that

$$\check{H}^p(\mathcal{U}, \epsilon^{-1}\mathcal{I}) = \operatorname{colim} \check{H}^p(\mathcal{U}_b, \epsilon^{-1}\mathcal{I}) = \operatorname{colim} \check{H}^p(\mathcal{U}_b, \mathcal{I})$$

the final equality by Lemma 19.2. Since each of the Čech complexes on the right hand side is acyclic in positive degrees (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-injective-trivial-cech`]) it follows that the one on the left is too. This proves condition (3) of Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-cech-vanish-collection`]. Since $X \in \mathcal{B}$ the lemma follows. □

Lemma 19.5. Let X be a scheme.

1. For an abelian sheaf $\mathcal{F}$ on $X_{étale}$ we have $R\epsilon_*(\epsilon^{-1}\mathcal{F}) = \mathcal{F}$.
2. For $K \in D^+(X_{étale})$ the map $K \to R\epsilon_*\epsilon^{-1}K$ is an isomorphism.

Proof. Let $\mathcal{I}$ be an injective abelian sheaf on $X_{étale}$. Recall that $R^q\epsilon_*(\epsilon^{-1}\mathcal{I})$ is the sheaf associated to $U \mapsto H^q(U_{pro\text{-}étale}, \epsilon^{-1}\mathcal{I})$, see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-higher-direct-images`]. By Lemma 19.4 we see that this is zero for $q > 0$ and U affine and étale over X. Since every object of $X_{étale}$ has a covering by affine objects, it follows that $R^q\epsilon_*(\epsilon^{-1}\mathcal{I}) = 0$ for $q > 0$.

Let $K \in D^+(X_{étale})$. Choose a bounded below complex $\mathcal{I}^\bullet$ of injective abelian sheaves on $X_{étale}$ representing K. Then $\epsilon^{-1}K$ is represented by $\epsilon^{-1}\mathcal{I}^\bullet$. By Leray's acyclicity lemma (Derived Categories, Lemma [original label: `derived-lemma-leray-acyclicity`]) we see that $R\epsilon_*\epsilon^{-1}K$ is represented by $\epsilon_*\epsilon^{-1}\mathcal{I}^\bullet$. By Lemma 19.2 we conclude that $R\epsilon_*\epsilon^{-1}\mathcal{I}^\bullet = \mathcal{I}^\bullet$ and the proof of (2) is complete. Part (1) is a special case of (2). □

Lemma 19.6. Let X be a scheme.

1. For an abelian sheaf $\mathcal{F}$ on $X_{étale}$ we have
$$H^i(X_{étale}, \mathcal{F}) = H^i(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{F})$$
for all i.
2. For $K \in D^+(X_{étale})$ we have
$$R\Gamma(X_{étale}, K) = R\Gamma(X_{pro\text{-}étale}, \epsilon^{-1}K)$$

Proof. Immediate consequence of Lemma 19.5 and the Leray spectral sequence (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-apply-Leray`]). □

Lemma 19.7. Let X be a scheme. Let $\mathcal{G}$ be a sheaf of (possibly noncommutative) groups on $X_{étale}$. We have
$$H^1(X_{étale}, \mathcal{G}) = H^1(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{G})$$
where H^1 is defined as the set of isomorphism classes of torsors (see Cohomology on Sites, Section [original label: `sites-cohomology-section-h1-torsors`]).

Proof. Since the functor ϵ^{-1} is fully faithful by Lemma 19.2 it is clear that the map $H^1(X_{étale}, \mathcal{G}) \to H^1(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{G})$ is injective. To show surjectivity it suffices to show that any $\epsilon^{-1}\mathcal{G}$-torsor $\mathcal{F}$ is étale locally trivial. To do this we may assume that X is affine. Thus we reduce to proving surjectivity for X affine.

Choose a covering $\{U \to X\}$ with (a) U affine, (b) $\mathcal{O}(X) \to \mathcal{O}(U)$ ind-étale, and (c) $\mathcal{F}(U)$ nonempty. We can do this by Proposition 9.1 and the fact that standard pro-étale coverings of X are cofinal among all pro-étale coverings of X (Lemma 12.5). Write $U = \lim U_i$ as a limit of affine schemes étale over X. Pick $s \in \mathcal{F}(U)$. Let $g \in \epsilon^{-1}\mathcal{G}(U \times_X U)$ be the unique section such that $g \cdot \mathrm{pr}_1^* s = \mathrm{pr}_2^* s$ in $\mathcal{F}(U \times_X U)$. Then g satisfies the cocycle condition
$$\mathrm{pr}_{12}^* g \cdot \mathrm{pr}_{23}^* g = \mathrm{pr}_{13}^* g$$
in $\epsilon^{-1}\mathcal{G}(U \times_X U \times_X U)$. By Lemma 19.3 we have
$$\epsilon^{-1}\mathcal{G}(U \times_X U) = \operatorname{colim} \mathcal{G}(U_i \times_X U_i)$$
and
$$\epsilon^{-1}\mathcal{G}(U \times_X U \times_X U) = \operatorname{colim} \mathcal{G}(U_i \times_X U_i \times_X U_i)$$
hence we can find an i and an element $g_i \in \mathcal{G}(U_i \times_X U_i)$ mapping to g satisfying the cocycle condition. The cocycle g_i then defines a torsor for $\mathcal{G}$ on $X_{étale}$ whose pullback is isomorphic to $\mathcal{F}$ by construction. Some details omitted (namely, the relationship between torsors and 1-cocycles which should be added to the chapter on cohomology on sites). □

Lemma 19.8. Let X be a scheme. Let Λ be a ring.

1. The essential image of the fully faithful functor $\epsilon^{-1} : \mathrm{Mod}(X_{étale}, \Lambda) \to \mathrm{Mod}(X_{pro\text{-}étale}, \Lambda)$ is a weak Serre subcategory $\mathcal{C}$.
2. The functor ϵ^{-1} defines an equivalence of categories of $D^+(X_{étale}, \Lambda)$ with $D^+_{\mathcal{C}}(X_{pro\text{-}étale}, \Lambda)$ with quasi-inverse given by $R\epsilon_*$.

Proof. To prove (1) we will prove conditions (1) – (4) of Homology, Lemma [original label: `homology-lemma-characterize-weak-serre-subcategory`]. Since ϵ^{-1} is fully faithful (Lemma 19.2) and exact, everything is clear except for condition (4). However, if
$$0 \to \epsilon^{-1}\mathcal{F}_1 \to \mathcal{G} \to \epsilon^{-1}\mathcal{F}_2 \to 0$$
is a short exact sequence of sheaves of Λ-modules on $X_{pro\text{-}étale}$, then we get
$$0 \to \epsilon_*\epsilon^{-1}\mathcal{F}_1 \to \epsilon_*\mathcal{G} \to \epsilon_*\epsilon^{-1}\mathcal{F}_2 \to R^1\epsilon_*\epsilon^{-1}\mathcal{F}_1$$
which by Lemma 19.5 is the same as a short exact sequence
$$0 \to \mathcal{F}_1 \to \epsilon_*\mathcal{G} \to \mathcal{F}_2 \to 0$$
Pulling pack we find that $\mathcal{G} = \epsilon^{-1}\epsilon_*\mathcal{G}$. This proves (1).

Part (2) follows from part (1) and Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-equivalence-bounded`]. □

Let Λ be a ring. In Modules on Sites, Section [original label: sites-modules-section-locally-constant] we have defined the notion of a locally constant sheaf of Λ-modules on a site. If M is a Λ-module, then $\underline{M}$ is of finite presentation as a sheaf of $\underline{\Lambda}$-modules if and only if M is a finitely presented Λ-module, see Modules on Sites, Lemma [original label: sites-modules-lemma-locally-constant-finite-type].

Lemma 19.9. Let X be a scheme. Let Λ be a ring. The functor ϵ^{-1} defines an equivalence of categories

$$\left\{\begin{array}{c}\text{locally constant sheaves} \\ \text{of } \Lambda\text{-modules on } X_{étale} \\ \text{of finite presentation}\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{locally constant sheaves} \\ \text{of } \Lambda\text{-modules on } X_{pro\text{-}étale} \\ \text{of finite presentation}\end{array}\right\}$$

Proof. Let $\mathcal{F}$ be a locally constant sheaf of Λ-modules on $X_{pro\text{-}étale}$ of finite presentation. Choose a pro-étale covering $\{U_i \to X\}$ such that $\mathcal{F}|_{U_i}$ is constant, say $\mathcal{F}|_{U_i} \cong \underline{M_i}_{U_i}$. Observe that $U_i \times_X U_j$ is empty if M_i is not isomorphic to M_j. For each Λ-module M let $I_M = \{i \in I \mid M_i \cong M\}$. As pro-étale coverings are fpqc coverings and by Descent, Lemma [original label: descent-lemma-open-fpqc-covering] we see that $U_M = \bigcup_{i \in I_M} \mathrm{Im}(U_i \to X)$ is an open subset of X. Then $X = \coprod U_M$ is a disjoint open covering of X. We may replace X by U_M for some M and assume that $M_i = M$ for all i.

Consider the sheaf $\mathcal{I} = \mathrm{Isom}(\underline{M}, \mathcal{F})$. This sheaf is a torsor for $\mathcal{G} = \mathrm{Isom}(\underline{M}, \underline{M})$. By Modules on Sites, Lemma [original label: sites-modules-lemma-locally-constant] we have $\mathcal{G} = \underline{G}$ where $G = \mathrm{Isom}_\Lambda(M, M)$. Since torsors for the étale topology and the pro-étale topology agree by Lemma 19.7 it follows that $\mathcal{I}$ has sections étale locally on X. Thus $\mathcal{F}$ is étale locally a constant sheaf which is what we had to show. □

Lemma 19.10. Let X be a scheme. Let Λ be a Noetherian ring. Let $D_{flc}(X_{étale}, \Lambda)$, resp. $D_{flc}(X_{pro\text{-}étale}, \Lambda)$ be the full subcategory of $D(X_{étale}, \Lambda)$, resp. $D(X_{pro\text{-}étale}, \Lambda)$ consisting of those complexes whose cohomology sheaves are locally constant sheaves of Λ-modules of finite type. Then

$$\epsilon^{-1} : D^+_{flc}(X_{étale}, \Lambda) \longrightarrow D^+_{flc}(X_{pro\text{-}étale}, \Lambda)$$

is an equivalence of categories.

Proof. The categories $D_{flc}(X_{étale}, \Lambda)$ and $D_{flc}(X_{pro\text{-}étale}, \Lambda)$ are strictly full, saturated, triangulated subcategories of $D(X_{étale}, \Lambda)$ and $D(X_{pro\text{-}étale}, \Lambda)$ by Modules on Sites, Lemma [original label: sites-modules-lemma-kernel-finite-locally-constant] and Derived Categories, Section [original label: derived-section-triangulated-sub]. The statement of the lemma follows by combining Lemmas 19.8 and 19.9. □

Lemma 19.11. Let X be a scheme. Let Λ be a Noetherian ring. Let K be an object of $D(X_{pro\text{-}étale}, \Lambda)$. Set $K_n = K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$. If K_1 is

1. in the essential image of $\epsilon^{-1} : D(X_{étale}, \Lambda/I) \to D(X_{pro\text{-}étale}, \Lambda/I)$, and
2. has tor amplitude in $[a, \infty)$ for some $a \in \mathbf{Z}$,

then (1) and (2) hold for K_n as an object of $D(X_{pro\text{-}étale}, \Lambda/I^n)$.

Proof. Assertion (2) for K_n follows from the more general Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-bounded]. Assertion (1) for K_n follows by induction on n from the distinguished triangles

$$K \otimes^{\mathrm{L}}_\Lambda \underline{I^n/I^{n+1}} \to K_{n+1} \to K_n \to K \otimes^{\mathrm{L}}_\Lambda \underline{I^n/I^{n+1}}[1]$$

and the isomorphism

$$K \otimes^{\mathrm{L}}_\Lambda \underline{I^n/I^{n+1}} = K_1 \otimes^{\mathrm{L}}_{\Lambda/I} \underline{I^n/I^{n+1}}$$

and the fact proven in Lemma 19.8 that the essential image of ϵ^{-1} is a triangulated subcategory of $D^+(X_{pro\text{-}étale}, \Lambda/I^n)$. □

Example 19.12. Let X be a scheme. Let A be an abelian group. Denote $fun(-, A)$ the sheaf on $X_{pro\text{-}étale}$ which maps U to the set of all maps $U \to A$ (of sets of points). Consider the sequence of sheaves

$$0 \to \underline{A} \to fun(-, A) \to \mathcal{F} \to 0$$

on $X_{pro\text{-}étale}$. Since the constant sheaf is the pullback from the final topos we see that $\underline{A} = \epsilon^{-1}\underline{A}$. However, if A has more than one element, then neither $fun(-, A)$ nor $\mathcal{F}$ are pulled back from the étale site of X. To work out the values of $\mathcal{F}$ in some cases, assume that all points of X are closed with separably closed residue fields and U is affine. Then all points of U are closed with separably closed residue fields and we have

$$H^1_{pro\text{-}étale}(U, \underline{A}) = H^1_{étale}(U, \underline{A}) = 0$$

by Lemma 19.6 and Étale Cohomology, Lemma [original label: etale-cohomology-lemma-affine-only-closed-points]. Hence in this case we have

$$\mathcal{F}(U) = fun(U, A)/\underline{A}(U)$$

20 Derived completion in the constant Noetherian case

We continue the discussion started in Algebraic and Formal Geometry, Section [original label: algebraization-section-derived-completion]; we assume the reader has read at least some of that section.

Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Recall from Modules on Sites, Lemma [original label: sites-modules-lemma-completion-flat] that

$$\underline{\Lambda}^\wedge = \lim \underline{\Lambda/I^n}$$

is a flat $\underline{\Lambda}$-algebra and that the map $\underline{\Lambda} \to \underline{\Lambda}^\wedge$ identifies quotients by I. Hence Algebraic and Formal Geometry, Lemma [original label: algebraization-lemma-restriction-derived-complete-equivalence] tells us that

$$D_{comp}(\mathcal{C}, \Lambda) = D_{comp}(\mathcal{C}, \underline{\Lambda}^\wedge)$$

In particular the cohomology sheaves $H^i(K)$ of an object K of $D_{comp}(\mathcal{C}, \Lambda)$ are sheaves of $\underline{\Lambda}^\wedge$-modules. For notational convenience we often work with $D_{comp}(\mathcal{C}, \Lambda)$.

Lemma 20.1. Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. The left adjoint to the inclusion functor $D_{comp}(\mathcal{C}, \Lambda) \to D(\mathcal{C}, \Lambda)$ of Algebraic and Formal Geometry, Proposition [original label: algebraization-proposition-derived-completion] sends K to

$$K^\wedge = R\lim(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$$

In particular, K is derived complete if and only if $K = R\lim(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$.

Proof. Choose generators $f_1, \ldots, f_r$ of I. By Algebraic and Formal Geometry, Lemma [original label: algebraization-lemma-derived-completion-koszul] we have

$$K^\wedge = R\lim(K \otimes^{\mathrm{L}}_\Lambda \underline{K_n})$$

where $K_n = K(\Lambda, f_1^n, \ldots, f_r^n)$. In More on Algebra, Lemma [original label: more-algebra-lemma-sequence-Koszul-complexes] we have seen that the pro-systems $\{K_n\}$ and $\{\Lambda/I^n\}$ of $D(\Lambda)$ are isomorphic. Thus the lemma follows. □

Lemma 20.2. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let $f : \mathrm{Sh}(\mathcal{D}) \to \mathrm{Sh}(\mathcal{C})$ be a morphism of topoi. Then

1. Rf_* sends $D_{comp}(\mathcal{D}, \Lambda)$ into $D_{comp}(\mathcal{C}, \Lambda)$,
2. the map $Rf_* : D_{comp}(\mathcal{D}, \Lambda) \to D_{comp}(\mathcal{C}, \Lambda)$ has a left adjoint $Lf^*_{comp} : D_{comp}(\mathcal{C}, \Lambda) \to D_{comp}(\mathcal{D}, \Lambda)$ which is Lf^* followed by derived completion,
3. Rf_* commutes with derived completion,
4. for K in $D_{comp}(\mathcal{D}, \Lambda)$ we have $Rf_*K = R\lim Rf_*(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$.
5. for M in $D_{comp}(\mathcal{C}, \Lambda)$ we have $Lf^*_{comp}M = R\lim Lf^*(M \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$.

Proof. We have seen (1) and (2) in Algebraic and Formal Geometry, Lemma [original label: algebraization-lemma-pushforward-derived-complete-adjoint]. Part (3) follows from Algebraic and Formal Geometry, Lemma [original label: algebraization-lemma-pushforward-commutes-with-derived-completion]. For (4) let K be derived complete. Then

$$Rf_*K = Rf_*(R\lim K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}) = R\lim Rf_*(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$$

the first equality by Lemma 20.1 and the second because Rf_* commutes with $R\lim$ (Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-Rf-commutes-with-Rlim]). This proves (4). To prove (5), by Lemma 20.1 we have

$$Lf^*_{comp}M = R\lim(Lf^*M \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$$

Since Lf^* commutes with derived tensor product by Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-pullback-tensor-product] and since $Lf^*\underline{\Lambda/I^n} = \underline{\Lambda/I^n}$ we get (5). □

21 Derived completion and weakly contractible objects

We continue the discussion in Section 20. In this section we will see how the existence of weakly contractible objects simplifies the study of derived complete modules.

Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Although the general theory concerning $D_{comp}(\mathcal{C}, \Lambda)$ is quite satisfactory it is hard to explicitly give examples of derived complete complexes. We know that

1. every object M of $D(\mathcal{C}, \Lambda/I^n)$ restricts to a derived complete object of $D(\mathcal{C}, \Lambda)$, and
2. for every $K \in D(\mathcal{C}, \Lambda)$ the derived completion $K^\wedge = R\lim(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$ is derived complete.

The first type of objects are trivially complete and perhaps not interesting. The problem with (2) is that derived completion in general is somewhat mysterious, even in case $K = \underline{\Lambda}$. Namely, by definition of homotopy limits there is a distinguished triangle

$$R\lim(\underline{\Lambda/I^n}) \to \prod \underline{\Lambda/I^n} \to \prod \underline{\Lambda/I^n} \to R\lim(\underline{\Lambda/I^n})[1]$$

in $D(\mathcal{C}, \Lambda)$ where the products are in $D(\mathcal{C}, \Lambda)$. These are computed by taking products of injective resolutions (Injectives, Lemma [original label: `injectives-lemma-derived-products`]), so we see that the sheaf $H^p(\prod \underline{\Lambda/I^n})$ is the sheafification of the presheaf

$$U \longmapsto \prod H^p(U, \Lambda/I^n).$$

As an explicit example, if $X = \mathrm{Spec}(\mathrm{C}[t, t^{-1}])$, $\mathcal{C} = X_{étale}$, $\Lambda = \mathrm{Z}$, $I = (2)$, and $p = 1$, then we get the sheafification of the presheaf

$$U \mapsto \prod H^1(U_{étale}, \mathrm{Z}/2^n\mathrm{Z})$$

for U étale over X. Note that $H^1(X_{étale}, \mathrm{Z}/m\mathrm{Z})$ is cyclic of order m with generator α_m given by the finite étale $\mathrm{Z}/m\mathrm{Z}$-covering given by the equation $t = s^m$ (see Étale Cohomology, Section [original label: `etale-cohomology-section-computation`]). Then the section

$$\alpha = (\alpha_{2^n}) \in \prod H^1(X_{étale}, \mathrm{Z}/2^n\mathrm{Z})$$

of the presheaf above does not restrict to zero on any nonempty étale scheme over X, whence the sheaf associated to the presheaf is not zero.

However, on the pro-étale site this phenomenon does not occur. The reason is that we have enough (quasi-compact) weakly contractible objects. In the following proposition we collect some results about derived completion in the Noetherian constant case for sites having enough weakly contractible objects (see Sites, Definition [original label: `sites-definition-w-contractible`]).

Proposition 21.1. Let $\mathcal{C}$ be a site. Assume $\mathcal{C}$ has enough weakly contractible objects. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal.

1. The category of derived complete sheaves Λ-modules is a weak Serre subcategory of $\mathrm{Mod}(\mathcal{C}, \Lambda)$.
2. A sheaf $\mathcal{F}$ of Λ-modules satisfies $\mathcal{F} = \lim \mathcal{F}/I^n\mathcal{F}$ if and only if $\mathcal{F}$ is derived complete and $\bigcap I^n\mathcal{F} = 0$.
3. The sheaf $\underline{\Lambda}^\wedge$ is derived complete.
4. If $\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$ is an inverse system of derived complete sheaves of Λ-modules, then $\lim \mathcal{F}_n$ is derived complete.
5. An object $K \in D(\mathcal{C}, \Lambda)$ is derived complete if and only if each cohomology sheaf $H^p(K)$ is derived complete.
6. An object $K \in D_{comp}(\mathcal{C}, \Lambda)$ is bounded above if and only if $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ is bounded above.
7. An object $K \in D_{comp}(\mathcal{C}, \Lambda)$ is bounded if $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has finite tor dimension.

Proof. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset such that every $U \in \mathcal{B}$ is weakly contractible and every object of $\mathcal{C}$ has a covering by elements of $\mathcal{B}$. We will use the results of Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-w-contractible`] and Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`] without further mention.

Recall that $R\lim$ commutes with $R\Gamma(U, -)$, see Injectives, Lemma [original label: `injectives-lemma-RF-commutes-with-Rlim`]. Let $f \in I$. Recall that $T(K, f)$ is the homotopy limit of the system

$$\ldots \xrightarrow{f} K \xrightarrow{f} K \xrightarrow{f} K$$

in $D(\mathcal{C}, \Lambda)$. Thus

$$R\Gamma(U, T(K, f)) = T(R\Gamma(U, K), f).$$

Since we can test isomorphisms of maps between objects of $D(\mathcal{C}, \Lambda)$ by evaluating at $U \in \mathcal{B}$ we conclude an object K of $D(\mathcal{C}, \Lambda)$ is derived complete if and only if for every $U \in \mathcal{B}$ the object $R\Gamma(U, K)$ is derived complete as an object of $D(\Lambda)$.

The remark above implies that items (1), (5) follow from the corresponding results for modules over rings, see More on Algebra, Lemmas [original label: `more-algebra-lemma-hom-from-Af`] and [original label: `more-algebra-lemma-serre-subcategory`]. In the same way (2) can be deduced from More on Algebra, Proposition [original label: `more-algebra-proposition-derived-complete-modules`] as $(I^n\mathcal{F})(U) = I^n \cdot \mathcal{F}(U)$ for $U \in \mathcal{B}$ (by exactness of evaluating at U).

Proof of (4). The homotopy limit $R\lim \mathcal{F}_n$ is in $D_{comp}(X, \Lambda)$ (see discussion following Algebraic and Formal Geometry, Definition [original label: `algebraization-definition-derived-complete`]). By part (5) just proved we conclude that $\lim \mathcal{F}_n = H^0(R\lim \mathcal{F}_n)$ is derived complete. Part (3) is a special case of (4).

Proof of (6) and (7). Follows from Lemma 20.1 and Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-bounded`] and the computation of homotopy limits in Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`]. □

22 Cohomology of a point

Let Λ be a Noetherian ring complete with respect to an ideal $I \subset \Lambda$. Let k be a field. In this section we "compute"

$$H^i(\mathrm{Spec}(k)_{pro\text{-}étale}, \underline{\Lambda}^\wedge)$$

where $\underline{\Lambda}^\wedge = \lim_m \underline{\Lambda/I^m}$ as before. Let k^{sep} be a separable algebraic closure of k. Then

$$\mathcal{U} = \{\mathrm{Spec}(k^{sep}) \to \mathrm{Spec}(k)\}$$

is a pro-étale covering of $\mathrm{Spec}(k)$. We will use the Čech to cohomology spectral sequence with respect to this covering. Set $U_0 = \mathrm{Spec}(k^{sep})$ and

$$\begin{aligned} U_n &= \mathrm{Spec}(k^{sep}) \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k^{sep}) \times_{\mathrm{Spec}(k)} \ldots \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k^{sep}) \\ &= \mathrm{Spec}(k^{sep} \otimes_k k^{sep} \otimes_k \ldots \otimes_k k^{sep}) \end{aligned}$$

($n+1$ factors). Note that the underlying topological space $|U_0|$ of U_0 is a singleton and for $n \geq 1$ we have

$$|U_n| = G \times \ldots \times G \quad (n \text{ factors})$$

as profinite spaces where $G = \mathrm{Gal}(k^{sep}/k)$. Namely, every point of U_n has residue field k^{sep} and we identify $(\sigma_1, \ldots, \sigma_n)$ with the point corresponding to the surjection

$$k^{sep} \otimes_k k^{sep} \otimes_k \ldots \otimes_k k^{sep} \longrightarrow k^{sep}, \quad \lambda_0 \otimes \lambda_1 \otimes \ldots \lambda_n \longmapsto \lambda_0 \sigma_1(\lambda_1) \ldots \sigma_n(\lambda_n)$$

Then we compute

$$\begin{aligned} R\Gamma((U_n)_{pro\text{-}étale}, \underline{\Lambda}^\wedge) &= R\lim_m R\Gamma((U_n)_{pro\text{-}étale}, \underline{\Lambda/I^m}) \\ &= R\lim_m R\Gamma((U_n)_{étale}, \underline{\Lambda/I^m}) \\ &= \lim_m H^0(U_n, \underline{\Lambda/I^m}) \\ &= \mathrm{Maps}_{cont}(G \times \ldots \times G, \Lambda) \end{aligned}$$

The first equality because $R\Gamma$ commutes with derived limits and as $\Lambda^\wedge$ is the derived limit of the sheaves $\underline{\Lambda/I^m}$ by Proposition 21.1. The second equality by Lemma 19.6. The third equality by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-affine-only-closed-points`]. The fourth equality uses Étale Cohomology, Remark [original label: `etale-cohomology-remark-constant-locally-constant-maps`] to identify sections of the constant sheaf $\underline{\Lambda/I^m}$. Then it uses the fact that Λ is complete with respect to I and hence equal to $\lim_m \Lambda/I^m$ as a topological space, to see that $\lim_m \mathrm{Map}_{cont}(G, \Lambda/I^m) = \mathrm{Map}_{cont}(G, \Lambda)$ and similarly for higher

powers of G. At this point Cohomology on Sites, Lemmas [original label: `sites-cohomology-lemma-cech-cohomology`] and [original label: `sites-cohomology-lemma-cech-spectral-sequence-application`] tell us that

$$\Lambda \to \mathrm{Maps}_{cont}(G, \Lambda) \to \mathrm{Maps}_{cont}(G \times G, \Lambda) \to \ldots$$

computes the pro-étale cohomology. In other words, we see that

$$H^i(\mathrm{Spec}(k)_{pro\text{-}étale}, \underline{\Lambda}^\wedge) = H^i_{cont}(G, \Lambda)$$

where the right hand side is Tate's continuous cohomology, see Étale Cohomology, Section [original label: `etale-cohomology-section-continuous-group-cohomology`]. Of course, this is as it should be.

Lemma 22.1. Let k be a field. Let $G = \mathrm{Gal}(k^{sep}/k)$ be its absolute Galois group. Further,

1. let M be a profinite abelian group with a continuous G-action, or
2. let Λ be a Noetherian ring and $I \subset \Lambda$ an ideal an let M be an I-adically complete Λ-module with continuous G-action.

Then there is a canonical sheaf $\underline{M}^\wedge$ on $\mathrm{Spec}(k)_{pro\text{-}étale}$ associated to M such that

$$H^i(\mathrm{Spec}(k), \underline{M}^\wedge) = H^i_{cont}(G, M)$$

as abelian groups or Λ-modules.

Proof. Proof in case (2). Set $M_n = M/I^nM$. Then $M = \lim M_n$ as M is assumed I-adically complete. Since the action of G is continuous we get continuous actions of G on M_n. By Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-equivalence-sheaves-point`] this action corresponds to a (locally constant) sheaf $\underline{M_n}$ of Λ/I^n-modules on $\mathrm{Spec}(k)_{étale}$. Pull back to $\mathrm{Spec}(k)_{pro\text{-}étale}$ by the comparison morphism ϵ and take the limit

$$\underline{M}^\wedge = \lim \epsilon^{-1}\underline{M_n}$$

to get the sheaf promised in the lemma. Exactly the same argument as given in the introduction of this section gives the comparison with Tate's continuous Galois cohomology. □

23 Functoriality of the pro-étale site

Let $f : X \to Y$ be a morphism of schemes. The functor $Y_{pro\text{-}étale} \to X_{pro\text{-}étale}$, $V \mapsto X \times_Y V$ induces a morphism of sites $f_{pro\text{-}étale} : X_{pro\text{-}étale} \to Y_{pro\text{-}étale}$, see Sites, Proposition [original label: `sites-proposition-get-morphism`]. In fact, we obtain a commutative diagram of morphisms of sites

$$\begin{array}{ccc} X_{pro\text{-}étale} & \xrightarrow{\epsilon} & X_{étale} \\ {\scriptstyle f_{pro\text{-}étale}}\downarrow & & \downarrow{\scriptstyle f_{étale}} \\ Y_{pro\text{-}étale} & \xrightarrow{\epsilon} & Y_{étale} \end{array}$$

where ϵ is as in Section 19. In particular we have $\epsilon^{-1}f_{étale}^{-1} = f_{pro\text{-}étale}^{-1}\epsilon^{-1}$. Here is the corresponding result for pushforward.

Lemma 23.1. Let $f : X \to Y$ be a morphism of schemes which is quasi-compact and quasi-separated.

1. Let $\mathcal{F}$ be a sheaf of sets on $X_{étale}$. Then we have $f_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F} = \epsilon^{-1}f_{étale,*}\mathcal{F}$.
2. Let $\mathcal{F}$ be an abelian sheaf on $X_{étale}$. Then we have $Rf_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F} = \epsilon^{-1}Rf_{étale,*}\mathcal{F}$.

Proof. Proof of (1). Let $\mathcal{F}$ be a sheaf of sets on $X_{étale}$. There is a canonical map $\epsilon^{-1}f_{étale,*}\mathcal{F} \to f_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F}$, see Sites, Section [original label: `sites-section-pullback`]. To show it is an isomorphism we may work (Zariski) locally on Y, hence we may assume Y is affine. In this case every object of $Y_{pro\text{-}étale}$ has a covering by objects $V = \lim V_i$ which are limits of affine schemes V_i étale over Y (by Proposition 9.1 for example). Evaluating the map $\epsilon^{-1}f_{étale,*}\mathcal{F} \to f_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F}$ on V we obtain a map

$$\mathrm{colim}\, \Gamma(X \times_Y V_i, \mathcal{F}) \longrightarrow \Gamma(X \times_Y V, \epsilon^*\mathcal{F})$$

See Lemma 19.3 for the left hand side. By Lemma 19.1 we have

$$\Gamma(X \times_Y V, \epsilon^*\mathcal{F}) = \Gamma(X \times_Y V, g_{étale}^{-1}\mathcal{F})$$

where $g : X \times_Y V \to X$ is the projection. Hence the result holds by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-directed-colimit-cohomology`].

Proof of (2). Arguing in exactly the same manner as above we see that it suffices to show that

$$\operatorname{colim} H^i_{étale}(X \times_Y V_i, \mathcal{F}) \longrightarrow H^i_{étale}(X \times_Y V, \mathcal{F})$$

which follows once more from Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-directed-colimit-cohomology`]. □

24 Finite morphisms and pro-étale sites

It is not clear that a finite morphism of schemes determines an exact pushforward on abelian pro-étale sheaves.

Lemma 24.1. Let $f : Z \to X$ be a finite morphism of schemes which is locally of finite presentation. Then $f_{pro\text{-}étale,*} : \mathrm{Ab}(Z_{pro\text{-}étale}) \to \mathrm{Ab}(X_{pro\text{-}étale})$ is exact.

Proof. The prove this we may work (Zariski) locally on X and assume that X is affine, say $X = \mathrm{Spec}(A)$. Then $Z = \mathrm{Spec}(B)$ for some finite A-algebra B of finite presentation. The construction in the proof of Proposition 11.3 produces a faithfully flat, ind-étale ring map $A \to D$ with D w-contractible. We may check exactness of a sequence of sheaves by evaluating on $U = \mathrm{Spec}(D)$ be such an object. Then $f_{pro\text{-}étale,*}\mathcal{F}$ evaluated at U is equal to $\mathcal{F}$ evaluated at $V = \mathrm{Spec}(D \otimes_A B)$. Since $D \otimes_A B$ is w-contractible by Lemma 11.6 evaluation at V is exact. □

25 Closed immersions and pro-étale sites

It is not clear (and likely false) that a closed immersion of schemes determines an exact pushforward on abelian pro-étale sheaves.

Lemma 25.1. Let $i : Z \to X$ be a closed immersion morphism of affine schemes. Denote X_{app} and Z_{app} the sites introduced in Lemma 12.21. The base change functor

$$u : X_{app} \to Z_{app}, \quad U \longmapsto u(U) = U \times_X Z$$

is continuous and has a fully faithful left adjoint v. For V in Z_{app} the morphism $V \to v(V)$ is a closed immersion identifying V with $u(v(V)) = v(V) \times_X Z$ and every point of $v(V)$ specializes to a point of V. The functor v is cocontinuous and sends coverings to coverings.

Proof. The existence of the adjoint follows immediately from Lemma 7.7 and the definitions. It is clear that u is continuous from the definition of coverings in X_{app}.

Write $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(A/I)$. Let $V = \mathrm{Spec}(\overline{C})$ be an object of Z_{app} and let $v(V) = \mathrm{Spec}(C)$. We have seen in the statement of Lemma 7.7 that V equals $v(V) \times_X Z = \mathrm{Spec}(C/IC)$. Any $g \in C$ which maps to an invertible element of $C/IC = \overline{C}$ is invertible in C. Namely, we have the A-algebra maps $C \to C_g \to C/IC$ and by adjointness we obtain an C-algebra map $C_g \to C$. Thus every point of $v(V)$ specializes to a point of V.

Suppose that $\{V_i \to V\}$ is a covering in Z_{app}. Then $\{v(V_i) \to v(V)\}$ is a finite family of morphisms of Z_{app} such that every point of $V \subset v(V)$ is in the image of one of the maps $v(V_i) \to v(V)$. As the morphisms $v(V_i) \to v(V)$ are flat (since they are weakly étale) we conclude that $\{v(V_i) \to v(V)\}$ is jointly surjective. This proves that v sends coverings to coverings.

Let V be an object of Z_{app} and let $\{U_i \to v(V)\}$ be a covering in X_{app}. Then we see that $\{u(U_i) \to u(v(V)) = V\}$ is a covering of Z_{app}. By adjointness we obtain morphisms $v(u(U_i)) \to U_i$. Thus the family $\{v(u(U_i)) \to v(V)\}$ refines the given covering and we conclude that v is cocontinuous. □

Lemma 25.2. Let $Z \to X$ be a closed immersion morphism of affine schemes. The corresponding morphism of topoi $i = i_{pro\text{-}étale}$ is equal to the morphism of topoi associated to the fully faithful cocontinuous functor $v : Z_{app} \to X_{app}$ of Lemma 25.1. It follows that

1. $i^{-1}\mathcal{F}$ is the sheaf associated to the presheaf $V \mapsto \mathcal{F}(v(V))$,
2. for a weakly contractible object V of Z_{app} we have $i^{-1}\mathcal{F}(V) = \mathcal{F}(v(V))$,
3. $i^{-1} : \mathrm{Sh}(X_{pro\text{-}étale}) \to \mathrm{Sh}(Z_{pro\text{-}étale})$ has a left adjoint $i^{Sh}_!$,
4. $i^{-1} : \mathrm{Ab}(X_{pro\text{-}étale}) \to \mathrm{Ab}(Z_{pro\text{-}étale})$ has a left adjoint $i_!$,

5. $\mathrm{id} \to i^{-1}i_!^{Sh}$, $\mathrm{id} \to i^{-1}i_!$, and $i^{-1}i_* \to \mathrm{id}$ are isomorphisms, and
6. i_*, $i_!^{Sh}$ and $i_!$ are fully faithful.

Proof. By Lemma 12.21 we may describe $i_{pro\text{-}étale}$ in terms of the morphism of sites $u : X_{app} \to Z_{app}$, $V \mapsto V \times_X Z$. The first statement of the lemma follows from Sites, Lemmas [original label: `sites-lemma-have-functor-other-way`] and [original label: `sites-lemma-have-functor-other-way-morphism`] (but with the roles of u and v reversed).

Proof of (1). By the description of i as the morphism of topoi associated to v this holds by the construction, see Sites, Lemma [original label: `sites-lemma-cocontinuous-morphism-topoi`].

Proof of (2). Since the functor v sends coverings to coverings by Lemma 25.1 we see that the presheaf $\mathcal{G} : V \mapsto \mathcal{F}(v(V))$ is a separated presheaf (Sites, Definition [original label: `sites-definition-separated`]). Hence the sheafification of $\mathcal{G}$ is $\mathcal{G}^+$, see Sites, Theorem [original label: `sites-theorem-plus`]. Next, let V be a weakly contractible object of Z_{app}. Let $\mathcal{V} = \{V_i \to V\}_{i=1,\ldots,n}$ be any covering in Z_{app}. Set $\mathcal{V}' = \{\coprod V_i \to V\}$. Since v commutes with finite disjoint unions (as a left adjoint or by the construction) and since $\mathcal{F}$ sends finite disjoint unions into products, we see that

$$H^0(\mathcal{V}, \mathcal{G}) = H^0(\mathcal{V}', \mathcal{G})$$

(notation as in Sites, Section [original label: `sites-section-sheafification`]; compare with Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-cech-complex`]). Thus we may assume the covering is given by a single morphism, like so $\{V' \to V\}$. Since V is weakly contractible, this covering can be refined by the trivial covering $\{V \to V\}$. It therefore follows that the value of $\mathcal{G}^+ = i^{-1}\mathcal{F}$ on V is simply $\mathcal{F}(v(V))$ and (2) is proved.

Proof of (3). Every object of Z_{app} has a covering by weakly contractible objects (Lemma 13.4). By the above we see that we would have $i_!^{Sh}h_V = h_{v(V)}$ for V weakly contractible if $i_!^{Sh}$ existed. The existence of $i_!^{Sh}$ then follows from Sites, Lemma [original label: `sites-lemma-existence-lower-shriek`].

Proof of (4). Existence of $i_!$ follows in the same way by setting $i_!\mathrm{Z}_V = \mathrm{Z}_{v(V)}$ for V weakly contractible in Z_{app}, using similar for direct sums, and applying Homology, Lemma [original label: `homology-lemma-partially-defined-adjoint`]. Details omitted.

Proof of (5). Let V be a contractible object of Z_{app}. Then $i^{-1}i_!^{Sh}h_V = i^{-1}h_{v(V)} = h_{u(v(V))} = h_V$. (It is a general fact that $i^{-1}h_U = h_{u(U)}$.) Since the sheaves h_V for V contractible generate $\mathrm{Sh}(Z_{app})$ (Sites, Lemma [original label: `sites-lemma-sheaf-coequalizer-representable`]) we conclude $\mathrm{id} \to i^{-1}i_!^{Sh}$ is an isomorphism. Similarly for the map $\mathrm{id} \to i^{-1}i_!$. Then $(i^{-1}i_*\mathcal{H})(V) = i_*\mathcal{H}(v(V)) = \mathcal{H}(u(v(V))) = \mathcal{H}(V)$ and we find that $i^{-1}i_* \to \mathrm{id}$ is an isomorphism.

The fully faithfulness statements of (6) now follow from Categories, Lemma [original label: `categories-lemma-adjoint-fully-faithful`]. □

Lemma 25.3. Let $i : Z \to X$ be a closed immersion of schemes. Then

1. $i_{pro\text{-}étale}^{-1}$ commutes with limits,
2. $i_{pro\text{-}étale,*}$ is fully faithful, and
3. $i_{pro\text{-}étale}^{-1}i_{pro\text{-}étale,*} \cong \mathrm{id}_{\mathrm{Sh}(Z_{pro\text{-}étale})}$.

Proof. Assertions (2) and (3) are equivalent by Sites, Lemma [original label: `sites-lemma-exactness-properties`]. Parts (1) and (3) are (Zariski) local on X, hence we may assume that X is affine. In this case the result follows from Lemma 25.2. □

Lemma 25.4. Let $i : Z \to X$ be an integral universally injective and surjective morphism of schemes. Then $i_{pro\text{-}étale,*}$ and $i_{pro\text{-}étale}^{-1}$ are quasi-inverse equivalences of categories of pro-étale topoi.

Proof. There is an immediate reduction to the case that X is affine. Then Z is affine too. Set $A = \mathcal{O}(X)$ and $B = \mathcal{O}(Z)$. Then the categories of étale algebras over A and B are equivalent, see Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-topological-invariance`] and Remark [original label: `etale-cohomology-remark-affine-inside-equivalence`]. Thus the categories of ind-étale algebras over A and B are equivalent. In other words the categories X_{app} and Z_{app} of Lemma 12.21 are equivalent. We omit the verification that this equivalence sends coverings to coverings and vice versa. Thus the result as Lemma 12.21 tells us the pro-étale topos is the topos of sheaves on X_{app}. □

Lemma 25.5. Let $i : Z \to X$ be a closed immersion of schemes. Let $U \to X$ be an object of $X_{pro\text{-}étale}$ such that

1. U is affine and weakly contractible, and
2. every point of U specializes to a point of $U \times_X Z$.

Then $i^{-1}_{pro\text{-}étale}\mathcal{F}(U \times_X Z) = \mathcal{F}(U)$ for all abelian sheaves on $X_{pro\text{-}étale}$.

Proof. Since pullback commutes with restriction, we may replace X by U. Thus we may assume that X is affine and weakly contractible and that every point of X specializes to a point of Z. By Lemma 25.2 part (1) it suffices to show that $v(Z) = X$ in this case. Thus we have to show: If A is a w-contractible ring, $I \subset A$ an ideal contained in the Jacobson radical of A and $A \to B \to A/I$ is a factorization with $A \to B$ ind-étale, then there is a unique retraction $B \to A$ compatible with maps to A/I. Observe that $B/IB = A/I \times R$ as A/I-algebras. After replacing B by a localization we may assume $B/IB = A/I$. Note that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective as the image contains $V(I)$ and hence all closed points and is closed under specialization. Since A is w-contractible there is a retraction $B \to A$. Since $B/IB = A/I$ this retraction is compatible with the map to A/I. We omit the proof of uniqueness (hint: use that A and B have isomorphic local rings at maximal ideals of A). □

Lemma 25.6. Let $i : Z \to X$ be a closed immersion of schemes. If $X \setminus i(Z)$ is a retrocompact open of X, then $i_{pro\text{-}étale,*}$ is exact.

Proof. The question is local on X hence we may assume X is affine. Say $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(A/I)$. There exist $f_1, \ldots, f_r \in I$ such that $Z = V(f_1, \ldots, f_r)$ set theoretically, see Algebra, Lemma [original label: `algebra-lemma-qc-open`]. By Lemma 25.4 we may assume that $Z = \mathrm{Spec}(A/(f_1, \ldots, f_r))$. In this case the functor $i_{pro\text{-}étale,*}$ is exact by Lemma 24.1. □

26 Extension by zero

The general material in Modules on Sites, Section [original label: `sites-modules-section-localize`] allows us to make the following definition.

Definition 26.1. Let $j : U \to X$ be a weakly étale morphism of schemes.

1. The restriction functor $j^{-1} : \mathrm{Sh}(X_{pro\text{-}étale}) \to \mathrm{Sh}(U_{pro\text{-}étale})$ has a left adjoint $j^{Sh}_! : \mathrm{Sh}(X_{pro\text{-}étale}) \to \mathrm{Sh}(U_{pro\text{-}étale})$.
2. The restriction functor $j^{-1} : \mathrm{Ab}(X_{pro\text{-}étale}) \to \mathrm{Ab}(U_{pro\text{-}étale})$ has a left adjoint which is denoted $j_! : \mathrm{Ab}(U_{pro\text{-}étale}) \to \mathrm{Ab}(X_{pro\text{-}étale})$ and called extension by zero.
3. Let Λ be a ring. The functor $j^{-1} : \mathrm{Mod}(X_{pro\text{-}étale}, \Lambda) \to \mathrm{Mod}(U_{pro\text{-}étale}, \Lambda)$ has a left adjoint $j_! : \mathrm{Mod}(U_{pro\text{-}étale}, \Lambda) \to \mathrm{Mod}(X_{pro\text{-}étale}, \Lambda)$ and called extension by zero.

As usual we compare this to what happens in the étale case.

Lemma 26.2. Let $j : U \to X$ be an étale morphism of schemes. Let $\mathcal{G}$ be an abelian sheaf on $U_{étale}$. Then $\epsilon^{-1}j_!\mathcal{G} = j_!\epsilon^{-1}\mathcal{G}$ as sheaves on $X_{pro\text{-}étale}$.

Proof. This is true because both functors are left adjoint to $j^{-1}_{pro\text{-}étale}\epsilon_* = \epsilon_* j^{-1}_{étale}$. The equality holds by the discussion in Section 23. □

Lemma 26.3. Let $j : U \to X$ be a weakly étale morphism of schemes. Let $i : Z \to X$ be a closed immersion such that $U \times_X Z = \emptyset$. Let $V \to X$ be an affine object of $X_{pro\text{-}étale}$ such that every point of V specializes to a point of $V_Z = Z \times_X V$. Then $j_!\mathcal{F}(V) = 0$ for all abelian sheaves on $U_{pro\text{-}étale}$.

Proof. Let $\{V_i \to V\}$ be a pro-étale covering. The lemma follows if we can refine this covering to a covering where the members have no morphisms into U over X (see construction of $j_!$ in Modules on Sites, Section [original label: `sites-modules-section-localize`]). First refine the covering to get a finite covering with V_i affine. For each i let $V_i = \mathrm{Spec}(A_i)$ and let $Z_i \subset V_i$ be the inverse image of Z. Set $W_i = \mathrm{Spec}(A^\sim_{i,Z_i})$ with notation as in Lemma 5.1. Then $\coprod W_i \to V$ is weakly étale and the image contains all points of V_Z. Hence the image contains all points of V by our assumption on specializations. Thus $\{W_i \to V\}$ is a pro-étale covering refining the given one. But each point in W_i specializes to a point lying over Z, hence there are no morphisms $W_i \to U$ over X. □

Lemma 26.4. Let $j : U \to X$ be an open immersion of schemes. Then $\mathrm{id} \cong j^{-1}j_!$ and $j^{-1}j_* \cong \mathrm{id}$ and the functors $j_!$ and j_* are fully faithful.

Proof. See Modules on Sites, Lemma [original label: `sites-modules-lemma-restrict-back`] (and Sites, Lemma [original label: `sites-lemma-restrict-back`] for the case of sheaves of sets) and Categories, Lemma [original label: `categories-lemma-adjoint-fully-faithful`]. □

Here is the relationship between extension by zero and restriction to the complementary closed subscheme.

Lemma 26.5. Let X be a scheme. Let $Z \subset X$ be a closed subscheme and let $U \subset X$ be the complement. Denote $i : Z \to X$ and $j : U \to X$ the inclusion morphisms. Assume that j is a quasi-compact morphism. For every abelian sheaf on $X_{pro\text{-}étale}$ there is a canonical short exact sequence

$$0 \to j_!j^{-1}\mathcal{F} \to \mathcal{F} \to i_*i^{-1}\mathcal{F} \to 0$$

on $X_{pro\text{-}étale}$ where all the functors are for the pro-étale topology.

Proof. We obtain the maps by the adjointness properties of the functors involved. It suffices to show that $X_{pro\text{-}étale}$ has enough objects (Sites, Definition [original label: `sites-definition-w-contractible`]) on which the sequence evaluates to a short exact sequence. Let $V = \mathrm{Spec}(A)$ be an affine object of $X_{pro\text{-}étale}$ such that A is w-contractible (there are enough objects of this type). Then $V \times_X Z$ is cut out by an ideal $I \subset A$. The assumption that j is quasi-compact implies there exist $f_1, \ldots, f_r \in I$ such that $V(I) = V(f_1, \ldots, f_r)$. We obtain a faithfully flat, ind-Zariski ring map

$$A \longrightarrow A_{f_1} \times \ldots \times A_{f_r} \times A^\sim_{V(I)}$$

with $A^\sim_{V(I)}$ as in Lemma 5.1. Since $V_i = \mathrm{Spec}(A_{f_i}) \to X$ factors through U we have

$$j_!j^{-1}\mathcal{F}(V_i) = \mathcal{F}(V_i) \quad \text{and} \quad i_*i^{-1}\mathcal{F}(V_i) = 0$$

On the other hand, for the scheme $V^\sim = \mathrm{Spec}(A^\sim_{V(I)})$ we have

$$j_!j^{-1}\mathcal{F}(V^\sim) = 0 \quad \text{and} \quad \mathcal{F}(V^\sim) = i_*i^{-1}\mathcal{F}(V^\sim)$$

the first equality by Lemma 26.3 and the second by Lemmas 25.5 and 11.7. Thus the sequence evaluates to an exact sequence on $\mathrm{Spec}(A_{f_1} \times \ldots \times A_{f_r} \times A^\sim_{V(I)})$ and the lemma is proved. □

Lemma 26.6. Let $j : U \to X$ be a quasi-compact open immersion morphism of schemes. The functor $j_! : \mathrm{Ab}(U_{pro\text{-}étale}) \to \mathrm{Ab}(X_{pro\text{-}étale})$ commutes with limits.

Proof. Since $j_!$ is exact it suffices to show that $j_!$ commutes with products. The question is local on X, hence we may assume X affine. Let $\mathcal{G}$ be an abelian sheaf on $U_{pro\text{-}étale}$. We have $j^{-1}j_*\mathcal{G} = \mathcal{G}$. Hence applying the exact sequence of Lemma 26.5 we get

$$0 \to j_!\mathcal{G} \to j_*\mathcal{G} \to i_*i^{-1}j_*\mathcal{G} \to 0$$

where $i : Z \to X$ is the inclusion of the reduced induced scheme structure on the complement $Z = X \setminus U$. The functors j_* and i_* commute with products as right adjoints. The functor i^{-1} commutes with products by Lemma 25.3. Hence $j_!$ does because on the pro-étale site products are exact (Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`]). □

27 Constructible sheaves on the pro-étale site

We stick to constructible sheaves of Λ-modules for a Noetherian ring. In the future we intend to discuss constructible sheaves of sets, groups, etc.

Definition 27.1. Let X be a scheme. Let Λ be a Noetherian ring. A sheaf of Λ-modules on $X_{pro\text{-}étale}$ is constructible if for every affine open $U \subset X$ there exists a finite decomposition of U into constructible locally closed subschemes $U = \coprod_i U_i$ such that $\mathcal{F}|_{U_i}$ is of finite type and locally constant for all i.

Again this does not give anything "new".

Lemma 27.2. Let X be a scheme. Let Λ be a Noetherian ring. The functor ϵ^{-1} defines an equivalence of categories

$$\left\{\begin{array}{c}\text{constructible sheaves of} \\ \Lambda\text{-modules on } X_{étale}\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{constructible sheaves of} \\ \Lambda\text{-modules on } X_{pro\text{-}étale}\end{array}\right\}$$

between constructible sheaves of Λ-modules on $X_{étale}$ and constructible sheaves of Λ-modules on $X_{pro\text{-}étale}$.

Proof. By Lemma 19.2 the functor ϵ^{-1} is fully faithful and commutes with pullback (restriction) to the strata. Hence ϵ^{-1} of a constructible étale sheaf is a constructible pro-étale sheaf. To finish the proof let $\mathcal{F}$ be a constructible sheaf of Λ-modules on $X_{pro\text{-}étale}$ as in Definition 27.1. There is a canonical map

$$\epsilon^{-1}\epsilon_*\mathcal{F} \longrightarrow \mathcal{F}$$

We will show this map is an isomorphism. This will prove that $\mathcal{F}$ is in the essential image of ϵ^{-1} and finish the proof (details omitted).

Since it suffices to prove this locally on X we may assume X is quasi-compact and quasi-separated and that we have a finite partition $X = \coprod_{i=1,\ldots,n} X_i$ by constructible locally closed strata such that $\mathcal{F}|_{X_i}$ is locally constant of finite type. We will use induction on n. The base case $n = 1$ follows from Lemma 19.9. Take a point $x \in X$; then $x \in X_k$ for some k. It suffices to show the displayed map is an isomorphism in an open neighbourhood of x. Hence we may assume X_k is closed in X. Set $Z = X_k$ and denote $U \subset X$ the complement of Z. Observe that the induction hypothesis applies to the restriction of $\mathcal{F}$ to Z and to U. By Lemma 26.5 we have a short exact sequence

$$0 \to j_!j^{-1}\mathcal{F} \to \mathcal{F} \to i_*i^{-1}\mathcal{F} \to 0$$

on $X_{pro\text{-}étale}$. Functoriality gives a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \epsilon^{-1}\epsilon_*j_!j^{-1}\mathcal{F} & \longrightarrow & \epsilon^{-1}\epsilon_*\mathcal{F} & \longrightarrow & \epsilon^{-1}\epsilon_*i_*i^{-1}\mathcal{F} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & j_!j^{-1}\mathcal{F} & \longrightarrow & \mathcal{F} & \longrightarrow & i_*i^{-1}\mathcal{F} & \longrightarrow & 0
\end{array}$$

By induction we know that on the one hand $\epsilon^{-1}\epsilon_*i^{-1}\mathcal{F} \to i^{-1}\mathcal{F}$ and $\epsilon^{-1}\epsilon_*j^{-1}\mathcal{F} \to j^{-1}\mathcal{F}$ are isomorphisms and on the other that $i^{-1}\mathcal{F} = \epsilon^{-1}\mathcal{A}$ and $j^{-1}\mathcal{F} = \epsilon^{-1}\mathcal{B}$ for some constructible sheaves of Λ-modules $\mathcal{A}$ on $Z_{étale}$ and $\mathcal{B}$ on $U_{étale}$. Then

$$\epsilon^{-1}\epsilon_*j_!j^{-1}\mathcal{F} = \epsilon^{-1}\epsilon_*j_!\epsilon^{-1}\mathcal{B} = \epsilon^{-1}\epsilon_*\epsilon^{-1}j_!\mathcal{B} = \epsilon^{-1}j_!\mathcal{B} = j_!\epsilon^{-1}\mathcal{B} = j_!j^{-1}\mathcal{F}$$

the second equality by Lemma 26.2, the third equality by Lemma 19.2, and the fourth equality by Lemma 26.2 again. Similarly, we have

$$\epsilon^{-1}\epsilon_*i_*i^{-1}\mathcal{F} = \epsilon^{-1}\epsilon_*i_*\epsilon^{-1}\mathcal{A} = \epsilon^{-1}\epsilon_*\epsilon^{-1}i_*\mathcal{A} = \epsilon^{-1}i_*\mathcal{A} = i_*\epsilon^{-1}\mathcal{A} = i_*i^{-1}\mathcal{F}$$

this time using Lemma 23.1. By the five lemma we conclude the vertical map in the middle of the big diagram is an isomorphism. □

Lemma 27.3. Let X be a scheme. Let Λ be a Noetherian ring. The category of constructible sheaves of Λ-modules on $X_{pro\text{-}étale}$ is a weak Serre subcategory of $\mathrm{Mod}(X_{pro\text{-}étale}, \Lambda)$.

Proof. This is a formal consequence of Lemmas 27.2 and 19.8 and the result for the étale site (Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-constructible-abelian`]). □

Lemma 27.4. Let X be a scheme. Let Λ be a Noetherian ring. Let $D_c(X_{étale}, \Lambda)$, resp. $D_c(X_{pro\text{-}étale}, \Lambda)$ be the full subcategory of $D(X_{étale}, \Lambda)$, resp. $D(X_{pro\text{-}étale}, \Lambda)$ consisting of those complexes whose cohomology sheaves are constructible sheaves of Λ-modules. Then

$$\epsilon^{-1} : D_c^+(X_{étale}, \Lambda) \longrightarrow D_c^+(X_{pro\text{-}étale}, \Lambda)$$

is an equivalence of categories.

Proof. The categories $D_c(X_{étale}, \Lambda)$ and $D_c(X_{pro\text{-}étale}, \Lambda)$ are strictly full, saturated, triangulated subcategories of $D(X_{étale}, \Lambda)$ and $D(X_{pro\text{-}étale}, \Lambda)$ by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-constructible-abelian`] and Lemma 27.3 and Derived Categories, Section [original label: `derived-section-triangulated-sub`]. The statement of the lemma follows by combining Lemmas 19.8 and 27.2. □

Lemma 27.5. Let X be a scheme. Let Λ be a Noetherian ring. Let $K, L \in D_c^-(X_{pro\text{-}étale}, \Lambda)$. Then $K \otimes^{\mathrm{L}}_{\Lambda} L$ is in $D_c^-(X_{pro\text{-}étale}, \Lambda)$.

Proof. Note that $H^i(K \otimes^{\mathrm{L}}_{\Lambda} L)$ is the same as $H^i(\tau_{\geq i-1}K \otimes^{\mathrm{L}}_{\Lambda} \tau_{\geq i-1}L)$. Thus we may assume K and L are bounded. In this case we can apply Lemma 27.4 to reduce to the case of the étale site, see Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-tensor-c`]. □

Lemma 27.6. Let X be a scheme. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let K be an object of $D(X_{pro\text{-}étale}, \Lambda)$. Set $K_n = K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. If K_1 is in $D^-_c(X_{pro\text{-}étale}, \Lambda/I)$, then K_n is in $D^-_c(X_{pro\text{-}étale}, \Lambda/I^n)$ for all n.

Proof. Consider the distinguished triangles

$$K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}} \to K_{n+1} \to K_n \to K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}}[1]$$

and the isomorphisms

$$K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}} = K_1 \otimes^{\mathrm{L}}_{\Lambda/I} \underline{I^n/I^{n+1}}$$

By Lemma 27.5 we see that this tensor product has constructible cohomology sheaves (and vanishing when K_1 has vanishing cohomology). Hence by induction on n using Lemma 27.3 we see that each K_n has constructible cohomology sheaves. □

28 Constructible adic sheaves

In this section we define the notion of a constructible Λ-sheaf as well as some variants.

Definition 28.1. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let X be a scheme. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{pro\text{-}étale}$.

1. We say $\mathcal{F}$ is a constructible Λ-sheaf if $\mathcal{F} = \lim \mathcal{F}/I^n\mathcal{F}$ and each $\mathcal{F}/I^n\mathcal{F}$ is a constructible sheaf of Λ/I^n-modules.
2. If $\mathcal{F}$ is a constructible Λ-sheaf, then we say $\mathcal{F}$ is lisse if each $\mathcal{F}/I^n\mathcal{F}$ is locally constant.
3. We say $\mathcal{F}$ is adic lisse[5] if there exists a I-adically complete Λ-module M with M/IM finite such that $\mathcal{F}$ is locally isomorphic to
$$\underline{M}^\wedge = \lim \underline{M/I^nM}.$$
4. We say $\mathcal{F}$ is adic constructible[6] if for every affine open $U \subset X$ there exists a decomposition $U = \coprod U_i$ into constructible locally closed subschemes such that $\mathcal{F}|_{U_i}$ is adic lisse.

The definition of a constructible Λ-sheaf is equivalent to the one in [5, Exposé VI, Definition 1.1.1] when $\Lambda = \mathrm{Z}_\ell$ and $I = (\ell)$. It is clear that we have the implications

$$\begin{array}{ccc} \text{lisse adic} & \Longrightarrow & \text{adic constructible} \\ \Downarrow & & \Downarrow \\ \text{lisse constructible } \Lambda\text{-sheaf} & \Longrightarrow & \text{constructible } \Lambda\text{-sheaf} \end{array}$$

The vertical arrows can be inverted in some cases (see Lemmas 28.2 and 28.5). In general neither the category of adic constructible sheaves nor the category of constructible Λ-sheaves is closed under kernels and cokernels.

Namely, let X be an affine scheme whose underlying topological space $|X|$ is homeomorphic to $\Lambda = \mathrm{Z}_\ell$, see Example 6.3. Denote $f : |X| \to \mathrm{Z}_\ell = \Lambda$ a homeomorphism. We can think of f as a section of $\underline{\Lambda}^\wedge$ over X and multiplication by f then defines a two term complex

$$\underline{\Lambda}^\wedge \xrightarrow{f} \underline{\Lambda}^\wedge$$

on $X_{pro\text{-}étale}$. The sheaf $\underline{\Lambda}^\wedge$ is adic lisse. However, the cokernel of the map above, is not adic constructible, as the isomorphism type of the stalks of this cokernel attains infinitely many values: $\mathrm{Z}/\ell^n\mathrm{Z}$ and Z_ℓ. The cokernel is a constructible Z_ℓ-sheaf. However, the kernel is not even a constructible Z_ℓ-sheaf as it is zero a non-quasi-compact open but not zero.

Lemma 28.2. Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a constructible Λ-sheaf on $X_{pro\text{-}étale}$. Then there exists a finite partition $X = \coprod X_i$ by locally closed subschemes such that the restriction $\mathcal{F}|_{X_i}$ is lisse.

[5] This may be nonstandard notation.
[6] This may be nonstandard notation.

Proof. Let $R = \bigoplus I^n/I^{n+1}$. Observe that R is a Noetherian ring. Since each of the sheaves $\mathcal{F}/I^n\mathcal{F}$ is a constructible sheaf of $\Lambda/I^n\Lambda$-modules also $I^n\mathcal{F}/I^{n+1}\mathcal{F}$ is a constructible sheaf of Λ/I-modules and hence the pullback of a constructible sheaf $\mathcal{G}_n$ on $X_{étale}$ by Lemma 27.2. Set $\mathcal{G} = \bigoplus \mathcal{G}_n$. This is a sheaf of R-modules on $X_{étale}$ and the map

$$\mathcal{G}_0 \otimes_{\Lambda/I} \underline{R} \longrightarrow \mathcal{G}$$

is surjective because the maps

$$\mathcal{F}/I\mathcal{F} \otimes \underline{I^n/I^{n+1}} \to I^n\mathcal{F}/I^{n+1}\mathcal{F}$$

are surjective. Hence $\mathcal{G}$ is a constructible sheaf of R-modules by Étale Cohomology, Proposition [original label: `etale-cohomology-proposition-constructible-over-noetherian`]. Choose a partition $X = \coprod X_i$ such that $\mathcal{G}|_{X_i}$ is a locally constant sheaf of R-modules of finite type (Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-constructible-quasi-compact-quasi-separated`]). We claim this is a partition as in the lemma. Namely, replacing X by X_i we may assume $\mathcal{G}$ is locally constant. It follows that each of the sheaves $I^n\mathcal{F}/I^{n+1}\mathcal{F}$ is locally constant. Using the short exact sequences

$$0 \to I^n\mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^n\mathcal{F} \to 0$$

induction and Modules on Sites, Lemma [original label: `sites-modules-lemma-kernel-finite-locally-constant`] the lemma follows. □

Lemma 28.3. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{pro\text{-}étale}$ such that

1. $\mathcal{F} = \lim \mathcal{F}/I^n\mathcal{F}$,
2. $\mathcal{F}/I^n\mathcal{F}$ is a constant sheaf of Λ/I^n-modules,
3. $\mathcal{F}/I\mathcal{F}$ is of finite type.

Then $\mathcal{F} \cong \underline{M}^\wedge$ where M is a finite $\Lambda^\wedge$-module.

Proof. Pick a Λ/I^n-module M_n such that $\mathcal{F}/I^n\mathcal{F} \cong \underline{M_n}$. Since we have the surjections $\mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^n\mathcal{F}$ we conclude that there exist surjections $M_{n+1} \to M_n$ inducing isomorphisms $M_{n+1}/I^nM_{n+1} \to M_n$. Fix a choice of such surjections and set $M = \lim M_n$. Then M is an I-adically complete Λ-module with $M/I^nM = M_n$, see Algebra, Lemma [original label: `algebra-lemma-limit-complete`]. Since M_1 is a finite type Λ-module (Modules on Sites, Lemma [original label: `sites-modules-lemma-locally-constant-finite-type`]) we see that M is a finite $\Lambda^\wedge$-module. Consider the sheaves

$$\mathcal{I}_n = \mathrm{Isom}(\underline{M_n}, \mathcal{F}/I^n\mathcal{F})$$

on $X_{pro\text{-}étale}$. Modding out by I^n defines a transition map

$$\mathcal{I}_{n+1} \longrightarrow \mathcal{I}_n$$

By our choice of M_n the sheaf $\mathcal{I}_n$ is a torsor under

$$\mathrm{Isom}(\underline{M_n}, \underline{M_n}) = \underline{\mathrm{Isom}_\Lambda(M_n, M_n)}$$

(Modules on Sites, Lemma [original label: `sites-modules-lemma-locally-constant`]) since $\mathcal{F}/I^n\mathcal{F}$ is (étale) locally isomorphic to $\underline{M_n}$. It follows from More on Algebra, Lemma [original label: `more-algebra-lemma-hom-systems-ML`] that the system of sheaves $(\mathcal{I}_n)$ is Mittag-Leffler. For each n let $\mathcal{I}'_n \subset \mathcal{I}_n$ be the image of $\mathcal{I}_N \to \mathcal{I}_n$ for all $N \gg n$. Then

$$\ldots \to \mathcal{I}'_3 \to \mathcal{I}'_2 \to \mathcal{I}'_1 \to *$$

is a sequence of sheaves of sets on $X_{pro\text{-}étale}$ with surjective transition maps. Since $*(X)$ is a singleton (not empty) and since evaluating at X transforms surjective maps of sheaves of sets into surjections of sets, we can pick $s \in \lim \mathcal{I}'_n(X)$. The sections define isomorphisms $\underline{M}^\wedge \to \lim \mathcal{F}/I^n\mathcal{F} = \mathcal{F}$ and the proof is done. □

Lemma 28.4. Let X be a connected scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. If $\mathcal{F}$ is a lisse constructible Λ-sheaf on $X_{pro\text{-}étale}$, then $\mathcal{F}$ is adic lisse.

Proof. By Lemma 19.9 we have $\mathcal{F}/I^n\mathcal{F} = \epsilon^{-1}\mathcal{G}_n$ for some locally constant sheaf $\mathcal{G}_n$ of Λ/I^n-modules. By Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-connected-locally-constant`] there exists a finite Λ/I^n-module M_n such that $\mathcal{G}_n$ is locally isomorphic to $\underline{M_n}$. Choose a covering $\{W_t \to X\}_{t \in T}$ with each W_t affine and weakly contractible. Then $\mathcal{F}|_{W_t}$ satisfies the assumptions of Lemma 28.3 and hence $\mathcal{F}|_{W_t} \cong \underline{N_t}^\wedge$ for some finite $\Lambda^\wedge$-module N_t. Note that $N_t/I^nN_t \cong M_n$ for all t and n. Hence $N_t \cong N_{t'}$ for all $t, t' \in T$, see More on Algebra, Lemma [original label: `more-algebra-lemma-isomorphic-completions`]. This proves that $\mathcal{F}$ is adic lisse. □

Lemma 28.5. Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a constructible Λ-sheaf on $X_{pro\text{-}étale}$. Then $\mathcal{F}$ is adic constructible.

Proof. This is a consequence of Lemmas 28.2 and 28.4, the fact that a Noetherian scheme is locally connected (Topology, Lemma [original label: `topology-lemma-locally-Noetherian-locally-connected`]), and the definitions. □

It will be useful to identify the constructible Λ-sheaves inside the category of derived complete sheaves of Λ-modules. It turns out that the naive analogue of More on Algebra, Lemma [original label: `more-algebra-lemma-derived-complete-finite`] is wrong in this setting. However, here is the analogue of More on Algebra, Lemma [original label: `more-algebra-lemma-derived-complete-zero`].

Lemma 28.6. Let X be a scheme. Let Λ be a ring and let $I \subset \Lambda$ be a finitely generated ideal. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{pro\text{-}étale}$. If $\mathcal{F}$ is derived complete and $\mathcal{F}/I\mathcal{F} = 0$, then $\mathcal{F} = 0$.

Proof. Assume that $\mathcal{F}/I\mathcal{F}$ is zero. Let $I = (f_1, \ldots, f_r)$. Let $i < r$ be the largest integer such that $\mathcal{G} = \mathcal{F}/(f_1, \ldots, f_i)\mathcal{F}$ is nonzero. If i does not exist, then $\mathcal{F} = 0$ which is what we want to show. Then $\mathcal{G}$ is derived complete as a cokernel of a map between derived complete modules, see Proposition 21.1. By our choice of i we have that $f_{i+1} : \mathcal{G} \to \mathcal{G}$ is surjective. Hence

$$\lim(\ldots \to \mathcal{G} \xrightarrow{f_{i+1}} \mathcal{G} \xrightarrow{f_{i+1}} \mathcal{G})$$

is nonzero, contradicting the derived completeness of $\mathcal{G}$. □

Lemma 28.7. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a derived complete sheaf of Λ-modules on $X_{pro\text{-}étale}$ with $\mathcal{F}/I\mathcal{F}$ a locally constant sheaf of Λ/I-modules of finite type. Then there exists an integer t and a surjective map

$$(\underline{\Lambda}^\wedge)^{\oplus t} \to \mathcal{F}$$

Proof. Since X is weakly contractible, there exists a finite disjoint open covering $X = \coprod U_i$ such that $\mathcal{F}/I\mathcal{F}|_{U_i}$ is isomorphic to the constant sheaf associated to a finite Λ/I-module M_i. Choose finitely many generators m_{ij} of M_i. We can find sections $s_{ij} \in \mathcal{F}(X)$ restricting to m_{ij} viewed as a section of $\mathcal{F}/I\mathcal{F}$ over U_i. Let t be the total number of s_{ij}. Then we obtain a map

$$\alpha : \underline{\Lambda}^{\oplus t} \longrightarrow \mathcal{F}$$

which is surjective modulo I by construction. By Lemma 20.1 the derived completion of $\underline{\Lambda}^{\oplus t}$ is the sheaf $(\underline{\Lambda}^\wedge)^{\oplus t}$. Since $\mathcal{F}$ is derived complete we see that α factors through a map

$$\alpha^\wedge : (\underline{\Lambda}^\wedge)^{\oplus t} \longrightarrow \mathcal{F}$$

Then $\mathcal{Q} = \mathrm{Coker}(\alpha^\wedge)$ is a derived complete sheaf of Λ-modules by Proposition 21.1. By construction $\mathcal{Q}/I\mathcal{Q} = 0$. It follows from Lemma 28.6 that $\mathcal{Q} = 0$ which is what we wanted to show. □

29 A suitable derived category

Let X be a scheme. It will turn out that for many schemes X a suitable derived category of ℓ-adic sheaves can be gotten by considering the derived complete objects K of $D(X_{pro\text{-}étale}, \Lambda)$ with the property that $K \otimes^{\mathrm{L}}_{\Lambda} \mathrm{F}_\ell$ is bounded with constructible cohomology sheaves. Here is the general definition.

Definition 29.1. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let X be a scheme. An object K of $D(X_{pro\text{-}étale}, \Lambda)$ is called constructible if

1. K is derived complete with respect to I,
2. $K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I}$ has constructible cohomology sheaves and locally has finite tor dimension.

We denote $D_{cons}(X, \Lambda)$ the full subcategory of constructible K in $D(X_{pro\text{-}étale}, \Lambda)$.

Recall that with our conventions a complex of finite tor dimension is bounded (Cohomology on Sites, Definition [original label: `sites-cohomology-definition-tor-amplitude`]). In fact, let's collect everything proved so far in a lemma.

Lemma 29.2. In the situation above suppose K is in $D_{cons}(X, \Lambda)$ and X is quasi-compact. Set $K_n = K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. There exist a, b such that

1. $K = R\lim K_n$ and $H^i(K) = 0$ for $i \notin [a, b]$,
2. each K_n has tor amplitude in $[a, b]$,
3. each K_n has constructible cohomology sheaves,
4. each $K_n = \epsilon^{-1}L_n$ for some $L_n \in D_{ctf}(X_{étale}, \Lambda/I^n)$ (Étale Cohomology, Definition [original label: `etale-cohomology-definition-ctf`]).

Proof. Since K is derived complete, we have $K = R\lim K_n$ by Lemma 20.1. By definition of locally having finite tor dimension, we can find a, b such that K_1 has tor amplitude in $[a, b]$. Part (2) follows from Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-bounded`]. Then (1) follows as K is derived complete by the description of limits in Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`] and the fact that $H^b(K_{n+1}) \to H^b(K_n)$ is surjective as $K_n = K_{n+1} \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. Part (3) follows from Lemma 27.6, Part (4) follows from Lemma 27.4 and the fact that L_n has finite tor dimension because K_n does (small argument omitted). □

Lemma 29.3. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let K be an object of $D_{cons}(X, \Lambda)$ such that the cohomology sheaves of $K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I}$ are locally constant. Then there exists a finite disjoint open covering $X = \coprod U_i$ and for each i a finite collection of finite projective $\Lambda^\wedge$-modules $M^a, \ldots, M^b$ such that $K|_{U_i}$ is represented by a complex

$$(\underline{M^a})^\wedge \to \ldots \to (\underline{M^b})^\wedge$$

in $D(U_{i,pro\text{-}étale}, \Lambda)$ for some maps of sheaves of Λ-modules $(\underline{M^i})^\wedge \to (\underline{M^{i+1}})^\wedge$.

Proof. We freely use the results of Lemma 29.2. Choose a, b as in that lemma. We will prove the lemma by induction on $b - a$. Let $\mathcal{F} = H^b(K)$. Note that $\mathcal{F}$ is a derived complete sheaf of Λ-modules by Proposition 21.1. Moreover $\mathcal{F}/I\mathcal{F}$ is a locally constant sheaf of Λ/I-modules of finite type. Apply Lemma 28.7 to get a surjection $\rho : (\underline{\Lambda}^\wedge)^{\oplus t} \to \mathcal{F}$.

If $a = b$, then $K = \mathcal{F}[-b]$. In this case we see that

$$\mathcal{F} \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I} = \mathcal{F}/I\mathcal{F}$$

As X is weakly contractible and $\mathcal{F}/I\mathcal{F}$ locally constant, we can find a finite disjoint union decomposition $X = \coprod U_i$ by affine opens U_i and Λ/I-modules $\overline{M}_i$ such that $\mathcal{F}/I\mathcal{F}$ restricts to $\underline{\overline{M}_i}$ on U_i. After refining the covering we may assume the map

$$\rho|_{U_i} \bmod I : \underline{\Lambda/I}^{\oplus t} \longrightarrow \underline{\overline{M}_i}$$

is equal to $\underline{\alpha_i}$ for some surjective module map $\alpha_i : \Lambda/I^{\oplus t} \to \overline{M}_i$, see Modules on Sites, Lemma [original label: `sites-modules-lemma-morphism-locally-constant`]. Note that each $\overline{M}_i$ is a finite Λ/I-module. Since $\mathcal{F}/I\mathcal{F}$ has tor amplitude in $[0, 0]$ we conclude that $\overline{M}_i$ is a flat Λ/I-module. Hence $\overline{M}_i$ is finite projective (Algebra, Lemma [original label: `algebra-lemma-finite-projective`]). Hence we can find a projector $\overline{p}_i : (\Lambda/I)^{\oplus t} \to (\Lambda/I)^{\oplus t}$ whose image maps isomorphically to $\overline{M}_i$ under the map α_i. We can lift $\overline{p}_i$ to a projector $p_i : (\Lambda^\wedge)^{\oplus t} \to (\Lambda^\wedge)^{\oplus t}$[7]. Then $M_i = \mathrm{Im}(p_i)$ is a finite I-adically complete $\Lambda^\wedge$-module with $M_i/IM_i = \overline{M}_i$. Over U_i consider the maps

$$\underline{M_i}^\wedge \to (\underline{\Lambda}^\wedge)^{\oplus t} \to \mathcal{F}|_{U_i}$$

By construction the composition induces an isomorphism modulo I. The source and target are derived complete, hence so are the cokernel $\mathcal{Q}$ and the kernel $\mathcal{K}$. We have $\mathcal{Q}/I\mathcal{Q} = 0$ by construction hence $\mathcal{Q}$ is zero by Lemma 28.6. Then

$$0 \to \mathcal{K}/I\mathcal{K} \to \underline{\overline{M}_i} \to \mathcal{F}/I\mathcal{F} \to 0$$

is exact by the vanishing of Tor_1 see at the start of this paragraph; also use that $\underline{\Lambda}^\wedge/I\underline{\Lambda}^\wedge$ by Modules on Sites, Lemma [original label: `sites-modules-lemma-completion-flat`] to see that $\underline{M_i}^\wedge/I\underline{M_i}^\wedge = \underline{\overline{M}_i}$. Hence $\mathcal{K}/I\mathcal{K} = 0$ by construction and we conclude that $\mathcal{K} = 0$ as before. This proves the result in case $a = b$.

If $b > a$, then we lift the map ρ to a map

$$\tilde{\rho} : (\underline{\Lambda}^\wedge)^{\oplus t}[-b] \longrightarrow K$$

[7]Proof: by Algebra, Lemma [original label: `algebra-lemma-lift-idempotents-noncommutative`] we can lift $\overline{p}_i$ to a compatible system of projectors $p_{i,n} : (\Lambda/I^n)^{\oplus t} \to (\Lambda/I^n)^{\oplus t}$ and then we set $p_i = \lim p_{i,n}$ which works because $\Lambda^\wedge = \lim \Lambda/I^n$.

in $D(X_{pro\text{-}étale}, \Lambda)$. This is possible as we can think of K as a complex of $\underline{\Lambda}^\wedge$-modules by discussion in the introduction to Section 20 and because $X_{pro\text{-}étale}$ is weakly contractible hence there is no obstruction to lifting the elements $\rho(e_s) \in H^0(X, \mathcal{F})$ to elements of $H^b(X, K)$. Fitting $\tilde{\rho}$ into a distinguished triangle

$$(\underline{\Lambda}^\wedge)^{\oplus t}[-b] \to K \to L \to (\underline{\Lambda}^\wedge)^{\oplus t}[-b+1]$$

we see that L is an object of $D_{cons}(X, \Lambda)$ such that $L \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has tor amplitude contained in $[a, b-1]$ (details omitted). By induction we can describe L locally as stated in the lemma, say L is isomorphic to

$$(\underline{M^a})^\wedge \to \ldots \to (\underline{M^{b-1}})^\wedge$$

The map $L \to (\underline{\Lambda}^\wedge)^{\oplus t}[-b+1]$ corresponds to a map $(\underline{M^{b-1}})^\wedge \to (\underline{\Lambda}^\wedge)^{\oplus t}$ which allows us to extend the complex by one. The corresponding complex is isomorphic to K in the derived category by the properties of triangulated categories. This finishes the proof. □

Motivated by what happens for constructible Λ-sheaves we introduce the following notion.

Definition 29.4. Let X be a scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $K \in D(X_{pro\text{-}étale}, \Lambda)$.

1. We say K is adic lisse[8] if there exists a finite complex of finite projective $\Lambda^\wedge$-modules $M^\bullet$ such that K is locally isomorphic to
$$\underline{M^a}^\wedge \to \ldots \to \underline{M^b}^\wedge$$
2. We say K is adic constructible[9] if for every affine open $U \subset X$ there exists a decomposition $U = \coprod U_i$ into constructible locally closed subschemes such that $K|_{U_i}$ is adic lisse.

The difference between the local structure obtained in Lemma 29.3 and the structure of an adic lisse complex is that the maps $\underline{M^i}^\wedge \to \underline{M^{i+1}}^\wedge$ in Lemma 29.3 need not be constant, whereas in the definition above they are required to be constant.

Lemma 29.5. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let K be an object of $D_{cons}(X, \Lambda)$ such that $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$ is isomorphic in $D(X_{pro\text{-}étale}, \Lambda/I^n)$ to a complex of constant sheaves of Λ/I^n-modules. Then

$$H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^n)$$

has the Mittag-Leffler condition.

Proof. Say $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$ is isomorphic to $\underline{E_n}$ for some object E_n of $D(\Lambda/I^n)$. Since $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has finite tor dimension and has finite type cohomology sheaves we see that E_1 is perfect (see More on Algebra, Lemma [original label: `more-algebra-lemma-perfect`]). The transition maps

$$K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^{n+1}} \to K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$$

locally come from (possibly many distinct) maps of complexes $E_{n+1} \to E_n$ in $D(\Lambda/I^{n+1})$ see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-locally-constant-map`]. For each n choose one such map and observe that it induces an isomorphism $E_{n+1} \otimes^{\mathrm{L}}_{\Lambda/I^{n+1}} \Lambda/I^n \to E_n$ in $D(\Lambda/I^n)$. By More on Algebra, Lemma [original label: `more-algebra-lemma-Rlim-perfect-gives-complete`] we can find a finite complex $M^\bullet$ of finite projective $\Lambda^\wedge$-modules and isomorphisms $M^\bullet/I^n M^\bullet \to E_n$ in $D(\Lambda/I^n)$ compatible with the transition maps.

Now observe that for each finite collection of indices $n > m > k$ the triple of maps

$$H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^n) \to H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^m) \to H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^k)$$

is isomorphic to

$$H^0(X, \underline{M^\bullet/I^n M^\bullet}) \to H^0(X, \underline{M^\bullet/I^m M^\bullet}) \to H^0(X, \underline{M^\bullet/I^k M^\bullet})$$

Namely, choose any isomorphism

$$\underline{M^\bullet/I^n M^\bullet} \to K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^n$$

[8] This may be nonstandard notation

[9] This may be nonstandard notation.

induces similar isomorphisms module I^m and I^k and we see that the assertion is true. Thus to prove the lemma it suffices to show that the system $H^0(X, \underline{M^\bullet/I^nM^\bullet})$ has Mittag-Leffler. Since taking sections over X is exact, it suffices to prove that the system of Λ-modules

$$H^0(M^\bullet/I^nM^\bullet)$$

has Mittag-Leffler. Set $A = \Lambda^\wedge$ and consider the spectral sequence

$$\mathrm{Tor}^A_{-p}(H^q(M^\bullet), A/I^nA) \Rightarrow H^{p+q}(M^\bullet/I^nM^\bullet)$$

By More on Algebra, Lemma [original label: `more-algebra-lemma-tor-strictly-pro-zero`] the pro-systems $\{\mathrm{Tor}^A_{-p}(H^q(M^\bullet), A/I^nA)\}$ are zero for $p > 0$. Thus the pro-system $\{H^0(M^\bullet/I^nM^\bullet)\}$ is equal to the pro-system $\{H^0(M^\bullet)/I^nH^0(M^\bullet)\}$ and the lemma is proved. □

Lemma 29.6. Let X be a connected scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. If K is in $D_{cons}(X, \Lambda)$ such that $K \otimes_\Lambda \underline{\Lambda/I}$ has locally constant cohomology sheaves, then K is adic lisse (Definition 29.4).

Proof. Write $K_n = K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$. We will use the results of Lemma 29.2 without further mention. By Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-locally-constant-bounded`] we see that K_n has locally constant cohomology sheaves for all n. We have $K_n = \epsilon^{-1}L_n$ some L_n in $D_{ctf}(X_{étale}, \Lambda/I^n)$ with locally constant cohomology sheaves. By Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-connected-ctf-locally-constant`] there exist perfect $M_n \in D(\Lambda/I^n)$ such that L_n is étale locally isomorphic to $\underline{M_n}$. The maps $L_{n+1} \to L_n$ corresponding to $K_{n+1} \to K_n$ induces isomorphisms $L_{n+1} \otimes^{\mathrm{L}}_{\Lambda/I^{n+1}} \underline{\Lambda/I^n} \to L_n$. Looking locally on X we conclude that there exist maps $M_{n+1} \to M_n$ in $D(\Lambda/I^{n+1})$ inducing isomorphisms $M_{n+1} \otimes_{\Lambda/I^{n+1}} \Lambda/I^n \to M_n$, see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-locally-constant-map`]. Fix a choice of such maps. By More on Algebra, Lemma [original label: `more-algebra-lemma-Rlim-perfect-gives-complete`] we can find a finite complex $M^\bullet$ of finite projective $\Lambda^\wedge$-modules and isomorphisms $M^\bullet/I^nM^\bullet \to M_n$ in $D(\Lambda/I^n)$ compatible with the transition maps. To finish the proof we will show that K is locally isomorphic to

$$\underline{M^\bullet}^\wedge = \lim \underline{M^\bullet/I^nM^\bullet} = R\lim \underline{M^\bullet/I^nM^\bullet}$$

Let $E^\bullet$ be the dual complex to $M^\bullet$, see More on Algebra, Lemma [original label: `more-algebra-lemma-dual-perfect-complex`] and its proof. Consider the objects

$$H_n = R\mathcal{H}\mathrm{om}_{\Lambda/I^n}(\underline{M^\bullet/I^nM^\bullet}, K_n) = \underline{E^\bullet/I^nE^\bullet} \otimes^{\mathrm{L}}_{\Lambda/I^n} K_n$$

of $D(X_{pro\text{-}étale}, \Lambda/I^n)$. Modding out by I^n defines a transition map $H_{n+1} \to H_n$. Set $H = R\lim H_n$. Then H is an object of $D_{cons}(X, \Lambda)$ (details omitted) with $H \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n} = H_n$. Choose a covering $\{W_t \to X\}_{t \in T}$ with each W_t affine and weakly contractible. By our choice of $\underline{M^\bullet}$ we see that

$$\begin{aligned} H_n|_{W_t} &\cong R\mathcal{H}\mathrm{om}_{\Lambda/I^n}(\underline{M^\bullet/I^nM^\bullet}, \underline{M^\bullet/I^nM^\bullet}) \\ &= \underline{\mathrm{Tot}(E^\bullet/I^nE^\bullet \otimes_{\Lambda/I^n} M^\bullet/I^nM^\bullet)} \end{aligned}$$

Thus we may apply Lemma 29.5 to $H = R\lim H_n$. We conclude the system $H^0(W_t, H_n)$ satisfies Mittag-Leffler. Since for all $n \gg 1$ there is an element of $H^0(W_t, H_n)$ which maps to an isomorphism in

$$H^0(W_t, H_1) = \mathrm{Hom}(\underline{M^\bullet/IM^\bullet}, K_1)$$

we find an element $(\varphi_{t,n})$ in the inverse limit which produces an isomorphism mod I. Then

$$R\lim \varphi_{t,n} : \underline{M^\bullet}^\wedge|_{W_t} = R\lim \underline{M^\bullet/I^nM^\bullet}|_{W_t} \longrightarrow R\lim K_n|_{W_t} = K|_{W_t}$$

is an isomorphism. This finishes the proof. □

Proposition 29.7. Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let K be an object of $D_{cons}(X, \Lambda)$. Then K is adic constructible (Definition 29.4).

Proof. This is a consequence of Lemma 29.6 and the fact that a Noetherian scheme is locally connected (Topology, Lemma [original label: `topology-lemma-locally-Noetherian-locally-connected`]), and the definitions. □

30 Proper base change

In this section we explain how to prove the proper base change theorem for derived complete objects on the pro-étale site using the proper base change theorem for étale cohomology following the general theme that we use the pro-étale topology only to deal with "limit issues" and we use results proved for the étale topology to handle everything else.

Theorem 30.1. Let $f : X \to Y$ be a proper morphism of schemes. Let $g : Y' \to Y$ be a morphism of schemes giving rise to the base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal such that Λ/I is torsion. Let K be an object of $D(X_{pro\text{-}étale})$ such that

1. K is derived complete, and
2. $K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$ is bounded below with cohomology sheaves coming from $X_{étale}$,
3. Λ/I^n is a perfect Λ-module[10].

Then the base change map

$$Lg^*_{comp} Rf_* K \longrightarrow Rf'_* L(g')^*_{comp} K$$

is an isomorphism.

Proof. We omit the construction of the base change map (this uses only formal properties of derived push-forward and completed derived pullback, compare with Cohomology on Sites, Remark [original label: `sites-cohomology-remark-base-change`]). Write $K_n = K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. By Lemma 20.1 we have $K = R\lim K_n$ because K is derived complete. By Lemmas 20.2 and 20.1 we can unwind the left hand side

$$Lg^*_{comp} Rf_* K = R\lim Lg^*(Rf_* K) \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n} = R\lim Lg^* Rf_* K_n$$

the last equality because Λ/I^n is a perfect module and the projection formula (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-projection-formula`]). Using Lemma 20.2 we can unwind the right hand side

$$Rf'_* L(g')^*_{comp} K = Rf'_* R\lim L(g')^* K_n = R\lim Rf'_* L(g')^* K_n$$

the last equality because Rf'_* commutes with $R\lim$ (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-Rf-commutes-with-Rlim`]). Thus it suffices to show the maps

$$Lg^* Rf_* K_n \longrightarrow Rf'_* L(g')^* K_n$$

are isomorphisms. By Lemma 19.8 and our second condition we can write $K_n = \epsilon^{-1} L_n$ for some $L_n \in D^+(X_{étale}, \Lambda/I^n)$. By Lemma 23.1 and the fact that ϵ^{-1} commutes with pullbacks we obtain

$$Lg^* Rf_* K_n = Lg^* Rf_* \epsilon^* L_n = Lg^* \epsilon^{-1} Rf_* L_n = \epsilon^{-1} Lg^* Rf_* L_n$$

and

$$Rf'_* L(g')^* K_n = Rf'_* L(g')^* \epsilon^{-1} L_n = Rf'_* \epsilon^{-1} L(g')^* L_n = \epsilon^{-1} Rf'_* L(g')^* L_n$$

(this also uses that L_n is bounded below). Finally, by the proper base change theorem for étale cohomology (Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-proper-base-change`]) we have

$$Lg^* Rf_* L_n = Rf'_* L(g')^* L_n$$

(again using that L_n is bounded below) and the theorem is proved. □

[10] This assumption can be removed if K is a constructible complex, see [2].

31 Change of partial universe

We advise the reader to skip this section: here we show that cohomology of sheaves in the pro-étale topology is independent of the choice of partial universe. Namely, the functor g_* of Lemma 31.2 below is an embedding of small pro-étale topoi which does not change cohomology. For big pro-étale sites we have Lemmas 31.3 and 31.4 saying essentially the same thing.

But first, as promised in Section 12 we prove that the topology on a big pro-étale site $\mathrm{Sch}_{pro\text{-}étale}$ is in some sense induced from the pro-étale topology on the category of all schemes.

Lemma 31.1. Let $\mathrm{Sch}_{pro\text{-}étale}$ be a big pro-étale site as in Definition 12.7. Let $T \in \mathrm{Ob}(\mathrm{Sch}_{pro\text{-}étale})$. Let $\{T_i \to T\}_{i\in I}$ be an arbitrary pro-étale covering of T. There exists a covering $\{U_j \to T\}_{j\in J}$ of T in the site $\mathrm{Sch}_{pro\text{-}étale}$ which refines $\{T_i \to T\}_{i\in I}$.

Proof. Namely, we first let $\{V_k \to T\}$ be a covering as in Lemma 13.3. Then the pro-étale coverings $\{T_i \times_T V_k \to V_k\}$ can be refined by a finite disjoint open covering $V_k = V_{k,1} \amalg \ldots \amalg V_{k,n_k}$, see Lemma 13.1. Then $\{V_{k,i} \to T\}$ is a covering of $\mathrm{Sch}_{pro\text{-}étale}$ which refines $\{T_i \to T\}_{i\in I}$. □

We first state and prove the comparison for the small pro-étale sites. Note that we are not claiming that the small pro-étale topos of a scheme is independent of the choice of partial universe; this isn't true in contrast with the case of the small étale topos (Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-etale-topos-independent-partial-universe`]).

Lemma 31.2. Let S be a scheme. Let $S_{pro\text{-}étale} \subset S'_{pro\text{-}étale}$ be two small pro-étale sites of S as constructed in Definition 12.8. Then the inclusion functor satisfies the assumptions of Sites, Lemma [original label: `sites-lemma-bigger-site`]. Hence there exist morphisms of topoi

$$\mathrm{Sh}(S_{pro\text{-}étale}) \xrightarrow{g} \mathrm{Sh}(S'_{pro\text{-}étale}) \xrightarrow{f} \mathrm{Sh}(S_{pro\text{-}étale})$$

whose composition is isomorphic to the identity and with $f_* = g^{-1}$. Moreover,

1. for $\mathcal{F}' \in \mathrm{Ab}(S'_{pro\text{-}étale})$ we have $H^p(S'_{pro\text{-}étale}, \mathcal{F}') = H^p(S_{pro\text{-}étale}, g^{-1}\mathcal{F}')$,
2. for $\mathcal{F} \in \mathrm{Ab}(S_{pro\text{-}étale})$ we have
$$H^p(S_{pro\text{-}étale}, \mathcal{F}) = H^p(S'_{pro\text{-}étale}, g_*\mathcal{F}) = H^p(S'_{pro\text{-}étale}, f^{-1}\mathcal{F}).$$

Proof. The inclusion functor is fully faithful and continuous. We have seen that $S_{pro\text{-}étale}$ and $S'_{pro\text{-}étale}$ have fibre products and final objects and that our functor commutes with these (Lemma 12.10). It follows from Lemma 31.1 that the inclusion functor is cocontinuous. Hence the existence of f and g follows from Sites, Lemma [original label: `sites-lemma-bigger-site`]. The equality in (1) is Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-cohomology-bigger-site`]. Part (2) follows from (1) as $\mathcal{F} = g^{-1}g_*\mathcal{F} = g^{-1}f^{-1}\mathcal{F}$. □

Next, we prove a corresponding result for the big pro-étale topoi.

Lemma 31.3. Suppose given big sites $\mathrm{Sch}_{pro\text{-}étale}$ and $\mathrm{Sch}'_{pro\text{-}étale}$ as in Definition 12.7. Assume that $\mathrm{Sch}_{pro\text{-}étale}$ is contained in $\mathrm{Sch}'_{pro\text{-}étale}$. The inclusion functor $\mathrm{Sch}_{pro\text{-}étale} \to \mathrm{Sch}'_{pro\text{-}étale}$ satisfies the assumptions of Sites, Lemma [original label: `sites-lemma-bigger-site`]. There are morphisms of topoi

$$\begin{aligned} g : \mathrm{Sh}(\mathrm{Sch}_{pro\text{-}étale}) &\longrightarrow \mathrm{Sh}(\mathrm{Sch}'_{pro\text{-}étale}) \\ f : \mathrm{Sh}(\mathrm{Sch}'_{pro\text{-}étale}) &\longrightarrow \mathrm{Sh}(\mathrm{Sch}_{pro\text{-}étale}) \end{aligned}$$

such that $f \circ g \cong \mathrm{id}$. For any object S of $\mathrm{Sch}_{pro\text{-}étale}$ the inclusion functor $(\mathrm{Sch}/S)_{pro\text{-}étale} \to (\mathrm{Sch}'/S)_{pro\text{-}étale}$ satisfies the assumptions of Sites, Lemma [original label: `sites-lemma-bigger-site`] also. Hence similarly we obtain morphisms

$$\begin{aligned} g : \mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale}) &\longrightarrow \mathrm{Sh}((\mathrm{Sch}'/S)_{pro\text{-}étale}) \\ f : \mathrm{Sh}((\mathrm{Sch}'/S)_{pro\text{-}étale}) &\longrightarrow \mathrm{Sh}((\mathrm{Sch}/S)_{pro\text{-}étale}) \end{aligned}$$

with $f \circ g \cong \mathrm{id}$.

Proof. Assumptions (b), (c), and (e) of Sites, Lemma [original label: `sites-lemma-bigger-site`] are immediate for the functors $\mathrm{Sch}_{pro\text{-}étale} \to \mathrm{Sch}'_{pro\text{-}étale}$ and $(\mathrm{Sch}/S)_{pro\text{-}étale} \to (\mathrm{Sch}'/S)_{pro\text{-}étale}$. Property (a) holds by Lemma 31.1. Property (d) holds because fibre products in the categories $\mathrm{Sch}_{pro\text{-}étale}$, $\mathrm{Sch}'_{pro\text{-}étale}$ exist and are compatible with fibre products in the category of schemes. □

Lemma 31.4. Let S be a scheme. Let $(\mathrm{Sch}/S)_{pro\text{-}étale}$ and $(\mathrm{Sch}'/S)_{pro\text{-}étale}$ be two big pro-étale sites of S as in Definition 12.8. Assume that the first is contained in the second. In this case

1. for any abelian sheaf $\mathcal{F}'$ defined on $(\mathrm{Sch}'/S)_{pro\text{-}étale}$ and any object U of $(\mathrm{Sch}/S)_{pro\text{-}étale}$ we have
$$H^p(U, \mathcal{F}'|_{(\mathrm{Sch}/S)_{pro\text{-}étale}}) = H^p(U, \mathcal{F}')$$
In words: the cohomology of $\mathcal{F}'$ over U computed in the bigger site agrees with the cohomology of $\mathcal{F}'$ restricted to the smaller site over U.
2. for any abelian sheaf $\mathcal{F}$ on $(\mathrm{Sch}/S)_{pro\text{-}étale}$ there is an abelian sheaf $\mathcal{F}'$ on $(\mathrm{Sch}/S)'_{pro\text{-}étale}$ whose restriction to $(\mathrm{Sch}/S)_{pro\text{-}étale}$ is isomorphic to $\mathcal{F}$.

Proof. By Lemma 31.3 the inclusion functor $(\mathrm{Sch}/S)_{pro\text{-}étale} \to (\mathrm{Sch}'/S)_{pro\text{-}étale}$ satisfies the assumptions of Sites, Lemma [original label: `sites-lemma-bigger-site`]. This implies (2) and (1) follows from Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-cohomology-bigger-site`]. □

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